

Next-Track Session Recommendation: Experiment Synthesis

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Abstract

This document is the living synthesis of the *next-track* sequence recommendation campaigns; the markdown reports in `experiments/` (filenames containing `nexttrack`) are the primary record. The task is next-item *retrieval*: given a listening *session* (an ordered sequence of tracks), rank the catalog and recover the next distinct track the user actually played, scored on a single leak-free chronological *next-distinct* split (7154 sessions / 19 402 items; 5723 train / 1431 test; cold-item rate 0.55). The headline metric is Recall@10, reported beside mean reciprocal rank (MRR) and the graded-relevance diagnostics artist@10 / genre@10 / artist_mrr. Twenty-one campaigns and one research memo are consolidated. A session GRU alone only *ties* the first-order Markov co-listening bar (R@10 0.1069) on Recall@10 and loses on MRR, and neither loss-tuning nor any network topology (depth, width, residual, direction) beats it. The breakthrough is *rank fusion*: an R+M z-blend (GRU $\times$ Markov) beats the bar by $\sim +61\%$ Recall@10 (0.1726, three-seed confirmed), because the two legs rank *different* correct tracks; a learned XGB stacker underperforms the fixed z-blend and adding redundant neural legs does not help. The one genuinely additive new signal is *content*: an item-representation sweep found latent *dimension* (not the compressor) to be the lever, with a peak at PCA-192, and on that rich space a three-leg R+M+C z-blend (GRU $\times$ Markov $\times$ content-kNN) is the current best-on-record row at Recall@10 0.1859, MRR 0.1025. A subsequent literature-fit campaign then superseded that crown: a supervised *learned content projection* ($R'+M+C'$), which fits a linear metric map of the PCA-192 item space toward next-track adjacency by InfoNCE and retrieves the GRU and content legs in it, reaches Recall@10 0.2117, MRR 0.1220 (paired Δ over R+M+C +0.0259, CI above zero), is *seed-robust* across three seeds and *cold-reaching*, and hardens on a second disjoint 0.64-cold split (paired Δ +0.0426, CI above zero) — resolving the two thin caveats (determinism-only confirm, +0.0007 margin) that had bounded the R+M+C crown. Unlike every earlier ranking predictor it is registered, promoted and *servable* (`blend-gru-markov-content-proj`, best-models rank 1), closing the productization gap; an item-vector representation exploration and a continuous “sounds-alike” *music@10* metric corroborate the champion without dethroning it. A final measurement campaign adds a display-only *session-holisticness* metric suite (artist_adj@10 / mood_coh@10 / ild@10) that fingerprints the champion as the *most* artist-eager model measured (artist_adj@10 0.610) — with the eagerness sourced from its Markov bigram leg, not the RNN, and the GRU and LSTM statistically tied — and finds an eval-time MMR re-rank ($\lambda \approx 0.9$) a near-free holisticness lever while a training-time anti-eager regularizer is weak; nothing is promoted and the primary metric stays Recall@10. Two final architecture campaigns then close the remaining architecture axes, and both are nulls: a jointly-trained *dual-tower* GRU over paired causal views of the prefix fails to beat a parameter-matched single tower — the fourth independent learned-combiner null, and the one that answers the “train the fusion end-to-end” rebuttal — and a learned per-step *pre-encoder* in front of the recurrence is not merely useless but actively *harmful* in a single tower (−0.0140 Recall@10, CI below zero), a loss traced by a counterfactual interpretability read to half-wave rectification of a zero-centred latent deleting exactly its negative half (`frac_exact_zero` 0.5327 = `frac_negative_coords` 0.5327). A measured power analysis finally reframes all five architecture nulls: at $n = 1431$ the paired-bootstrap

CI half-width is 0.0124 on Recall@10 while a whole architecture family spans only 0.028, so large harms were detected reliably while moderate wins were never detectable at all — and *holisticness@10* exists in three mutually non-comparable code generations, which is why this synthesis keeps its board denominated in Recall@10. A final measurement campaign then changes the *surface* rather than the model: the shipped product serves an *autoregressive walk*, while *holisticness@10* scores a one-shot top-10 whose *mood_coh* anchors to a *growing* prefix centroid, so journey drift was invisible by construction (and the composite is $5.5\times$ more sensitive to its relevance factor than to its eagerness factor). Four walk-level metrics calibrated against the user’s own listening (73 632 real consecutive transitions, median step cosine 0.261, 27.0 % of them negative) show the shipped engine walking $2.5\times$ tighter than the listener it imitates (median step 0.661), and three deterministic retrieval-time controls — a seed anchor, a stride term, and a hard per-artist cap that *cannot be outscored* (two disjoint splits: $\Delta_{\text{artist_conc}} -0.527$ with $\Delta_{\text{music@10}} +0.026$, both CI clear of zero) — make a real-listening stride and a held vibe simultaneously reachable. On 40 identical held-out sessions under paired bootstrap the *dual-tower* family that three campaigns had refuted on recall is the winner on the stated objective (*latent/cummean*: stride error -0.2545 , genres $+1.50$, drift $+0.0351$, all CI clear of zero, vibe *held*), while pushing the single GRU or the champion to the same stride *costs* vibe — but *ground* straddles zero for every arm, so only journey *shape* is established, and the primary metric is unchanged. A closing architecture campaign then follows that reversal up on its own stated mechanism and closes the axis: a *bank* of N parallel bare-GRU towers under one linear head, dosed from $N = 2$ to $N = 4$ at a matched $3.00\times$ parameter budget, wins on neither surface — every arm ties the h256 baseline on one-shot recall while *five of eight lose* to a parameter-matched single GRU, the two arms that earn a walk trade at the primary cell both flip to *vibe* below zero at the deployed cell, and the fixed-reference *cummean* towers the dose hypothesis was built on are the *worst* arms in the batch. The decisive datapoint is a zero: four h211 towers and one h512 GRU return the identical 186 hits of 1431, paired Δ 0.000000, so topology and width are interchangeable at this budget. The batch’s one CI-clear result is a by-product — at byte-identical parameters dropout 0.1 beats dropout 0.0 by $+0.0049$ recall@10 — which matters because the *fusion_layers* = 0 path constructs *zero* dropout modules, so every f0 arm on record, including the deployed engine, trained under-regularized. Six further campaigns close the score-time and depth questions. An eval-time *MMR* λ frontier, run natively on both anchors, is the **first confirmed crown lever on record** — three single-GRU arms clear the prior all-time *holisticness* bar at a recall whose paired CI straddles zero, with three of the four crown factors (diversity, eagerness, *and* graded relevance) improving at once — and it resolves its pre-registered mechanism question unanimously: sonic diversification de-concentrates *artists* 1.59 to $3.06\times$ faster than it reduces seed-echo in 8 of 8 arms, so an explicit artist penalty is not the indicated sequel. A purpose-built *rollout* metric (*hold@N*, calibrated on 2135 real train sessions with an anti-gaming term inside the metric) then returns a sixth architecture null in the deployment regime, and finds that *per-seed* variance dominates model identity by two orders of magnitude — which is a product recommendation (serve an ensemble) rather than a modelling one. The largest single recall gain in the whole arc then comes from neither: *markov_gate*, a nine-line score-time change that makes the Markov leg *abstain* on the 55.6 % of queries where it has no bigram evidence, because the blend’s scale-invariant normalizer had been re-inflating an evidence-free back-off to a full one-third vote. It buys $+0.02027$ [$+0.01048, +0.03075$] Recall@10 on the canonical split and $+0.01957$ on a disjoint 64 %-cold one (R@10 0.2320, 332 of 1431 — the best on record), with the two warm evidence buckets *byte-for-byte identical*, which is a mechanical proof that no global static leg weight could have found it and which narrows the standing learned-combiner pitfall to *learned score-level fusion* rather than per-query evidence gating. Composed with MMR it is also the first crown arm that is recall-neutral against the champion at $3.6\times$ its *holisticness*. Against that, the *training-time* anti-eager regularizer is refuted with a mechanism — its margins were calibrated from the item→item transition tail while the hinge acts on prediction→current, whose ceiling sits below the quantiles used, so four arms are *provably inert* (loss traces bit-identical to the control at β up to the schema cap) and the arms that do fire destroy relevance before they help. And a

stage-stack depth campaign closes the serial branch of the architecture axis on *both* surfaces, with the sharpest instance yet of a dissociation that should change how future searches are run: its best-fitting arm holds the *lowest* validation loss in the batch and retrieves 55 fewer hits.

1 Introduction

This document is the living synthesis of the next-track sequence-recommendation campaigns; the markdown reports in **experiments/** are the primary record. Each section transcribes (never re-computes) the quantitative results of a campaign, preserves its run and dataset identifiers verbatim for traceability, and adds the rationale and interpretation the terse reports omit. This instance was cloned from a sibling taste-fit / rotation lab and retargeted to a sequence ranking goal; the rotation campaigns are a separate line of work and are *not* covered here.

Task and split. The lab predicts, within a listening *session*, which track the user plays *next* — a next-item retrieval task. The model consumes the ordered session prefix and scores every catalog item; candidates are ranked and the held-out next *distinct* track is the target (a leave-last-out evaluation that excludes prefix items, matching the Markov parity convention). All work runs on one sequence artifact, **seq-20260713-014226** (7154 sessions / 19 402 items), split *chronologically by session start* into 5723 train / 1431 test at cut 2024-08-24. The defining property of this split is its **cold-item rate of 0.55**: more than half of test-session target tracks never appear in any training session. This is the leak-free forward reality, and it caps absolute recall by construction — every result is read against the incumbent baseline bar, never an absolute number.

Metrics. The headline metric is **Recall@10** (R@10, the fraction of test sessions whose held-out next track appears in the top ten retrieved), reported beside **mean reciprocal rank** (MRR, the average inverse rank of the true next track) and Recall@20. Since 2026-07-14 the evaluation also emits first-class *graded-relevance* metrics over the same prefix-excluded ranking (Section 7): **artist@k** (a top-*k* candidate shares the true next track’s artist), **genre@k** (likewise on primary genre), and **artist_mrr** (MRR of the first same-artist candidate). All confidence intervals are bootstrap 95% CIs over 2000 resamples of the 1431 test sessions, and the decisive statistic is the *paired* per-session Δ against a named reference. A board *win* requires beating the reference on *both* R@10 and MRR beyond non-overlapping marginal CIs (the strict bar), then a reproduction confirm; the paired- Δ CI-entirely-above-zero test is the more powerful alternative statistic (it crowned both champions) and can disagree with the strict bar. The graded metrics are the honest scorecard: exact recall is modest by construction, but the champion lands the right *artist* in the top ten $\sim 32\%$ of the time and the right *genre* $\sim 40\%$ — far more useful than the $\sim 17\%$ exact number the cold-item rate imposes. Since 2026-07-20 the leaderboard also carries **music@10**, a continuous “sounds-alike” graded metric (Section 14): the mean best cosine, in a balanced intrinsic musical-distance space, between the true next track and the top-ten retrieved candidates — a smooth superset of Recall@10 that credits landing something that *sounds like* the truth on an exact miss. It is display-only; the primary metric stays Recall@10.

The standing champion. The current best-on-record *row* — and, since 2026-07-18, a *registered, promoted and servable* model — is a supervised *learned content projection*, $R' + M + C'$: a session GRU $\times$ a first-order Markov co-occurrence model $\times$ a content-kNN leg, where the GRU (R') and the content leg (C') retrieve in an InfoNCE metric map of the PCA-192 item space fit toward

next-track adjacency, at R@10 0.2117, MRR 0.1220 (Sections 11–12). It supersedes the three-leg $R+M+C$ z-blend on the same PCA-192 space (R@10 0.1859; Section 10), which had itself dethroned the two-leg $R+M$ z-blend on the 64-dim song-autoencoder space (AE-64, R@10 0.1726; Section 6). The projection is *seed-robust* and *two-split hardened*, and it is deployable via `POST /api/models/blend-gru-markov-content-proj/predict` — closing the productization gap the earlier off-server fusion drivers left open. The prior champions remain useful references (R+M is still the leanest two-leg blend), but they are no longer the deployable anchor. One qualification has since been added: the best *recall* figure on record is no longer the champion as promoted but the same model with its Markov leg *gated* to abstain where it has no bigram evidence (R@10 0.2320; Section 22), a nine-line score-time change carrying a promotion recommendation that has not yet been executed.

2 Background: the sequence task, the baselines, and the metrics

This section gives the minimum theory for a reader who knows machine learning but not this repo. It is keyed to the project’s living facts file and grows only when a new predictor family or metric enters the reports.

The models and baselines in play.

Session GRU / LSTM (seq-nexttrack). A recurrent neural network that reads the ordered session prefix one item at a time (each item represented by its item-latent vector) and emits a predicted next-item latent; candidates are ranked by cosine similarity to that prediction. The GRU (gated recurrent unit) and LSTM (long short-term memory) are the two recurrent cells tried. The recurrence captures *session context* — the trajectory of the listening session — rather than a single previous item.

Training loss: InfoNCE versus cosine. The sequence model can be trained to regress the next-item latent by *cosine* loss (pull the prediction toward the true next latent) or by **InfoNCE**, a contrastive objective that pulls the prediction toward the true next item and pushes it away from a batch of negatives. InfoNCE is the single most important lever in the whole arc: it is $\approx 2.3\times$ cosine on Recall@10, because ranking retrieval rewards a representation that *separates* the true next item from distractors, which a contrastive loss optimizes directly and a pointwise regression does not.

First-order Markov co-occurrence (the bar, B2). A train-only transition model: a `track_bigram` counting “what track follows what” with artist/genre back-off for unseen pairs (the transition term the sibling app already ships). It directly *memorizes* adjacency, which is exactly what exact next-track ranking rewards, and it is the reference bar every arm is measured against. A second-order Markov (trigram) back-off was also tried and is essentially neutral — sessions are too short/sparse for a trigram to fire, so it falls through to the bigram.

item2vec. A skip-gram (SGNS) co-listening embedding trained on the *train* sessions only (a leak-safe analogue of word2vec, where “words” are tracks and “sentences” are sessions); candidates are ranked by cosine to the mean-pooled prefix. On 5723 train sessions it is too sparse to localize and is the weakest leg on record; the full-history co-listening vectors are deliberately *not* used because they leak across the split.

content-kNN. A training-free retrieval leg: cosine of each candidate’s item latent to the prefix latents (max-aggregated). It is weak standalone on exact recall but captures *vibe* similarity the counts miss (its genre@10 already edges the Markov bar), and it is the one genuinely additive third leg for the fusion blend.

Rank-fusion z-blend versus a learned stacker. Two ways to combine legs. The **z-blend** normalizes each leg’s candidate scores (z-score) and takes a weighted sum ($\alpha \cdot z(\text{leg}) + \dots$) — a fixed, parameter-light fuser. The **XGB stacker** is a gradient-boosted meta-learner over per-candidate features (leg scores + popularity + recency + artist/genre-match), trained pointwise with sampled negatives. Across two campaigns the fixed z-blend *beats* the learned stacker decisively, because the stacker’s pointwise binary training does not optimize the ranking objective as well as directly fusing normalized scores.

Learned content projection (supervised metric map). A linear map W of the item-latent space fit by **InfoNCE** over the train consecutive pairs ($u \rightarrow v$) (self-loops dropped, Markov convention), so cosine in W -space reflects “likely to follow.” It is leak-free (train pairs + intrinsic content latents only, never the split). Both the content-kNN leg (C') and the GRU (R') then retrieve in the projected space; Markov is unchanged. It is the one *supervised* alternative to the unsupervised, variance-ordered PCA geometry, and — because content is the only channel with cold reach — it lifts *both* legs that share the space, making it the current champion (Section 11).

Dual-tower recurrent fusion (seq-dualgru). Two *independent* recurrent towers (separate weights) read the prefix in parallel over configurable strictly-causal *views* of it — **latent** (the raw item vector), **delta** (the step-to-step difference) or **cummean** (the running session centroid) — and their concatenated per-step hidden states are read out by a fusion head (**fusion_layers** = 0 for a pure linear read, = 1 for a 256-wide MLP). It reuses the **seq-nexttrack** objective, batching, early stopping and evaluation verbatim, so it isolates architecture as the only variable. The motivating idea is that two towers should help only if they see genuinely different information; Section 16 tests that and refutes it.

Per-step pre-encoder. A nonlinear re-embedding placed *in front of* the recurrence — [Linear, ReLU, Dropout] applied at each timestep, exposed as **pre_hidden/pre_layers**. It is an unusually sharp test object: a GRU’s own input transform is *already* linear, so a linear pre-encoder at width $\geq$ the input dimension is exactly as expressive as no pre-encoder at all, and the *rectifier* is the only change — which is what makes Section 17’s null mechanistically attributable rather than merely negative.

Item-latent spaces: song-AE versus PCA, and the wrong-collection trap. The sequence model predicts *into*, and retrieves *over*, a per-item latent space, so the choice of that space is a first-class axis. Three families appear:

AE-64 (spotify_tracks_song_ae). A 64-dim nonlinear *song-autoencoder* latent, the original champion item space. An autoencoder compresses each item by reconstructing it through a bottleneck; its equal-weight-per-block reconstruction loss spreads capacity evenly across content, acoustic and catalog blocks.

PCA- k (spotify_tracks_song_pca k). A *linear* principal components compression at dim $k \in \{32, 64, 128, 192, 256\}$, which orders dimensions by explained variance and decorrelates them. As Sections 8–9 show, this variance-ordered, decorrelated geometry is *easier* for the GRU to

regress into than the AE’s entangled equal-weight latents, so PCA beats the AE at every $\text{dim} \geq 64$.

The wrong-collection trap. Both of the above are fit *only* from the leak-safe `spotify_tracks_content` source (intrinsic content/acoustic/catalog fields; no behavioral or target field). The default `spotify_tracks` 200-dim collection is a *behavioral* co-listening embedding that soft-leaks replay behavior and must **never** be used as an item space.

3 Next-track leaderboard

Table 1 is the standing best-on-record board. It is on the single leak-free next-distinct split and sorted by Recall@10. Since 2026-07-18 the sequence family carries a ranking serving contract, so the champion learned-projection blend is a *promoted, servable* model (`blend-gru-markov-content-proj`, best-models rank 1), not merely a board row (Section 12); the earlier blends below it, and the leg standalones, remain off-server references. The item spaces are all fit from the leak-safe content source (Section 2). Since 2026-07-22 the evaluation also carries a *display-only session-holisticness* suite (`artist_adj@10` / `mood_coh@10` / `ild@10`; Section 15); it does not reorder this board (the primary metric stays Recall@10), but it fingerprints the champion as the most artist-eager row. Since 2026-07-25 the board also carries three *refuted* architecture datapoints — the widest single-GRU capacity control, the best dual-tower arm and the best single-tower pre-encoder arm (Sections 16–17). They are listed because they *bound* their axes, not because they are candidates: none beats the single-GRU baseline by the paired statistic and none was registered. Since 2026-08-02 the board carries two more such bounding rows, the widest single-GRU point (h512) and the best parallel-tower-bank arm, which return the *identical* hit count at a matched parameter budget — neither is significant against the h256 baseline, and together they close the tower-bank axis (Section 26). Since 2026-07-27c the board’s top row is the *evidence-gated* champion (Section 22): making the Markov leg abstain on the 55.6% of queries where it has no bigram evidence lifts Recall@10 to 0.2320 (332 of 1431), paired CI above zero on two disjoint splits, with MRR and both graded metrics up. It is the best recall figure on record, and it is a promotion *recommendation* only — nothing was registered or promoted — so the servable champion below it is unchanged. On the server-effective primary metric, and on why this board stays denominated in Recall@10 across three holisticness generations, see Section 18.2.

4 The sequence sweep: a GRU-infonce ties the Markov bar

Rationale. The instance’s opening campaign asked the founding question: does a session-context sequence model — a GRU/LSTM over the ordered prefix, predicting the next track’s 64-dim song-AE latent and retrieving by cosine — beat the first-order Markov transition model the sibling app already ships, *forward*, on the leak-free session split? Four arms crossed the recurrent cell (GRU, LSTM), capacity (`hidden` 128, 256) and, decisively, the training loss (cosine versus InfoNCE), at 60 epochs.

Results. No arm beat the bar. The best arm (M3, GRU `hidden`128 InfoNCE) reached a statistical *tie* with the Markov bar on R@10 (0.0894, paired- Δ CI includes zero) but *lost* on MRR (0.0388 versus 0.0694) and Recall@20. The dominant lever was the loss objective: InfoNCE was $\approx 2.3\times$ cosine on R@10 (0.0894 versus 0.0391), while capacity (`h256`) and the LSTM cell barely moved the cosine arms (all ~ 0.038 – 0.041). The popularity and recency floors sat near zero (R@10 0.0007).

Interpretation. A latent-plus-single-cosine-hop is a harder route to the *exact* next track than a co-listening bigram: the bigram directly memorizes “what follows what,” while the GRU must land near the right point in a 64-d content space and then retrieve, losing on precise ranking (MRR). The finding is consistent with a taste-drift reality — 0.55 cold items, identity/adjacency-driven behavior — and it set the agenda: InfoNCE is the base loss, and the natural next arm is a *blend* of the GRU’s candidate scores with the Markov bigram. *Source: 2026-07-13-nexttrack-sequence-sweep.md.*

5 Topology scan: depth, width and direction are a spent lever

Rationale. With the loss fixed at InfoNCE, this scan isolated network *topology*: does depth (stacked layers), width (`hidden 256`), an inter-layer residual stack or bidirectionality lift the sequence model over the Markov bar? Because a bidirectional RNN under per-step teacher forcing would leak future items, the bidirectional arm used a leak-free last-item objective (encode the full prefix, predict only the held-out next item).

Results. Refuted. No arm beat the bar on both metrics. The best arm (width `h256`, R@10 0.0957) is a *tie* on R@10 and loses on MRR (0.0411 versus 0.0694). Depth is flat 1→2 layers (0.0894→0.0922, in-band) and *hurts* at 3 layers (0.0825, out-of-band below the bar); the residual stack over depth-2/h256 added nothing. The bidirectional arm collapsed (0.0482), but this is an *objective* artifact, not a directionality verdict: the leak-free last-item objective supplies ~ 1 training target per session versus teacher forcing’s ~ 13 next-step pairs, so it trained on $\sim 13\times$ less signal.

Interpretation. Architecture is not the lever for next-track on this corpus. The transferable latent-retrieval signal caps at R@10 ~ 0.09 –0.10 across depth, width, residual and direction, below the co-listening bigram — consistent with the whole arc that the loss objective moved the needle and capacity/topology do not. The one remaining unexplored lever was the candidate framing / fusion, which the next campaign pulled decisively. *Source: 2026-07-13-nexttrack-topology-scan.md.*

6 The GRU×Markov fusion champion, and why more ensembling fails

Rationale. The sweep and the topology scan agreed that the best single GRU only *ties* the Markov bar on R@10 while losing on MRR. This campaign asked whether *combining* models beats the bar, and how far ensembling goes. Three legs were defined, each emitting full-vocab candidate scores: **M** = the train-only Markov bigram (the bar), **R** = the GRU-infonce-h256 recurrence, and **A** = an ANN bag-of-prefix MLP (a feed-forward twin of R). They were combined by a weighted z-blend and, as a learned alternative, an XGB stacker.

Results. The **R+M z-blend** ($\alpha=0.5$) *beats the bar decisively*: R@10 0.1726 [0.154, 0.192], MRR 0.0963, a paired Δ over Markov of +0.0657 [+0.049, +0.083] (CI entirely above zero) — $\sim +61\%$ R@10, +39% MRR — on a robust plateau across $\alpha \in [0.25, 0.75]$ and reproduced across three seeds (0.1705 / 0.1726 / 0.1726). It was registered as the **seq-blend** predictor and the **blend-gru-markov** definition, the first next-track champion. Further ensembling then *failed* to improve on it (Table 2): R+A+M (0.1684) is slightly worse, R+A ($\approx$ GRU alone, 0.0894) is much worse, and both XGB stacker variants (M+R 0.1020; M+R+A 0.0887) barely clear Markov alone.

Interpretation. The win is about signal *diversity*, not model count. The GRU (latent similarity) and Markov (co-occurrence counts) rank *different* correct next-tracks, so fusing them roughly doubles the hit rate; piling on more of the *same* kind of signal — another neural latent pre-

dicator (A), redundant with R — adds nothing. The learned XGB stacker *underperforms* the simple z-blend, counter to the usual “stacking beats blending” intuition: its pointwise binary-with-negatives training does not optimize the ranking objective as well as directly fusing normalized scores. The remaining headroom is bounded by the 0.55 cold-item rate and requires *genuinely new signal*, not more model fusion — the hypothesis the next campaign tests. *Source: 2026-07-14-nexttrack-blend-and-ensemble.md.*

7 Graded-relevance metrics and new base learners: content is the only additive signal

Rationale. Two threads in one campaign. (A) Make the evaluation credit landing the right *band/vibe* even when the exact track is wrong — the observation that “the first choice is usually not the exact track, but the artist is usually right” — by promoting *artist@k*, *genre@k* and *artist_mrr* to first-class metrics wired end-to-end (`seq_common.eval_from_scores` → Rust Metrics / MetricsSpec → `domain.toml` → UI). (B) Try genuinely new leak-free base learners — **C** = content-kNN, **M2** = a second-order Markov trigram with back-off, and **I** = a train-only item2vec SGNS — scored on both the exact and the new graded metrics.

Results. The exact-metric path was byte-identical (purely additive keys); the champion reproduced R@10 0.1726 / MRR 0.0963 exactly. No new base learner beat the bar standalone (M2 0.1076 ≈ M 0.1069 — the trigram falls through to the bigram; item2vec 0.0517 is the weakest leg). Blended into the champion, **content is the only additive signal: R+M+C** (adding the content leg) is the top blend on MRR (0.0963→0.1014), artist@10 (0.3215→0.3298) and artist_mrr, and ties the champion on exact R@10 (paired Δ +0.0014, CI straddles zero). Item2vec *drags* the blend (R+M+I 0.1349); the trigram is neutral.

Interpretation. Content-kNN ranks *different* near-correct tracks (latent vibe similarity) than the GRU (session recurrence) and Markov (co-occurrence), exactly the diversity the fusion campaign predicted new signal would need. On the AE-64 space the content leg only *tied* the champion on exact recall while lifting the graded metrics, so it was carried forward as the graded-preferred third leg, not a new exact champion. That “tie” becomes a *win* once the item space is richer (Section 10): on PCA-192 the same R+M+C construction is the strongest on record. Co-listening item2vec and the trigram Markov added nothing on this data. *Source: 2026-07-14-nexttrack-graded-relevance-and-new-legs.md.*

8 The item-representation compression bake-off

Rationale. The champion’s item space — a 64-dim AE latent — was one arbitrary early choice, and the sequence GRU predicts *into* and retrieves *over* whatever geometry that space imposes. This campaign asks which *compression* is the best space for the GRU: linear PCA versus the nonlinear autoencoder, each at latent dim 32 / 64 / 128. Six alternative spaces were baked from the *same* leak-safe 384-dim content source (`spotify_tracks_content`), each fit unsupervised over all 23 529 items using only intrinsic content/acoustic/catalog fields, then scored under a *frozen* GRU readout (`gru-infonce-h256`), so the item latent space is the only axis that moves. The design bet was that the AE’s equal-weight-per-block reconstruction loss would *preserve* the small acoustic/catalog blocks that PCA’s text-dominated global variance would starve; the null was that dim, not the compressor, sets the ceiling.

Results. The parity gate passed: the AE-64 control reproduced its on-record R@10 0.0957.

All six spaces trained cleanly (Table 3; Figure 1). **PCA-128** is the single-GRU winner at R@10 0.1146, a paired Δ of +0.0189 [+0.0042, +0.0335] over the AE-64 control — a CI *entirely above zero*, the only arm that clears the control significantly — and it also tops MRR, artist@10 and artist_mrr. PCA-64 is the runner-up (0.1041, tie). The two 32-dim spaces sit far below the 64-dim control (both CIs below zero). The training-free content-kNN probe is flat at ~ 0.06 across all six spaces and does *not* track the GRU ranking.

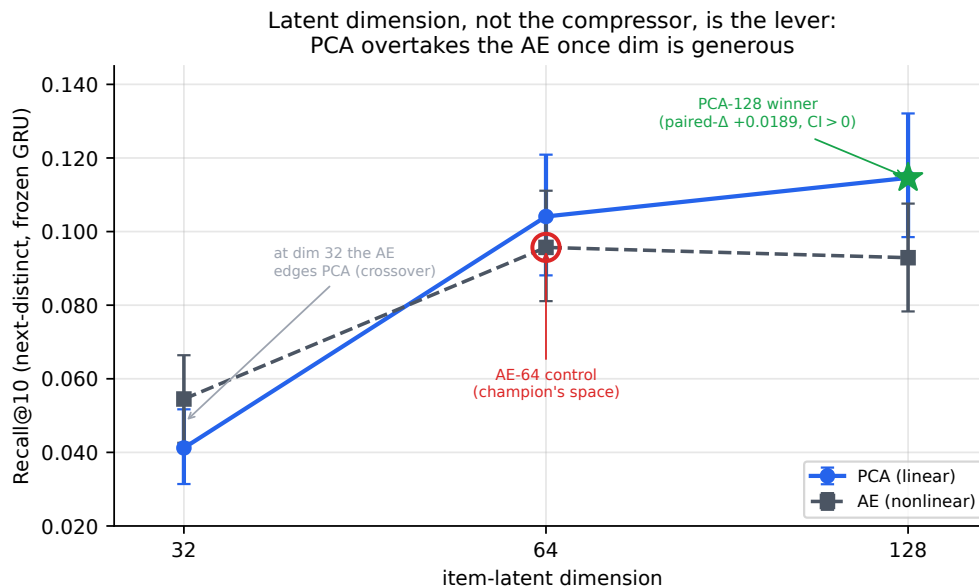


Figure 1: Latent dimension, not the compressor, is the lever. Both series climb steeply from dim 32 to 64; PCA (blue) then keeps climbing to 128 and *overtakes* the AE, while the AE (grey) plateaus — its 128-dim point actually dips below its own 64-dim control. At dim 32 the AE edges PCA, a low-capacity crossover that reverses once dim is generous. The AE-64 control (red ring) is the champion’s item space; the PCA-128 winner (green star) is the only arm whose paired- Δ CI clears zero. Error bars are bootstrap 95% CIs; the neighbouring CIs overlap, so the win rests on the *paired* test.

The single-GRU win does **not** survive into the production blend: re-fusing the winning PCA-128 GRU leg with the Markov leg scores R@10 0.1670, *below* the AE-64 champion blend (0.1726) with overlapping CIs. Nothing was promoted; PCA-128 was carried forward only as the sanctioned Phase-2 base space.

Interpretation. Latent dimensionality is the dominant lever and the compressor is secondary. The design’s acoustic-preservation hypothesis is *refuted*: at generous dim the linear PCA beats the nonlinear AE, and although AE-32 preserves the numeric/acoustic blocks near-perfectly it ranks near the bottom. What helps is a *variance-ordered, decorrelated* target geometry that the GRU finds easier to regress into than the AE’s entangled equal-weight latents. The AE’s edge at dim 32 is a purely low-capacity phenomenon that reverses once dim is generous. The flat content-kNN probe that fails to track the GRU ranking is a flagged divergence: the win comes from the *learnability* of PCA’s target geometry, not a demonstrably richer space. And the load-bearing caution is the blend result: a clearly stronger GRU leg bought *nothing* at the blend, because the Markov leg is representation-independent and the extra tracks the better GRU recovers overlap what Markov already supplies. *Source: 2026-07-15-nexttrack-compression-representation-bakeoff.md.*

9 The dimension ceiling: PCA peaks at 192

Rationale. With dimension established as the lever and PCA still climbing at 128, this leg extended the PCA sweep upward (PCA-192, PCA-256) to locate the real optimum before Phase 2 committed a base space, and added AE-192 to test whether $\text{PCA} \geq \text{AE}$ holds at higher dimension. The protocol was identical (same frozen `gru-infonce-h256` readout, same parity gate); the reference is PCA-128 (run-20260715-042100-a1d9d).

Results. The PCA R@10-versus-dimension curve **peaks at ~ 192** (R@10 0.1230) and *turns over* by 256 (0.1167); it was not still climbing. The full trend is $32 \rightarrow 0.041$, $64 \rightarrow 0.104$, $128 \rightarrow 0.115$, **$192 \rightarrow 0.123$** , $256 \rightarrow 0.117$ (Figure 2). PCA-192 is the point-optimum, but its paired Δ versus PCA-128 is $+0.0084$ $[-0.0007, +0.0189]$ — it misses the “CI entirely above zero” bar by 0.0007, a whisker — so **PCA-128 stands as the Phase-2 base**, with PCA-192 flagged as the confirm-and-promote candidate. $\text{PCA} \geq \text{AE}$ holds strongly and the gap is widest at 192 (PCA-192 0.123 versus AE-192 0.098); the AE plateaus flat at ~ 0.09 – 0.10 from dim 64 upward.

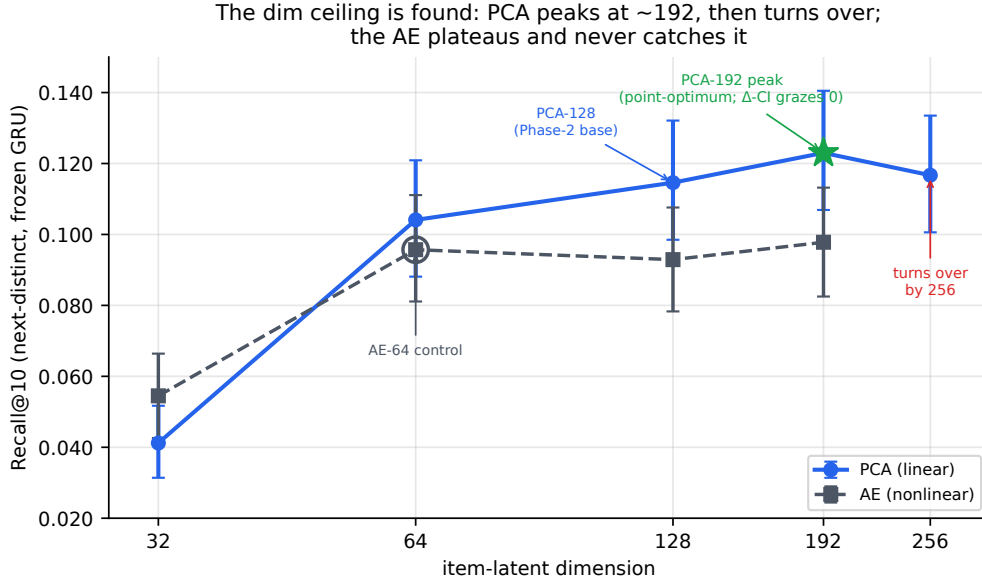


Figure 2: The dimension ceiling is found. Extending the compression sweep of Figure 1 to PCA-192/256 and AE-192, the PCA curve (blue) peaks at ~ 192 (R@10 0.123) and turns over by 256, while the AE (grey) plateaus flat at ~ 0.09 – 0.10 and never catches it — the $\text{PCA} \geq \text{AE}$ gap is widest at 192. PCA-192 is the point-optimum (green star) but its paired Δ over PCA-128 grazes zero, so PCA-128 stays the Phase-2 single-model base. Error bars are bootstrap 95% CIs over the 1431 test sessions.

Interpretation. Extra variance past ~ 192 (PCA-256 captures cumulative explained-variance ratio 0.997) is low-signal / noise dimensions the GRU cannot exploit and that slightly dilute the target geometry. The AE’s flat plateau reconfirms the refuted acoustic-preservation hypothesis at higher dimension: its equal-weight reconstruction objective keeps spending capacity on the tiny numeric/acoustic blocks (reconstruction $R^2 \approx 0.999$ at 192) rather than the high-variance text/catalog structure the GRU needs. As before, the training-free content-kNN probe stays flat and does not reproduce the GRU’s peak, so the dimension optimum is about target *learnability*, not intrinsic retrieval richness. *Source: 2026-07-15b-nexttrack-compression-dim-extension.md.*

10 The model-family sweep and the new champion, R+M+C on PCA-192

Rationale. With the compression axis settled (PCA-128 base, PCA-192 candidate), this campaign swept the model *family* on the chosen spaces: does a different recurrent cell (LSTM), a non-recurrent pooled MLP (ANN), or a *learned* XGB stacker beat the fixed GRU×Markov z-blend — and does adding the content-kNN third leg (R+M+C) on the *rich* PCA space finally beat the champion? A frozen held config ran across all arms; only the model family and the representation varied, and the server was serialized (`--max-runs 1`) to avoid a thread-oversubscription throughput trap.

Results. A **new best-on-record row**: the **R+M+C z-blend on PCA-192** (GRU × train-only Markov × content-kNN, equal thirds) reaches R@10 0.1859 [0.166, 0.207], MRR 0.1025, a paired Δ over the prior champion of +0.0133 [+0.0007, +0.0259] — entirely above zero, with no MRR regression (Table 4; Figure 3). The content leg is genuinely additive on the rich space, beating same-space R+M by +0.0154 (PCA-128) / +0.0147 (PCA-192), both CIs above zero — *stronger* than the AE-64 R+M+C that only tied (Section 7), because the richer, higher-EVR space makes the content leg pay off. The learned **XGB stacker badly underperforms** the fixed z-blend (0.080–0.087, losing even to its own single GRU base leg); the **LSTM is the best single model** (PCA-128 0.1272, paired Δ over the GRU +0.0126, CI above zero) but far below the blends; and the Phase-2 single base **stays PCA-128**.

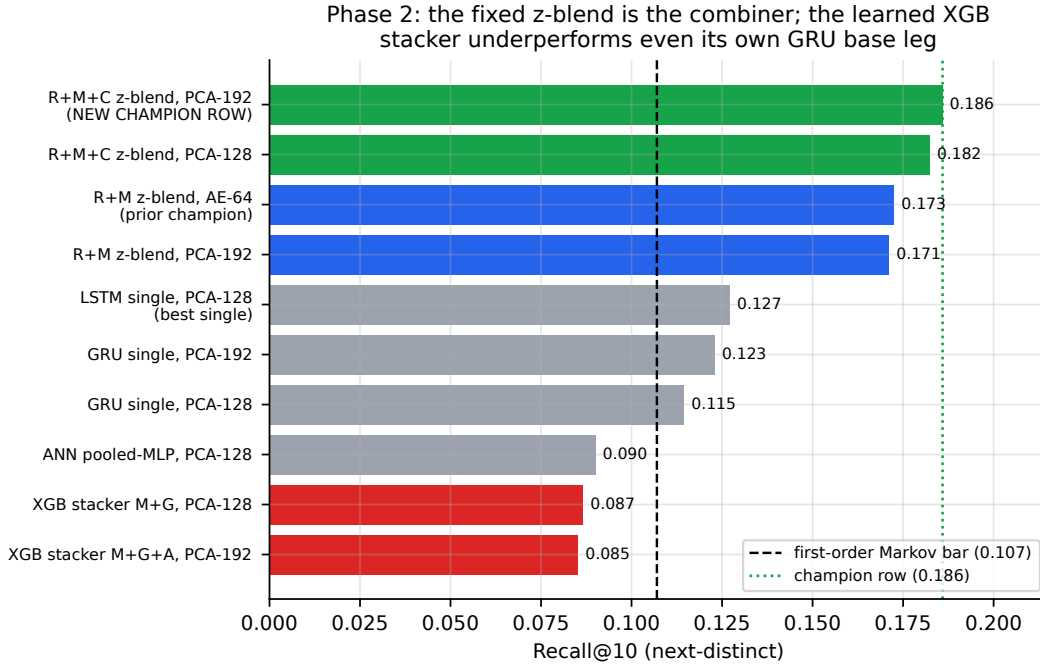


Figure 3: The Phase-2 model-family sweep. The fixed z-blend is the combiner of record: R+M+C on PCA-192 (green) is the new champion row, the content leg is genuinely additive on the rich space, and the learned XGB stacker (red) badly underperforms — it loses even to its own single GRU base leg. LSTM edges GRU as the best single model (grey) but sits far below the blends. The first-order Markov bar (0.107) is the incumbent reference; the champion row is at 0.186.

The crown, and its two honest caveats. The board-event candidate was confirmed by repro-

ducing the GRU-192 leg five times (the original plus four reruns) and recomputing R+M+C each time; the paired Δ over the prior champion was +0.0133 [+0.0007, +0.0259] *every* rerun, with the MRR lead intact, so R+M+C on PCA-192 was recorded as the new champion row. Two caveats travel with the crown and must not be omitted. First, **the reproduction spread was exactly zero** because `seq-nexttrack` training is deterministic on this CPU — all five GRU-192 checkpoints were byte-identical — so the confirm verified *determinism, not seed-robustness*; the only uncertainty quantified is the single-split bootstrap. Second, **the paired Δ lower bound is +0.0007**: the crown clears zero by a whisker on a single split. The crown is real by the pre-agreed criterion, but the margin over the prior champion is thin, and hardening it needs a second leak-free test split (or seed-varied legs once the blend is generalized to three legs).

Interpretation. The compression-plus-model-family thread is otherwise closed: PCA-128 is the single-model base, the LSTM is the best single model, the learned stacker is dead as a combiner, and the fixed z-blend is the combiner of record. The load-bearing result is that *content is the one leg that adds genuinely new ranking signal, and it pays off more on the higher-EVR PCA space* — the exception to the standing rule that single-model / representation gains do not carry to the blend (the R+M pair on the rich PCA spaces only ties the AE-64 champion). The productization gap remains: the champion is train-only and off-server, so the prior champion R+M on AE-64 stays the deployable anchor until the three-leg blend is registered with a `predict` subcommand. *Source: 2026-07-15c-nexttrack-phase2-model-sweep.md.*

11 The learned content projection: a new champion from the literature-fit campaign

Rationale. The R+M+C champion (Section 10) carried two thin caveats — a determinism-only reproduction confirm and a paired- Δ lower bound of only +0.0007 — and its headroom was capped by the 0.55 cold-item rate, which only *new* signal can attack. Guided by a survey of the established session-recommendation literature (the Jannach–Ludewig session-kNN line, sequential rules / factorized Markov, GRU4Rec training tricks, rank fusion and learning-to-rank, content / cold-start-aware session rec), this campaign pre-registered three tiers of classic methods and asked which actually improve the fit to *this* problem — a single-user, session-based, next-distinct *retrieval* task with 55% cold targets. All arms were measured off-server on the PCA-192 champion artifact `seq-20260715-131139`, reusing the sanctioned driver that reproduces the champion (R+M+C 0.1859) exactly; the decision statistic is the paired per-session Δ against that champion.

Results — the one win. A supervised **learned content projection** is a new best-on-record row. Fitting a linear metric map W of the PCA-192 content space toward next-track adjacency (InfoNCE over train consecutive pairs, self-loops dropped) and then retrieving *both* the content leg (C') and the GRU (R') in that projected space gives $R'+M+C'$ **z-blend R@10** 0.2117 [0.191, 0.233], **MRR** 0.1220 — a paired Δ over the champion of +0.0259 [+0.012, +0.041], entirely above zero (Table 5). Unlike the R+M+C crown it is **seed-robust** (three seeds; paired Δ +0.026 / +0.025 / +0.029, every CI > 0 — a genuine seed-varied confirm, since the projection has random init and shuffled batches) and **cold-reaching**: the projected content leg C' scores cold targets 0.152 $\approx$ warm 0.151, versus Markov’s warm 0.226 / cold 0.010, so it reaches never-seen items as well as seen ones — genuine content generalization, the one channel that reaches the 55% cold majority. The symmetric single map beats the directional two-tower, and rank-192 = rank-64 (the gain is the supervision, not the capacity). As a side effect the projected single GRU R' (~ 0.176) nearly matches the *old* champion blend and crushes the prior best single (LSTM 0.127).

The lanes that closed. Every other literature-fit direction failed to beat the fixed z-blend, ex-

actly as the survey predicted for a single-user, cold-heavy corpus (Table 6). Principled Markov smoothing (absolute-discounting + a Kneser–Ney continuation floor) is a real but tiny standalone estimator gain (0.1069→0.1104) that washes out in the blend (−0.0021, CI straddles 0 — the content leg already covers the marginal cold mass it recovers). A ranking-aware combiner has no honest headroom: even the *oracle* weights, fit directly on the test labels, reach only 0.1901 (+0.0042, straddles 0), and RRF rank fusion *hurts* recall (−0.029, CI<0, discarding the score magnitude the z-blend keeps). A gated V-SKNN session-kNN is warm-only against 55% cold targets and *drags* the blend (−0.038, CI<0) — the single-user setting strips the cross-user collaborative neighborhood that drives its multi-user benchmark wins. And GRU4Rec importance negatives (a larger support-weighted negative bank) give a within-noise standalone tick (+0.005) that also washes out (−0.0014, tie). **The only additive gain comes from new signal, not smarter fusion.**

Interpretation. The supervised content projection is the first genuinely-additive new signal since the content leg itself, and it works for the reason the survey flagged: content is the only channel with cold reach, and replacing the unsupervised, variance-ordered PCA geometry with a target-fit one lifts *both* legs that share it. The flashy borrowed methods — session-kNN, learned combiners, GRU-side loss tricks — are dead ends here because they either re-shuffle the same three legs (no headroom over the z-blend, even for the oracle) or import a collaborative assumption the single-user corpus violates. The one movable secondary axis is graded relevance: the oracle / RRF reweights and the projection nudge artist@10 / genre@10, a lever the exact-recall axis does not share. *Source: 2026-07-18-nexttrack-literature-fit-campaign.md.*

12 Registration, the objective / rank / τ scan, and second-split hardening

Rationale. The projection win landed as an off-server driver — the same productization gap the R+M+C crown had. This follow-through implemented the projection as a first-class capability of the seq-blend predictor (hyperparameters `projection`, `projection_rank`, `projection_objective`, `projection_tau`, ...: it fits the metric map from train consecutive pairs, saves it as `projection.npz` in the run dir, and bakes it into the promoted model for serving), then (1) reproduced the win on-server and registered + promoted it, and (2) ran a pre-approved objective / rank / τ ablation scan to confirm the default is the robust config to promote.

Registration. The verify run (run-20260718-191539-1b89c-seq-blend, definition default = InfoNCE / full-rank / τ 0.07) reproduced the off-server driver to four decimals — R@10 0.21174, MRR 0.12199 — and was promoted via the API as `blend-gru-markov-content-proj`, the projection matrix baked into the model directory for serving. GET `/api/best-models` confirms it at **rank 1** (0.21174), above the prior servable champion `blend-gru-markov-content` (rank 2, 0.18588). With the sequence family’s ranking serving contract this is a genuinely deployable model, servable via POST `/api/models/blend-gru-markov-content-proj/predict`.

The scan. The decision rule — re-promote only a config beating the default on *both* R@10 and MRR by more than the $\sim\pm 0.015$ noise band — confirms the default (Table 7). **InfoNCE** $\gg$ **BPR** is the sharpest axis: both BPR cells (0.1579, 0.1523) fall ~ 0.055 below the InfoNCE default and land *below even the old R+M+C champion* (0.1859), so the pairwise BPR map produces a materially worse geometry for both shared-latent legs. **Full-rank** $\geq$ **rank-64** (−0.006, a tie — reconfirming “the gain is the supervision, not the capacity”). And τ is **mildly sensitive and monotone up** over the tested range (0.05→0.1929, 0.07→0.2117, 0.10→0.2229) — but the nominal τ -0.10 top cell beats the default by only +0.0112 on recall, *inside* the noise band, so it is a

tie and the default **HELD**. Unlike the reweighting axis of Section 11, the graded metrics here *track* recall (best and worst together), because the scan moves the *signal quality* rather than re-fusing fixed legs.

12.1 Second-split hardening

Rationale. The projection crown was measured on a single leak-free split. This validation-only pass re-ran it — plus the two reference blends — on a **second, disjoint, leak-free split** to settle whether the win is split-specific. Nothing is promoted or re-promoted. The split seq-20260718-211238 is an *earlier-holdout* 60–80% chronological band on the same PCA-192 space (train 4,292 [earliest 60%] / test 1,431 [60–80% band]); the item space is byte-identical to the champion split, and the test window is **disjoint in both index and time** from the champion split’s 80–100% test. Its **cold-item rate** is 0.64 (versus 0.55), a stronger, independent cold-reaching stress test — so absolute R@10 lands *below* the first-split 0.212 by construction, and the verdict is the paired per-session Δ on *this* split, not absolute reproduction.

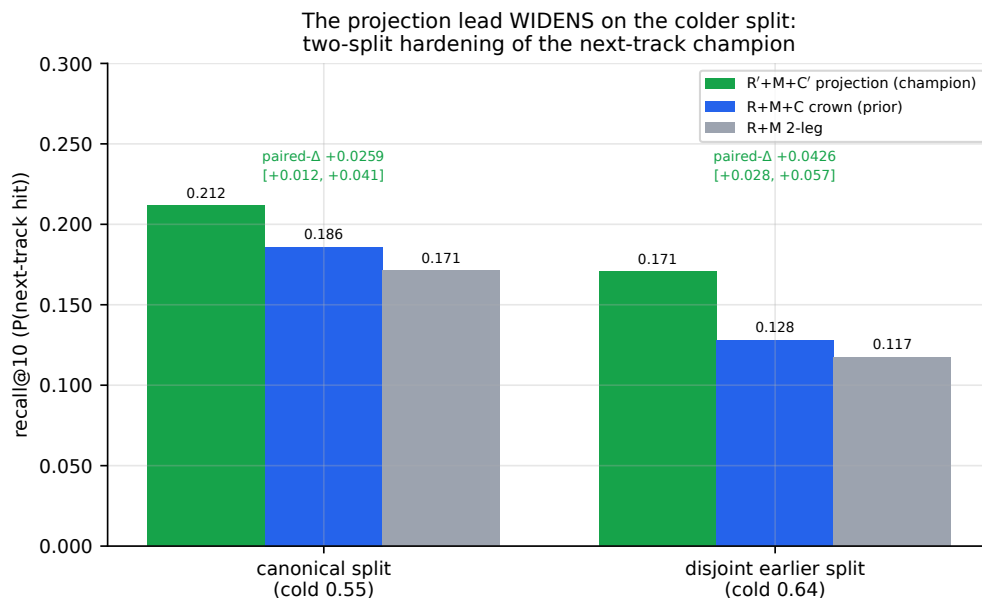


Figure 4: The projection lead *widens* on the colder split. On the canonical 0.55-cold split the projection champion beats the R+M+C crown by paired Δ +0.0259; on a disjoint, colder 0.64-cold earlier-holdout split the margin *grows* to +0.0426 (both CIs entirely > 0). Absolute R@10 is lower on the colder split by construction (more targets are cold); the verdict is the paired Δ , which is unambiguous on both. The mechanism: on a colder split the Markov / content-kNN legs of the crown reach fewer targets, and the projection’s cold-reaching C' leg carries proportionally more of the win.

Verdict — crown hardened. The decision rule (paired Δ vs R+M+C with 95% CI entirely > 0 and MRR not regressing) is met: paired Δ +0.0426 [+0.0280, +0.0573], MRR +0.0234 (no regression). The projection’s advantage is not split-specific — in fact the margin is *wider* here (+0.0426) than on the first split (+0.0259), consistent with the cold-reaching mechanism (Figure 4). Combined with the prior three-seed robustness and the cold-reach decomposition, the crown is now hardened on two disjoint leak-free splits — resolving both caveats that travelled with the R+M+C crown. *Source: 2026-07-18b-nexttrack-projection-registration-and-scan.md.*

13 Item-vector representation: is the geometry the bottleneck?

Rationale. The compression axis (Sections 8–9) found latent *dimension* to dominate but never asked whether the *composition* of the 192-d item vector — the equal-weighted 4-block PCA (text / numeric / acoustic / categorical) — is itself a handicap. An explained-variance-ratio (EVR) preflight said it might be: the raw input variance budget is 44% acoustic + 43% numeric versus only 2.5% text (text is L2-normalized, so its per-dimension scale is tiny), so full-matrix PCA-192 spends **87% of its variance budget** reconstructing the numeric + acoustic blocks to $R^2 = 1.000$ while the signal-carrying text block reaches only $R^2 = 0.717$; per-column standardize (whitening) flips the captured budget to 73% text / 21% categorical (text R^2 0.905) (Figure 5). Eight alternative item-latent spaces (whitening $\times$ acoustic-drop, text-only, text+categorical, and the musical-distance metric geometry) were built parity-gated to the champion split (vocab / split byte-identical; only `item_latents.f32` differs) and each evaluated under *two* readouts: **(A)** the champion projection `blend-gru-markov-content-proj` (the InfoNCE map re-fits on that space — the space’s ceiling under the winning recipe) and **(C)** the frozen `gru-infonce-h256` single (raw representation quality). Validation-only; nothing promoted.

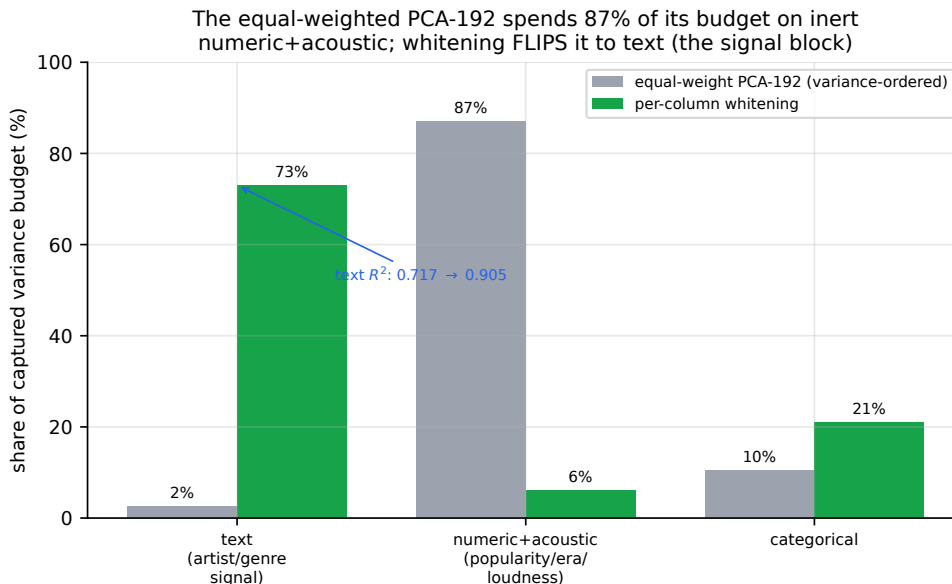


Figure 5: Why the item vector is a handicap for a raw model — and why the projection is the fix. The equal-weighted PCA-192 orders dimensions by raw variance, so 87% of its captured budget reconstructs the behaviorally-inert numeric + acoustic blocks (popularity / era / loudness) while the signal-carrying text block (2.5% of raw input variance, L2-normalized) is under-preserved (R^2 0.717). Per-column whitening flips the captured budget to 73% text / 21% categorical (text R^2 0.905). The learned InfoNCE projection learns this same reweighting from data.

Verdict — exact recall. **Readout A: no variant beats the PCA-192 champion with paired Δ CI entirely > 0 .** The two nominally-positive spaces (V3 std-noaco +0.0049, V5 balanced-pca192 +0.0042) straddle zero; two spaces are significantly *worse* (V4 textcat -0.0168 , dropping numeric+acoustic costs the projection real signal; V6 sonic-64 -0.0182 , the expected 64-d handicap). **Readout C: six of eight variants beat the frozen GRU with paired Δ CI entirely > 0** (+0.025 to +0.036; best V5 balanced +0.0356, V3 std-noaco +0.0342). So the raw full-matrix PCA-192 geometry *is* a handicap — suppressing the loud acoustic / numeric directions

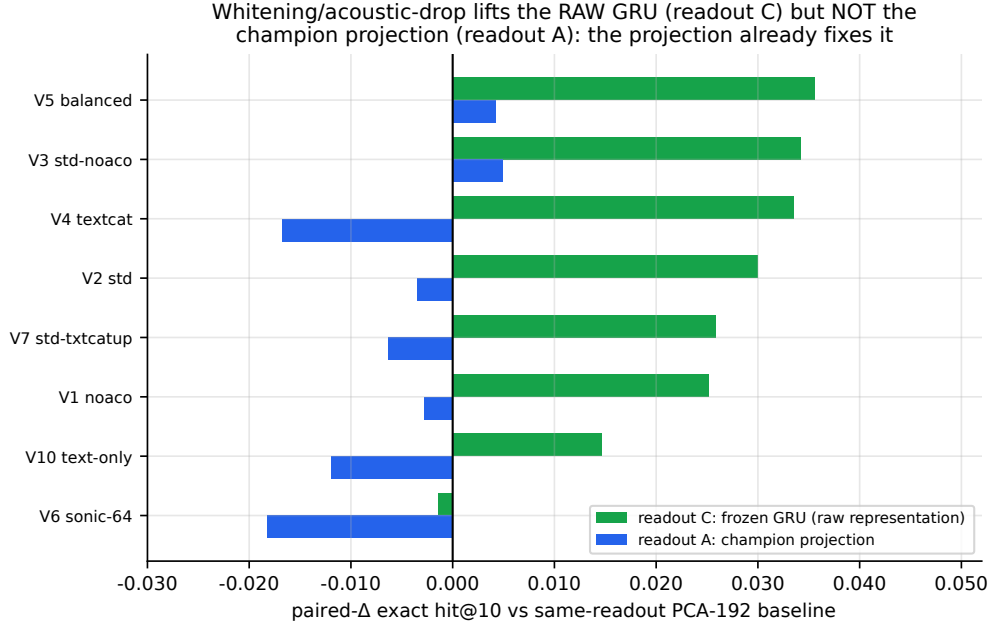


Figure 6: The representation handicap is real for the *raw* GRU but *absorbed* by the champion projection. Six of eight whitening / acoustic-drop / metric spaces lift the frozen GRU (readout C, green; all six with paired-Δ CI entirely > 0 , $+0.025$ to $+0.036$), but none lifts the champion projection (readout A, blue; the two nominally-positive spaces straddle 0, and V4/V6 significantly regress). The learned InfoNCE map already re-weights away the loud acoustic / numeric directions, so whitening the input first is redundant at the champion ceiling: whitening and the projection are the same fix.

lifts the raw GRU by $\sim 20\text{--}29\%$ relative — **but the gain is fully absorbed by the projection:** whitening lifts readout C, not readout A, because the learned InfoNCE map already re-weights away those directions from data. Whitening and the learned projection are the *same fix*, and the champion has already pulled it (Figure 6).

Graded and music@10. The representation levers move the *graded* axis more cleanly than exact recall: whitening lifts the frozen GRU’s genre@10 to ~ 0.47 (V7 std-txtcatup 0.4717, V3 std-noaco 0.4710 vs baseline 0.4500), and under the projection a text-only space (V10) posts the readout-A maxima on artist@10 (0.3466) and genre@10 (0.4144) while trading -0.0119 exact recall. The continuous **music@10** metric (Section 14) *corroborates* the exact-recall verdict rather than overturning it: the projection champion leads on it (0.4538) over the R+M+C crown (0.4491), and under the champion projection it is flat across the eight representations ($\sim 0.447\text{--}0.460$). Re-basing the geometry moves band / vibe matching, but exact next-track adjacency at the champion ceiling is immovable — which is precisely the axis the supervised projection owns.

Interpretation. The equal-weighted PCA-192 item vector is a genuine handicap for a *raw* model, and the learned projection is exactly the right response to it — this campaign independently validates *why* the projection is the champion (it is the representation fix, learned) and confirms there is no free lunch left in re-basing the input geometry. The one caution: dropping numeric+acoustic *entirely* (V4 textcat) hurts the champion even though it helps the raw GRU, so the numeric block carries something the projection uses — keep numeric, whiten / drop only acoustic. The most promising open direction is composing the two fixes: fitting the projection *on* a whitened / std-noaco / metric space to see whether it reaches further or is redundant. *Source:*

14 The music@10 metric: the recall champion is not the musically-closest

Rationale. Exact Recall@10 is unforgiving on a 55%-cold single-user corpus, and the artist@10 / genre@10 diagnostics are coarse category hits. This campaign introduces a *continuous* “sounds-alike” graded metric, **music@10**, and back-fills it across every next-track run. It is defined as the mean over test cases of the best (max) cosine, in a *balanced* intrinsic musical-distance space (spotify_tracks_content_metric: a block-weighted, partial-whitened fusion of the whitened 64-d song-AE latent, the 384-d name/tag text embedding, five metadata numerics, and a per-genre AE prototype), between the true next track and the top-ten retrieved candidates. A top-ten that contains the exact truth scores 1.0, so music@10 is a *smooth superset* of Recall@10 — it credits landing something that *sounds like* the truth even on an exact / artist / genre miss. It is display-only; the primary metric stays Recall@10.

Results. music@10 was recomputed offline and non-destructively across all 53 next-track runs (from each run’s stored `predictions.json` and its dataset `items.json`, on the shared 1431-case split). It **correlates** with Recall@10 ($r = 0.609$) — a better ranker also lands closer misses — but it is far from redundant, and it crowns a *different* winner (Table 10; Figure 7). The Recall@10 champion is a **seq-blend** run (0.223), but the *musically-closest* predictions come from **seq-nexttrack** — which regresses the next track’s content latent directly and retrieves by cosine — at best music@10 0.482 and the highest family mean (0.448). When a next-track model misses the exact track, it still tends to surface something that *sounds* right; the **seq-popularity** floor is correctly worst (music@10 0.185), confirming the metric is not trivially saturated.

Interpretation. music@10 is a genuinely distinct, non-saturated view of next-track quality: it rewards *sonic* proximity, which the exact-hit metric ignores. The recall-versus-music decoupling has a clean mechanism — a model that regresses the content latent (**seq-nexttrack**) lands *near* the truth in content space even when it misses the exact next track, so its misses “sound right” more often than the blend’s, which optimizes exact rank. This is why the metric is display-only and must not silently become a selection objective: promotion still ranks by Recall@10. The metric inherits a mild circularity from the *balanced* dial (the text block embeds the artist name; the genre block is a genre prototype), so it partly rewards same-artist / genre proximity — intended for “sounds / feels alike,” not a perceptual ground truth. A companion GET `/api/pathfinder/similar` endpoint serves “songs like this” over the same index. *Source: 2026-07-20-musical-distance-metric.md.*

15 Session holistictness metrics and the anti-eager levers (MMR λ / eager β)

Rationale. A recurring qualitative complaint about the top blends is that their predictions feel “too album-eager / bigram-like” — they parrot the current artist rather than curating a holistic continuation. This campaign quantifies that symptom with a newly-added *display-only session-holistictness* metric suite, tests the user’s hypothesis that *the GRU beats the LSTM because it is more eager*, and measures whether two new Phase-2 levers reduce eagerness at an acceptable recall cost: an eval-time MMR re-rank (`mmr_lambda`) and an anti-eager training regularizer (`eager_beta`). It is a measurement / exploratory pass: nothing is promoted, the primary metric stays Recall@10, and the holistictness objective decision is explicitly deferred to the user. Every arm runs on the

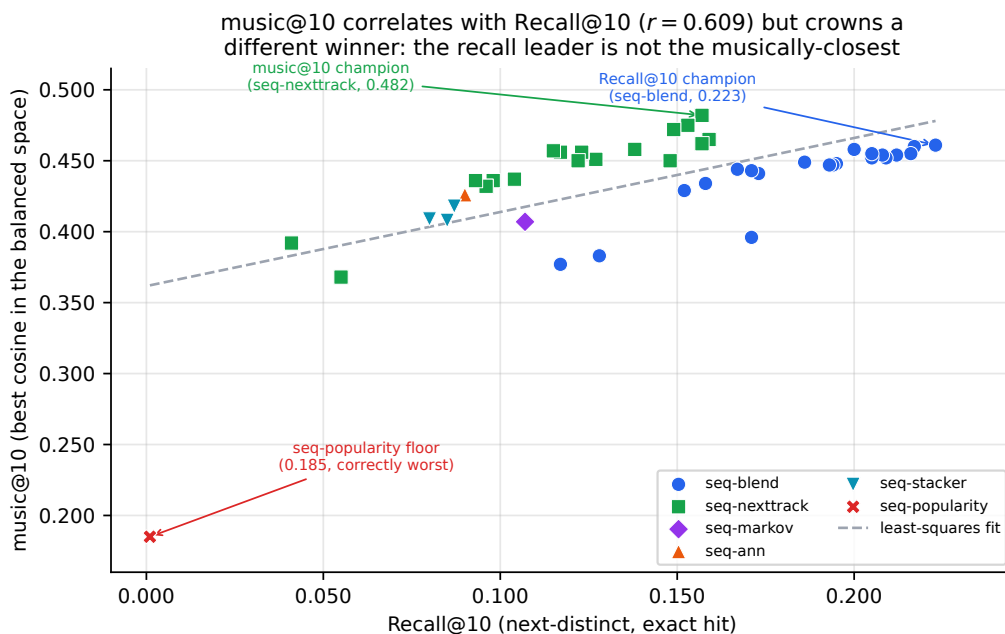


Figure 7: music@10 versus Recall@10 across all 53 next-track runs, one point per run, coloured by predictor family. The two axes correlate ($r = 0.609$: a better ranker also lands closer misses) but crown different winners — the Recall@10 champion (a **seq-blend** run at 0.223) is not the musically-closest, which is a **seq-nexttrack** run (music@10 0.482) that regresses the content latent and retrieves by cosine. The **seq-popularity** floor is correctly worst on both. music@10 is display-only; Recall@10 stays the primary metric.

canonical PCA-192 champion split seq-20260715-131139 (1431 test sessions; cold-item rate 0.55), and the champion reproduced its on-record Recall@10 0.2117 exactly (0.21174).

The new display-only holisticness metrics. Wired end-to-end alongside music@10 (seq_common.eval_from_scores → Rust → domain.toml → UI); the primary metric is unchanged.

artist_adj@10 (↓). Fraction of the top-10 sharing the *seed* track’s artist — the “parrots the current artist” eagerness rate. **Lower is better.**

album_adj@10. The same on (artist, album); **omitted on this dataset** because its items.json carries no album field (expected, not an error — covered by a separate album-carrying dataset).

mood_coh@10 (↑). Mean cosine of the top-10 to the prefix mood centroid in the content-metric space; higher is more mood-coherent.

ild@10 (↑). Intra-list diversity (mean pairwise sonic distance) of the top-10; higher is less duplicative.

suffix_recall@10 / cont_prec@10. Multi-step continuation coverage, produced only by the on-demand harness seq_continuation_eval.py (Phase E below), not by normal runs.

The sanity gate passed: music@10 / mood_coh@10 / ild@10 populate on all blend and GRU runs, so the content-metric vectors loaded and the MMR / mood / ild levers are live, not no-ops. Both new levers are OFF by default and verified byte-identical when off.

Results — the holisticness fingerprint (Phase A). Table 11 fingerprints the model family. **Eagerness tracks Recall@10 inside the blend family** — the champion (Recall@10 0.2117) is the *most* artist-eager model measured (artist_adj@10 0.610), with R+M next (0.579) — but the eagerness lives in the *Markov* leg, not the RNN: the standalone first-order Markov bar is already at 0.543, while the single GRU / LSTM sit far lower at artist_adj@10 $\approx$ 0.345. So the blends buy their Recall@10 lead by being artist-adjacent (bigram-like), which is exactly what the holisticness suite flags. The single RNNs are conversely the *mood-coherent, low-eager* models (highest mood_coh@10, lowest real-model artist_adj@10). Crucially the metrics are *not trivially gamed*: the popularity floor maxes ild@10 (0.898) and minimizes artist_adj@10 (0.003) yet scores $\approx$ 0 recall and *negative* mood_coh@10 — you maximize them by being random, so the meaningful low-eager / diverse models are the real single RNNs.

GRU versus LSTM (Phase B) — the hypothesis is not supported. With hyperparameters matched (h256, InfoNCE, 40 epochs, lr 1×10^{-3}), the GRU and LSTM are statistically *tied*: Recall@10 0.1230 versus 0.1188 (Δ +0.0042, inside the ± 0.015 band), and artist_adj@10 0.347 versus 0.344 (a negligible +0.003). The GRU is not measurably more eager than the LSTM on this split, so the “GRU wins because it parrots the current artist harder” hypothesis is **refuted here**.

The MMR λ lever (Phase C) — the strong knob. MMR is a pure eval-time re-rank, so the four originally-launched per- λ training runs were *cancelled* (not failed experiments); the sweep was computed offline from the single champion checkpoint (the λ =off cell reproduced the champion’s Recall@10 0.2117 exactly). As λ drops from off toward 0.3 (Table 12; Figure 8), Recall@10 falls — the *only* quantity traded — while ild@10 rises and artist_adj@10 falls (de-eagering). λ =0.9 is **essentially free**: the recall Δ is -0.0014 , well inside the ± 0.015 band (a tie), yet it already lifts ild@10 0.406 $\rightarrow$ 0.436 and trims eagerness 0.610 $\rightarrow$ 0.603. $\lambda \approx 0.7$ trades ~ 2 recall points ($\Delta -0.0181$, just outside the band) for a large ild@10 gain (0.406 $\rightarrow$ 0.520) and a real eagerness drop (0.610 $\rightarrow$ 0.571). A notable side effect: **music@10 rises monotonically** as you diversify (0.454 $\rightarrow$ 0.492) — sonic closeness and list diversity are not in tension; only exact recall is traded.

The anti-eager regularizer β (Phase D) — the weak knob. A training-time penalty on near-copy prediction of the current track’s latent (**eager_margin** 0.0) barely moves the needle (Table 13): across β 0 to 0.2, artist_adj@10 shifts only 0.347 $\rightarrow$ 0.340 ($\Delta -0.0074$) and ild@10 only 0.454 $\rightarrow$ 0.458, with Recall@10 flat (all four inside ± 0.015). The single GRU is already low-eager (0.347), so there is little copy behaviour left to penalize — β is not a useful holisticness knob at these settings.

Multi-step continuation (Phase E). The on-demand harness (`seq_continuation_eval.py -m 3 -k 10`, 1172 continuation cases) shows the GRU *beats* the Markov bigram on both multi-step metrics (Table 14): the RNN’s advantage *widens* past the immediate next track, because the bigram’s single-step co-occurrence does not compound into good 3-step continuations whereas the GRU’s learned sequence state does — the mirror image of the single-step board, where the bigram-carrying blend wins raw Recall@10.

Interpretation and the deferred decision. The campaign delivers a clean mechanistic account of the eagerness symptom: it is sourced from the Markov bigram leg (which parrots the current artist via raw co-occurrence), not the neural leg, so any future de-eagering should target the bigram / the fusion, not the RNN. Of the two new levers, the eval-time MMR re-rank is the *strong* knob and the training-time β regularizer is *weak*. Nothing is promoted — the primary

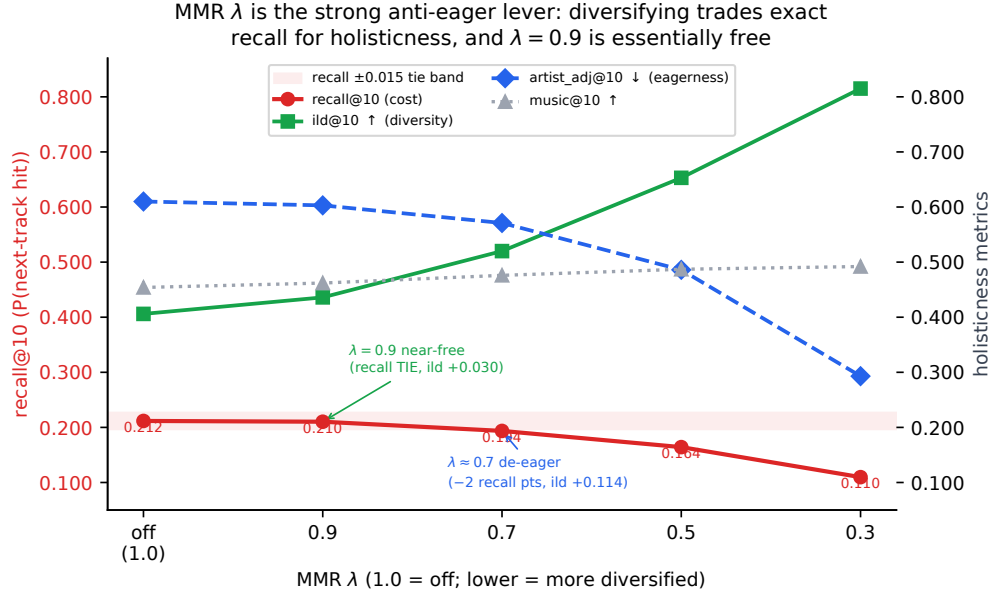


Figure 8: The MMR λ trade-off on the next-track champion. As λ drops from off toward 0.3, Recall@10 (red, left axis) falls — the only quantity traded — while ild@10 rises and artist_adj@10 falls (right axis; more diverse, less artist-eager) and music@10 even *rises*. $\lambda=0.9$ sits inside the ± 0.015 recall tie band yet already lifts ild@10 by 0.030; $\lambda \approx 0.7$ buys a large ild@10 gain and real de-eagering for ~ 2 recall points. MMR is the strong holisticness lever; β (Table 13) is weak.

metric stays Recall@10 and the champion is unchanged — but $\lambda \approx 0.9$ is **flagged as a near-free holisticness upgrade for a future user promotion decision** (Recall@10 tie within noise, ild@10 0.406 $\rightarrow$ 0.436, eagerness 0.610 $\rightarrow$ 0.603), with $\lambda \approx 0.7$ the “meaningful de-eagering for ~ 2 recall points” operating point; adopting either requires a user decision plus a 3-seed / native-run confirm. The β regularizer is not recommended as a lever at these settings. *Source: 2026-07-22-session-holisticness-and-antieager-levers.md.*

16 Dual-tower fusion refuted: the fourth combiner null

Rationale. By 2026-07-25 the record held three independent *learned-combiner* nulls — the XGB stacker (0.080 to 0.087, *below* its own GRU base leg; Section 10), reciprocal-rank fusion (-0.029) and even *oracle* test-fit blend weights ($+0.004$, CI straddling zero; Section 11). All three were **post-hoc, score-level** combiners, which left one obvious rebuttal standing: *of course* a bolted-on combiner fails — train the fusion end-to-end so the parts co-adapt. This campaign runs exactly that rebuttal, at the *representation* level, jointly, from scratch.

The vehicle is a new predictor, **seq-dualgru**: two *independent* recurrent towers (separate weights) read the prefix in parallel over configurable *causal views*, their per-step hidden states are concatenated, and one MLP head fuses them into the predicted next-item latent. Three views are exposed, all strictly causal at step t : **latent** (the raw item vector), **delta** (the step-to-step difference, $x_{-1} = 0$), and **cummean** (the running session centroid — the quantity the **seq-ann** pooled-MLP leg was built from). Crucially it reuses **seq_nexttrack.make_batches** and **step_loss** verbatim, so objective, batching, early stopping and evaluation are byte-for-byte the family’s, and it carries no bidirectional mode (a backward pass would leak the teacher-forcing

target).

The hypothesis was sharp: two towers should beat one *only if they see genuinely different views*; two towers on the same `latent` view should reduce to a capacity / implicit-ensemble effect a wider single GRU can buy more cheaply. The design therefore crosses the *complete upper triangle* of the $\{\text{latent}, \text{delta}, \text{cummean}\}^2$ view matrix (six cells at `fusion_layers` = 1) with a *fusion-depth* axis (two cells at `fusion_layers` = 0, a pure linear read of the concatenated states) and — the axis the earlier topology scan lacked (Section 5) — a **parameter-matched single-GRU capacity control**. Parameter counts were re-measured by instantiating the real classes at `latent_dim` = 192: the control C1 (h425, 871 017 parameters) matches the six `fusion_layers` = 1 dual arms (871 872) to 0.098 %, and is 10 % *larger* than the `fusion_layers` = 0 arms (789 696) — deliberately generous to the single tower.

The honest bar. Two boards were pre-registered *separately*, so that the scientific verdict and the board event could not be traded off after the fact. On **Recall@10** the bar for a standalone single model is the single-model board 0.123, *not* the three-leg blend champion 0.212 — comparing one trained model against a fusion of three legs would be a category error, and dual-tower-as-the- R' -leg is a **seq-blend** code change explicitly out of scope. On **holisticness@10** the crown rule carries an anti-degenerate recall floor (see below). All ten runs are on the canonical split seq-20260715-131139 at seed 1337, executed one at a time on a single remote worker, and the baseline C0 reproduced its on-record Recall@10 0.12299 (176 of 1431) *exactly*.

Results. Table 15 is the batch. **No arm is bolded as a winner because there is no winner**: the single-GRU baseline holds the batch’s best Recall@10, and not one of the eight dual arms beats it by paired- Δ CI above zero — six lose outright and two (the `fusion_layers` = 0 pair) tie. The pre-registered refutation rule fired.

Capacity is not the missed lever either. The param-matched control C1 (h425, 871 017 parameters — $2.2\times$ C0) scores 0.12089 against C0’s 0.12299, a paired Δ of -0.0021 straddling zero. This fills a real gap: single-GRU *width* had only ever been scanned on the 64-dim AE space (Section 5), and the rich PCA-192 space was untested. It behaves the same way, so **single-GRU width is now closed on PCA-192 as well as AE-64**. Because C1 is a genuine 0.098 % parameter match to the `fusion_layers` = 1 arms, the null cannot be re-attributed to “capacity would have helped”: it demonstrably would not.

View attribution — complementarity is real, but lands on the wrong pair. The complete view triangle is what makes this mechanically decidable rather than a judgment call, and the answer is genuinely surprising (Table 16, Figure 9). The pair the design nominated as its best hope — **latent + cummean** — is *not* complementary: it ties its latent/latent parent and beats only the centroid-only one, i.e. the raw-latent tower carries the cell and the centroid tower is dead weight. Same for **latent + delta**. But **delta + cummean** is complementary — it beats *both* parents with CI above zero. Read together: the co-adaptation mechanism the hypothesis proposed genuinely **works**; it is simply that **the raw latent view already subsumes everything the other two carry**. A GRU over raw latents integrates step-to-step movement and the running mean internally, so pairing it with an explicit computation of either is redundant — while two *impoverished* views do have non-overlapping information to pool, at a hopeless absolute level (0.08665). That is a clean mechanistic explanation for the null, not merely an absence of effect.

The MLP fusion head is actively harmful, not merely useless. Both `fusion_layers` = 0 arms beat their `fusion_layers` = 1 twins by paired- Δ CI above zero (+0.0161 on latent/latent, +0.0203 on latent/cummean) at 10 % *fewer* parameters — so removing the head makes the model measurably better in *both* view regimes. Mechanistically this fits the whole family arc: the objective

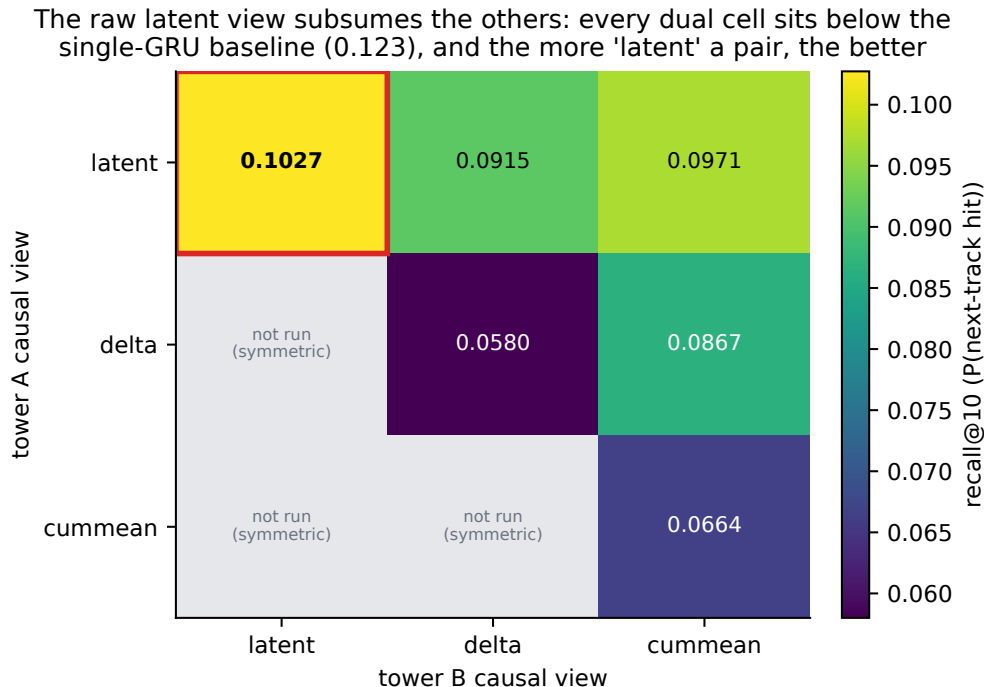


Figure 9: The complete view-pairing matrix at `fusion_layers = 1`, one run per cell. Every dual cell sits *below* the single-GRU baseline (0.12299), and the ordering is monotone in how much raw `latent` the pair contains: both-latent is best, both-delta worst. The lower triangle was never run (the pairing is symmetric by construction) and is masked in grey rather than mirrored, so the colour scale reflects only measured cells. The gradient is the finding: a recurrent tower over raw latents already computes what `delta` and `cummean` make explicit.

is cosine/InfoNCE retrieval *in* the item-latent space, so the head’s job is to land *in* that geometry, and an extra 256-wide non-linear layer between the recurrent state and a metric read-out just adds a warp the loss must then undo. The trade-off runs the other way on *holisticness* (both `f0` arms have *lower* `H`, ΔH CI below zero): the `f1` arms’ extra warp diffuses the top-ten (higher `ild`, lower `artist_adj`) at the cost of exact hits. The rule adopted is **default `fusion_layers` = 0**, later confirmed a third time (Table 19).

Holisticness and recall are anti-correlated, and the recall floor earned its place. Five arms lift `holisticness@10` by paired- ΔH CI above zero, topping out at D-CC’s 0.19715 (+0.055) — which on the naked crown metric would have ranked second on the entire board. But D-CC lands 95 of 1431 exact hits against the baseline’s 176. The ordering of the batch by `H` is almost the *reverse* of its ordering by recall (Figure 10), so **all five ΔH winners fail the pre-registered ± 0.015 recall floor** and the crown rule never fired. Without that floor the campaign would have “won the crown” with a model 46 % worse at its actual job — precisely the degenerate **mood-session** failure mode (`H` 0.2332 at Recall@10 0.0238) the floor was written to block. On the present evidence, **unconstrained holisticness is maximized by predicting less well**, so any future *holisticness* objective on this corpus must be a *constrained* one.

Interpretation. Nothing was registered and nothing promoted; the champion, the leaderboard and the pins are untouched. What the campaign buys is a *closure*: the combiner null is now robust across the whole combiner design space on this corpus — post-hoc score-level (three ways) *and* representation-level jointly-trained with a parameter-matched control — which materially

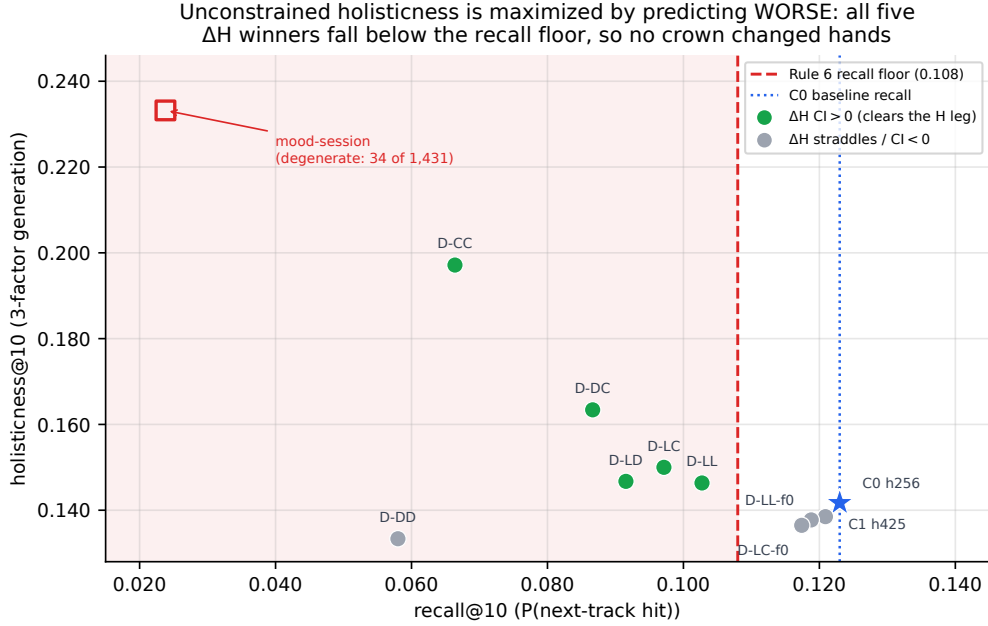


Figure 10: Why the crown rule needs a hard recall floor. Across the ten arms, holistiness@10 runs almost exactly *backwards* along Recall@10: the five arms that clear the ΔH CI-above-zero leg (green) are the five furthest *below* the baseline on exact recall, and all five fall outside the ± 0.015 floor (shaded). The open square is the incumbent `mood-session` model that motivated the floor — the highest holistiness on record at 34 of 1431 correct next tracks. H values are the 3-factor generation (Section 18).

consolidates “the fixed equal-thirds z-blend is the combiner of record.” Two genuine sub-findings survive the null (the harmful MLP fusion head, and `delta+cummean` complementarity), and the machinery itself is sound: 10 of 10 runs succeeded, the new predictor trains, converges and evaluates at exact parity with the family, and D-LL-f0 even posts the batch’s best MRR (0.05631 against C0’s 0.05341) — the one metric on which a dual arm leads. Per the noise-band discipline that is “at least equal, likely neither better nor worse,” not a win.

A provenance finding that revises two earlier numbers. Before the batch, the live remote worker directory was found *stale*: its `seq_common.py` had no `holistiness_at_k` at all, `seq_nexttrack.py` differed from the hub by md5, and three `seq_*.py` modules were missing entirely. After refreshing and verifying the whole family md5-identical to the hub, C0 returned 0.12299, bit-identical to the hub’s on-record value — *not* the 0.1201957 that two earlier remote runs (run-20260723-002044-af11a, -17cf8) had returned. The “cross-host reduction-order offset of -0.0028 ” those runs had suggested is therefore **withdrawn: it was a code difference, not a host effect**. Their recall numbers belong to a superseded code state and are not code-comparable to later batches; they remain valid as the β -eagerness measurements they were taken for (Table 13), since those conclusions rest on within-batch comparisons. The general lesson — a divergent remote checkout produces a *wrong number*, not a crash — is now a standing pre-flight gate. *Source: 2026-07-25-nexttrack-dual-tower-fusion-scan.md*.

17 The per-step pre-encoder refuted, and the rectifier mechanism

Rationale. The dual-tower null closed *deterministic* view asymmetry: `delta` and `cummean` are lossy functions of what a raw-latent tower already holds, so pairing them buys nothing. This campaign asks the strictly stronger version — does a **learned** per-step re-embedding, which is *not* a function of information the tower already has, enlarge the hypothesis space usefully? The axis is a nonlinear re-embedding in front of the recurrence, $[\text{Linear}(d_{\text{in}}, w), \text{ReLU}, \text{Dropout}] \times \text{pre_layers}$, applied per timestep (strictly causal), with the GRU’s `input_size` becoming w .

The hypothesis has an unusually sharp analytic edge. A GRU’s own input transform is *already linear* ($W_i x$ per gate), and a *linear* pre-encoder at $w \geq d_{\text{in}}$ is exactly as expressive as a bare GRU, since $W_i(Vx) = (W_i V)x$ with no rank constraint. **The ReLU is therefore the only expressivity change**, which means no linear-pre control run is needed and a null cannot be explained away as “the pre-encoder was redundant by construction.” The design’s decisive move is the **single-tower ablation S1** — a MLP256→GRU256 against a parameter-matched plain GRU — because without it any dual-tower result is uninterpretable: pre-encoder, or second tower?

Ten runs, same canonical split, same single remote worker, zero failures. Two reproduction gates both passed *bit-identically*: C0 returned Recall@10 0.12299091544374564 (176 of 1431, identical to the last digit, a seventh reproduction), confirming the `pre_hidden` edit did not perturb the family’s default code path; and C3, the bare dual-tower arm, returned 0.10272536687631029, exactly its on-record value, validating every dual verdict.

Results. Table 17 is the batch and Figure 11 is the verdict at a glance. Not one of the six pre-MLP arms beats the baseline by paired- Δ CI above zero: five lose outright, one ties. **And the decisive ablation does not merely tie — it loses**: S1 at 0.10901 against C0’s 0.12299, $\Delta -0.0140$ with the CI entirely below zero, 156 hits against 176. Since a linear pre-encoder here is provably expressivity-neutral, **the ReLU re-embedding is destroying prefix information that the GRU’s own gated linear input transform preserved**.

The dose curve is flat, and placement is irrelevant. Sweeping `pre_hidden_b` through $\{0, 128, 256, 384\}$ at `fusion_layers` = 1 (Table 18), every adjacent pair straddles zero and so do the endpoints — a **flat** axis, which by the pre-registered clause would have blocked registration even had a win rule fired. But note the *nominal* ordering is downward-then-flat, never upward: every dosed arm sits below the bare dual. Taken with S1’s significant loss, the sign of this axis is consistently negative across both towers and all four widths, so there is no interior peak to chase. Symmetry is likewise a non-factor: T1 (asymmetric, one bare tower) against T4 (both towers rectified) straddles zero on recall, so keeping tower A bare is not load-bearing at this dose.

The one solid positive: the fusion-depth regression, measured a third time and bigger. T1-f0 beats T1 by +0.0266 with the CI above zero, against the +0.016 and +0.020 measured in the two dual-tower view regimes (Table 19). The mechanism holds and sharpens: every extra nonlinear layer between the recurrent state and the metric read-out adds a warp the InfoNCE loss must undo, and *the more nonlinearity already added on the input side, the more the output-side warp costs*. Note too that T1-f0 — the only pre arm that ties the baseline — is exactly the arm with *both* a bare tower A *and* `fusion_layers` = 0: the shortest un-warped path from raw latents to the read-out. `fusion_layers` = 0 is the family default, full stop.

The mechanism, measured rather than inferred. The analytic argument named the rectifier as the only candidate; a follow-up interpretability read then *measured* it (Figure 12). The decisive control was a counterfactual tap the sparse-autoencoder engine cannot express — the model’s *own* affine map with the ReLU deleted. Reading next-item genre decodability at the same step (11 184 probe test rows, 25 classes, paired standard error ≤ 0.0065): the raw latent scores 0.4033, the affine

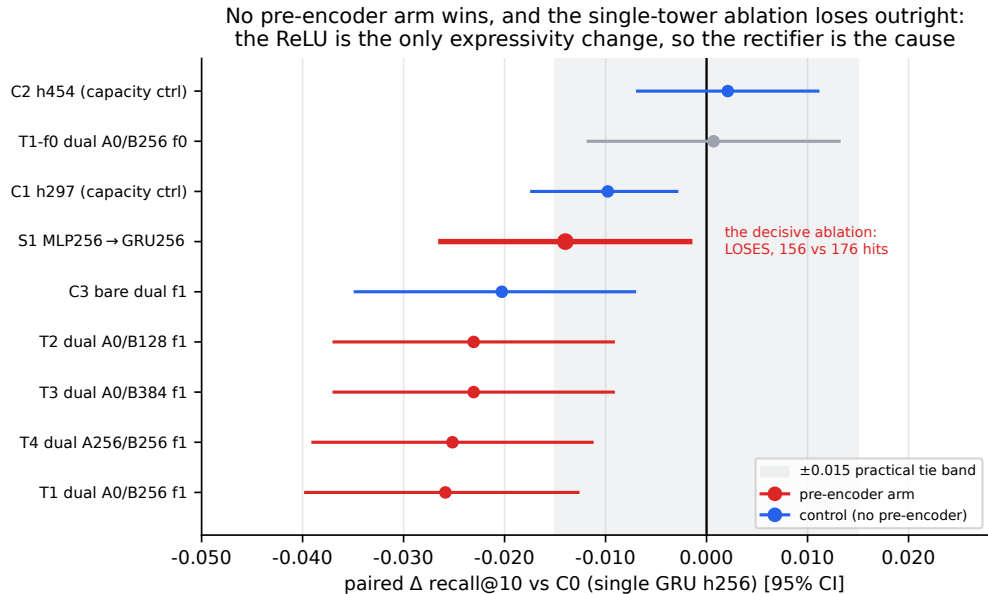


Figure 11: The pre-encoder verdict. Paired per-session bootstrap Δ Recall@10 against the reproduced baseline, one row per arm; the shaded strip is the ± 0.015 practical tie band. Not one pre-encoder arm reaches a CI entirely above zero. The decisive row is S1 — the single tower, the only arm whose result is unambiguously attributable to the pre-encoder — which loses outright. Because a *linear* pre-encoder at this width is provably expressivity-neutral, the rectifier is the only candidate cause.

map 0.4084 (**neutral**, $\sim 0.8\sigma$, exactly as $W_i(Vx) = (W_iV)x$ predicts), and the rectified tap 0.3700 ($\sim 5.9\sigma$ — **the ReLU is the whole loss**). The smoking gun is arithmetic: `frac_exact_zero` on the rectified tap is 0.5327, matching the affine tap’s `frac_negative_coords` 0.5327 to four digits. The rectifier deletes *precisely* the negative half, leaving ~ 120 live units for 192 signed directions — *fewer live coordinates than the input has dimensions*. PCA-192 latents are zero-centred and near-symmetric by construction, and at $w = 256$ (only $1.33\times$ the input dimension) there is nowhere near enough over-completeness to re-encode the discarded half-space, unlike the wide embeddings where ReLU MLPs normally live. The damage survives the recurrence: within one model, `tower_b` (fed rectified) scores 0.3929 against its *identical twin* `tower_a` (fed raw) at 0.4183, $\sim 3.9\sigma$ — and `tower_a` reproduces the standalone plain GRU (0.4235), which is exactly what “the bare tower *rescues*” requires.

Interpretation, and one honest caveat about the controls. This is the fifth consecutive architecture null, and it closes the *input side* as the view null closed the input-view direction: together with the four learned-combiner nulls, adding representation machinery *around* a single GRU over a good latent space does not help on this corpus, whereas every real gain on record came from improving the *latent space itself* (the InfoNCE content projection, PCA-192) or from the fixed z-blend. The axis is now quantitatively bounded — best pre arm 0.12369, a tie; dose flat to twice the input dimension; placement irrelevant; the fusion-depth interaction resolved. The caveat is that the *parameter-matched width control was a weaker instrument than assumed*: C1 (h297) came in at 0.11321, itself losing to C0, and the batch’s width points run $176 \rightarrow 162 \rightarrow 173 \rightarrow 179$ hits at h256 / h297 / h425 / h454 — non-monotone, spanning 0.0119 in recall with no trend. “Single-GRU width is flat on PCA-192” survives as a *trend* claim (h454 ties h256), but per-width jitter is

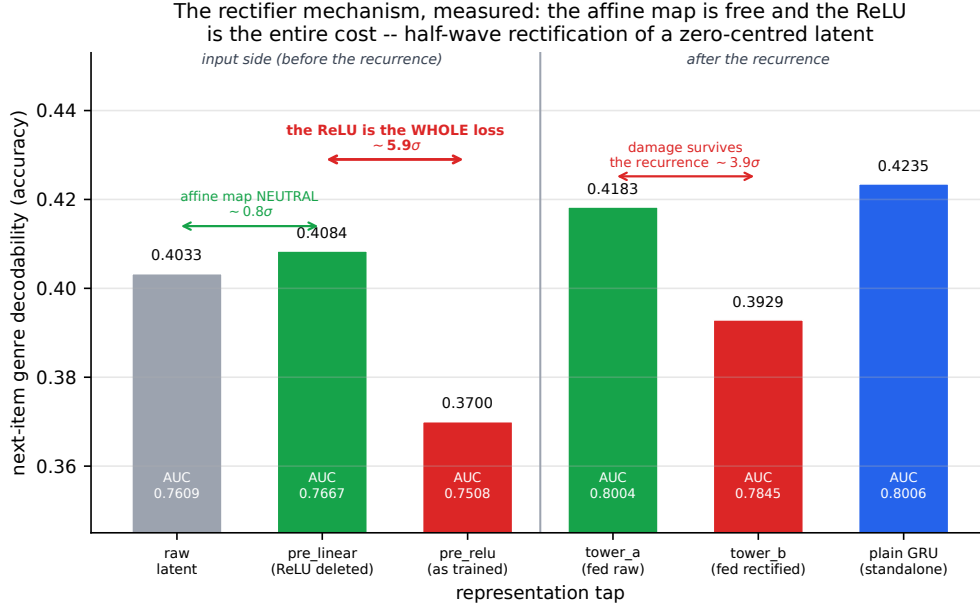


Figure 12: The rectifier mechanism, measured. Next-item genre decodability at each representation tap, including one counterfactual the interpretability engine cannot express: the model’s own affine map with the ReLU deleted. On the input side the affine map is *free* ($\sim 0.8\sigma$) and the ReLU is the *entire* cost ($\sim 5.9\sigma$); after the recurrence the damage persists ($\sim 3.9\sigma$ between two identical towers that differ only in whether their input was rectified), while the raw-fed tower reproduces the standalone plain GRU. `frac_exact_zero` on the rectified tap (0.5327) matches the affine tap’s `frac_negative_coords` (0.5327) to four digits: half-wave rectification of a zero-centred latent deletes exactly the negative half, leaving ~ 120 live units for 192 signed directions.

$\sim \pm 0.01$ on a single split, roughly two-thirds of the noise band. **A width-matched control that is itself a single unreplicated run is not a reliable capacity anchor at this effect size** — which is why it matters that S1’s verdict rests on the C0 comparison, reproduced to the last digit. A second caveat is un-disentangled: the pre-MLP ends in `Dropout`, so pre arms receive GRU-*input* dropout that the bare arms lack, and S1’s loss could be partly over-regularization rather than the rectifier — separating them needs a `pre_dropout` parameter that does not yet exist. *Source: 2026-07-26-nexttrack-pre-encoder-scan.md.*

18 Reading the record: statistical power and the holistiness generations

Two cross-cutting facts emerged from the 2026-07-25/26 batches that change how every number above should be read. Neither is a result about a model; both are results about the *measuring instrument*, and they belong here rather than buried in one campaign.

18.1 The nulls are power-limited, and that is a different claim

Every verdict in this document rests on a paired per-session bootstrap over the same 1431 test sessions. That n has a measurable resolution. Averaged over the nine paired comparisons of the pre-encoder batch the 95% CI half-width is 0.0124 on Recall@10 (0.0128 over the dual-tower batch’s

nine, so the figure is stable), and ~ 0.0018 on `holisticness@10`. Now compare those to the *entire spread* of an architecture family: 0.028 on `Recall@10` and 0.0031 on `holisticness@10` (Table 20). The ratio is the whole story — a family’s full range is only ~ 2.3 resolution widths wide on recall and ~ 1.9 on `holisticness`, so of eight adjacent arms roughly **two tiers** are separable on `Recall@10` and **one** on `holisticness@10`. Resolving a 0.005 `Recall@10` difference at this precision would require $n \approx 1431 \cdot (0.0124/0.005)^2 \approx 8800$ test sessions, six times what the split provides (Figure 13).

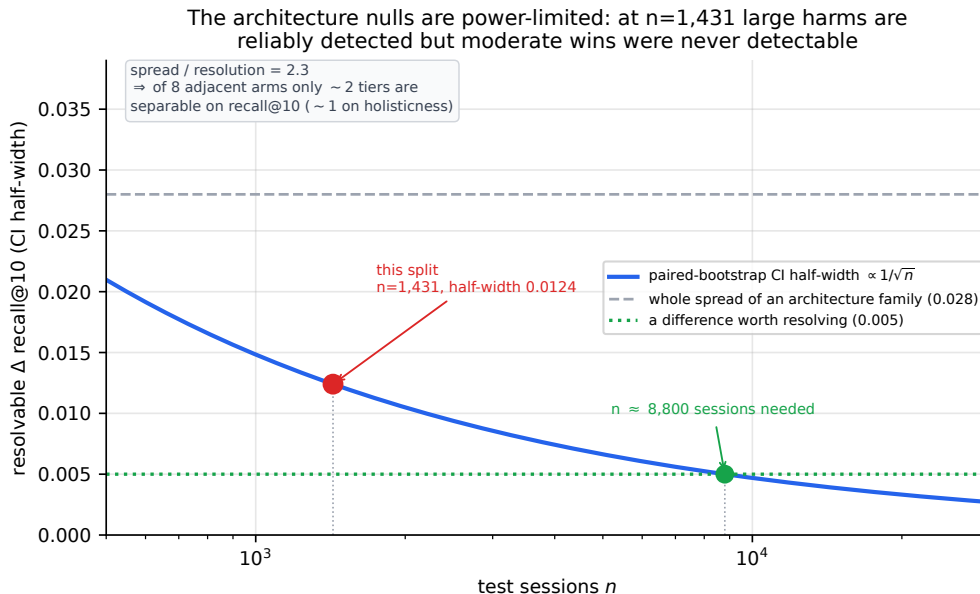


Figure 13: The architecture nulls are power-limited. The paired-bootstrap CI half-width falls as $1/\sqrt{n}$; at the split’s $n = 1431$ it is 0.0124 on `Recall@10`, while a whole architecture family spans only 0.028. Resolving a 0.005 difference would need ~ 8800 test sessions. The honest reading of the five architecture nulls follows directly: *large harms were detected reliably, moderate wins were never detectable at all*.

The honest reading of the five architecture nulls follows directly, and it is not the reading a casual glance gives. What these campaigns established firmly is that the tested architectures do not deliver *large* gains, and that several of them do real *harm* — the harms are the well-powered results, because they are big (S1 -0.0140 ; the `fusion_layers` regression $+0.0266$; six of eight dual arms below the baseline). What the campaigns *cannot* establish is the absence of a moderate gain: a genuine $+0.005$ improvement — which on a 0.123 baseline would be a respectable 4% relative lift — would have been invisible by construction. So “dual-tower fusion is refuted” and “the pre-encoder is refuted” should be read as *no large effect, and no effect of the hypothesized magnitude*, not as “these architectures do not work.” The practical consequence is a prioritization rule rather than a caveat: on this split, further architecture arms are a poor use of runs no matter how they come out, because the design space’s whole dynamic range is barely twice the measurement error. Either enlarge n or move to axes whose effects are large — which is, empirically, exactly what the representation and supervision axes have delivered (the content projection’s $+0.0259$ is twice the half-width, and the second-split hardening’s $+0.0426$ is three times it).

18.2 Three generations of `holisticness@10`, and why the board is denominated in recall

The composite *holisticness@10* was revised twice in four days, and stored values are **not comparable across the cuts**:

Gen. 1 — 3-factor (to 2026-07-24). The bare product

$$\text{clamp}(\text{mood_coh}, 0) \times \text{ild} \times (1 - \text{artist_adj}),$$

carrying *no correctness term at all*. Values land in 0.063 to 0.233 — the column in Table 15.

Gen. 2 — 4-factor, grounded (2026-07-25). Multiplies in a `music@10` grounding factor,

$$\text{clamp}((\text{music@10} - 0.185)/0.815, 0, 1).$$

Because `music@10` falls in lockstep with recall, this cancels the de-eagering “wins” the 3-factor form rewarded; values land in 0.044 to 0.050 — the column in Table 17.

Gen. 3 — concentration-aware (2026-07-26). Eagerness becomes `max(artist_adj, artist_conc)` and the grounding floor rises 0.185→0.34. Values land near 0.009 to 0.025.

Every facet is stored per row, so any generation is recomputable after the fact (the helper `rescore_holisticness.py` does this for the runs it can reach), and Table 21 re-derives all three for a representative slice. Two consequences matter for reading this document. First, **a `holisticness` column must always be labelled with its generation**, because the generation effect dwarfs the model effect: every 3-factor value mechanically outranks every 4-factor value, since the grounding factor multiplies by ~ 0.3 . Second, and more importantly, **the recall and `holisticness` boards point in opposite directions at every generation**. The champion blend — 0.21174 Recall@10, 303 of 1431 — ranks *last* in Table 21 under all three formulas, because it is the most artist-eager model on record; the degenerate `cummean/cummean` arm and `mood-session` lead. Generation 3 demoted `mood-session` from first to a genuine zero, but it did *not* fix the dual-tower degenerate arm, so the ± 0.015 recall floor remains mandatory on any `holisticness` claim.

What is primary, and what this document sorts on. For completeness: the `[metrics].primary` key in `domain.toml`, and the best-models group’s `primary_metric`, have both read `holisticness@10` since 2026-07-23, so the *server-effective* primary metric is no longer Recall@10. Statements in the earlier sections that “the primary metric stays Recall@10” were true when written and are superseded *for the server*. This document nevertheless continues to denominate its leaderboard (Table 1) and every campaign verdict in **Recall@10**, deliberately: it is the record’s comparability currency, it is stable across all three metric generations, and — as Table 21 shows — a `holisticness` ordering would place the servable champion last. `Holisticness` is reported throughout as a labelled, generation-tagged diagnostic column. *Sources: 2026-07-25-nexttrack-dual-tower-fusion-scan.md, 2026-07-26-nexttrack-pre-encoder-scan.md.*

19 The MMR λ frontier — the first confirmed crown lever

Rationale. Section 15 left one standing flag: an eval-time MMR re-rank at $\lambda \approx 0.9$ was “the only near-free `holisticness` lever on record”, measured in an *offline* sweep over a stored checkpoint and

explicitly deferred to a user decision. This campaign is the native confirm, and it was designed to settle a second, pre-registered *mechanism* question rather than merely to re-measure the knob: does sonic diversification de-concentrate **artists** (H1, “responsive”), or are same-artist tracks far enough apart in the content-metric space that the diversity penalty leaves eagerness untouched (H2, “sticky”)? The answer decides whether an explicit artist-diversity penalty is the indicated sequel or a redundant one.

What is being maximized. All ten arms are scored under the *live* four-factor crown, taken verbatim from `predictors/seq_common.py`:

$$\begin{aligned}
H = & \max(\text{mood_coh}, 0) \times \text{ild} \\
& \times (1 - \text{clamp}(\max(\text{artist_adj}, \text{artist_conc}), 0, 1)) \\
& \times \text{clamp}((\text{music@10} - 0.34)/0.66, 0, 1),
\end{aligned}$$

each facet the *mean over rows* and the product taken of the *aggregates* — generation 3 of Section 18.2. Two anchors frame the batch: **A**, the recall champion $R'+M+C'$ (H 0.008603, its eagerness factor bound by `artist_conc` 0.691653), and **B**, the best *single* model on the live crown, `gru-infonce-h256` (H 0.024538). The bar to beat is the all-time healthy-arm crown maximum $H_{\text{bar}} = 0.024872$, held by an h425 capacity control (`run-20260725-143502-be11b-seq-nexttrack`). All arms ride the canonical split `seq-20260715-131139`; no dataset work was needed.

Why this batch is unusually sharp. λ is an *eval-only* parameter: the trained weights are bit-identical across the arms of one model, so $\lambda = 1.0$ is a bit-identity gate and every Δ is a maximally paired comparison. Both controls reproduced their anchors exactly (303 and 176 hits of 1431, MRR agreeing to 3.1×10^{-11}), and every arm’s H was *recomputed* from the stored `predictions.json` under one code path, so the per-row rounding bias cancels in every Δ (recomputed versus stored agreed to 7.4×10^{-8} and 2.1×10^{-7}). A third gate — MMR-active — required each $\lambda < 1$ arm’s `ild` to differ from its own control by far more than 1×10^{-6} , which is the registered detector for MMR silently no-opping when the content-metric index is unreachable. No arm was a no-op.

Results. Table 22 is the frontier, and Figure 14 plots it. Arms are tiered by a pre-registered rule: **TIER-1 FREE** means ΔH above zero with the recall Δ straddling zero and $|\Delta \text{recall}| \ll 0.015$; **TIER-2 PRICED** means a real H gain at a recall cost the CI can see; **TIER-3 REJECTED** means Δrecall below -0.03 .

Interpretation. **MMR is a real crown lever, and the first one on record.** Every one of the eight $\lambda < 1$ arms lifts H with the CI entirely above zero. What makes the gain legitimate rather than an artefact of the composite is that *three* of the four factors improve at once: `ild` rises (0.4536→0.5314 on the GRU, 0.4064→0.8146 on the champion), eagerness falls, and — the load-bearing part — `music@10` *also* rises (0.4558→0.4747). A lever that raised H by degrading relevance would be cancelled by the grounding factor; this one is not, which is precisely the distinction the floor was written to enforce after the `cummean/cummean` and `seq-mood` pathologies.

H1 is confirmed unanimously and H2 is refuted, which *cancels* the sequel the previous campaign had pre-registered. In 8 of 8 arms $\Delta \text{artist_conc} \leq \Delta \text{artist_adj}$ (both negative) at a ratio of 1.59 to 3.06. Same-artist tracks *are* near-neighbours in the content-metric space, so a max-similarity penalty de-concentrates artists as a side effect, with no explicit notion of artist anywhere in it. Sonic MMR therefore *is* a de-eagering lever, and porting `seq-mood`’s explicit `artist_penalty` to the blend — the pre-registered H2 follow-up — is no longer the indicated next move. It stays on file as a *complement*, because `artist_adj` falls $\sim 2.5\times$ slower than `artist_conc` and becomes the binding facet at low λ .

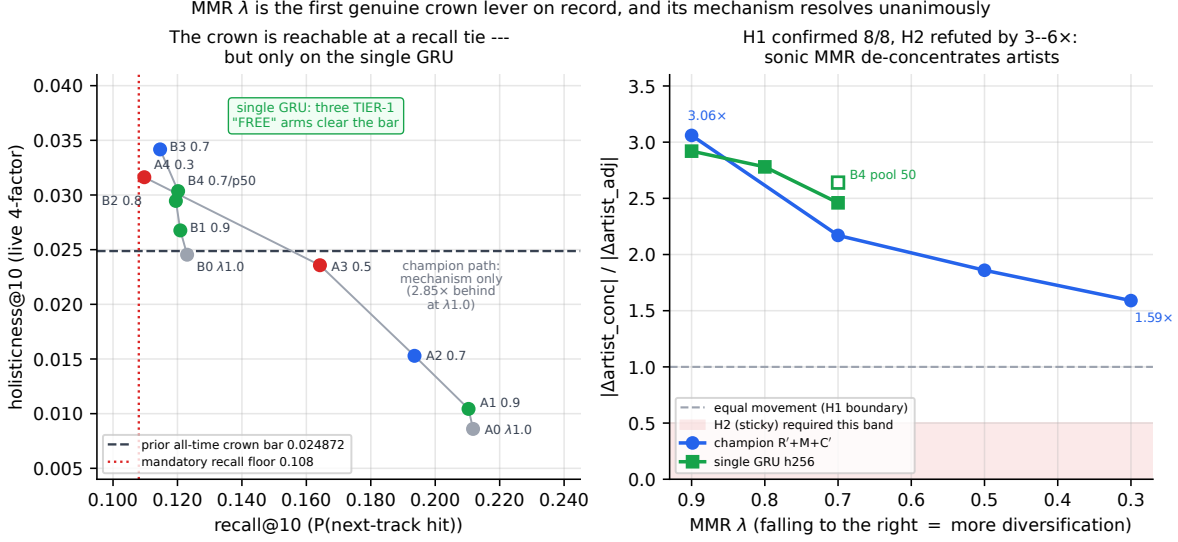


Figure 14: **Left:** the λ frontier on the recall $\times$ crown plane. Each engine traces a path as λ falls from 1.0; colour is the pre-registered tier (green TIER-1 FREE, blue TIER-2 PRICED, red TIER-3 REJECTED, grey control). Three single-GRU arms clear the prior all-time crown bar 0.024872 at a recall whose paired CI straddles zero — the first genuinely free crown movement on record. The champion’s path reaches comparable H only after paying recall the CI can see, and its two best- H points are rejected as degenerate. **Right:** the mechanism question. H1 required $|\Delta \text{artist_conc}| \geq |\Delta \text{artist_adj}|$; H2 required the ratio below 0.5 (shaded). Measured 1.59 to 3.06 in 8 of 8 arms, so H2 is refuted by a factor of 3 to 6 in the wrong direction — and the ratio shrinks monotonically as λ falls, i.e. the easy de-concentration is bought first and the mechanism saturates.

The binding facet switches, and that locates the end of the lever. On the champion the switch from `artist_conc` to `artist_adj` happens between $\lambda 0.7$ and $\lambda 0.5$; on the GRU it has already happened by $\lambda 0.9$. Once `artist_adj` binds, the $(1 - \text{eager})$ factor is the stiff one and each further recall point buys progressively less crown — the structural reason the GRU’s frontier flattens where it does, and a prediction that pushing below $\sim \lambda 0.7$ on the GRU is poor value. B3 is the confirmation: the highest H measured anywhere (0.034171) at the cost of losing the recall tie.

The promotable operating point is the pool probe. B4 ($\lambda 0.7$, `mmr_pool 50`) is the single best TIER-1 arm at H 0.030355 — +22.1 % over H_{bar} and 5.4 measured half-widths clear of it — and it beats B2 on H by +0.000889 with the CI above zero at a recall that straddles, so its rank is *resolved* rather than nominal. B2 ($\lambda 0.8$) was the design’s pre-registered promotion hypothesis and fired almost exactly as projected (0.0321 predicted against 0.029458 measured, with a *smaller* recall cost than projected). Nothing was promoted: six definitions were registered, each carrying the note “NOT PROMOTED — robustness confirm + user approval pending”. The recall board is untouched — **no arm improves recall**; MMR only ever costs it.

The second axis, `mmr_pool`, closes as a fine-tuning knob rather than an axis. Capping the candidate pool at 50 recovers recall (Δ vs pool 200 +0.005584, CI above zero, as hypothesized — a cap bounds how far MMR can reach for diversity) but gives H back (−0.003838, CI below zero). It does not win at fixed λ ; it moves along the *same* frontier. What earns it a line in the record is the cross-comparison above: the cap reaches a slightly *better* point than the nearest

pure- λ setting. Prefer λ for coarse movement, pool for fine-tuning at a chosen λ .

The champion/GRU contrast resolves in favour of factor health, not de-eagering headroom. The design framed this as the campaign-carrying question: the champion is $1.9\times$ more eager, so a given absolute de-eagering is worth $\sim 2.1\times$ more relative H there — yet it starts $2.85\times$ behind, and the gap *widens* under diversification (H 0.015289 against 0.034171 at $\lambda 0.7$). The facets say why: the champion’s **mood_coh collapses** as it diversifies ($0.3982 \rightarrow 0.2378$ at $\lambda 0.3$, -40%) while the GRU’s barely moves (-4%). The champion’s recall comes from artist-adjacency — the Markov bigram leg, per Section 15 — so diversifying away from the seed artist also walks it out of the prefix’s mood neighbourhood. It pays twice. **MMR’s value scales with the health of the other three factors, not with room to de-eager**, and no λ closes the champion’s crown gap.

Two instrument findings, one of them a warning about the record itself. First, the **shared-weights instrument is 3 to $33\times$ sharper than the architecture-family noise floor**: measured half-widths run 0.000382 to 0.002476 on ΔH against the family’s conservative 0.0018, with the number of sessions whose hit@10 changed at all as low as 9. Arm A1 had been pre-registered as “likely UNRESOLVABLE” and resolved at 4.8 measured half-widths. Eval-time levers must therefore budget their minimum detectable effect from the *pairing*, not from the architecture figure of Section 18.1 — the two regimes differ by more than an order of magnitude, and conflating them under-powers a whole class of cheap experiments. The corollary is reassuring in the other direction: the 2026-07-22 *offline* λ sweep projected recall 0.2103 / 0.1936 / 0.1642 / 0.1097 and the native runs returned 0.21034 / 0.19357 / 0.16422 / 0.10971 — four for four to four decimals, which validates the offline driver as a *screening* instrument.

Second, a new infrastructure pitfall was found the hard way and it retroactively qualifies part of the record: **a remote worker running a (deleted) binary image silently strips metric keys its struct does not know**. The metric-completeness gate failed on the first attempt with `artist_conc_at_k` missing, although the predictors were md5-identical to the hub and the worker’s own `metrics.json` carried the field: the worker *daemon* had started a day before the field existed, `rsync` had replaced the file on disk but not the running image, and the old image deserialized the predictor’s metrics into a struct lacking the field, dropped it, and posted the filtered set **without erroring**. `holisticness@10` was nevertheless correct throughout, because the predictor computes H itself before the server sees it — which is exactly what made the defect invisible. Every remote run in a four-hour window on 2026-07-26 lost `artist_conc_at_k`, which is why anchor B’s quoted concentration was a *reconstruction* from `top_k_ids` and the dataset artist map. That reconstruction is now independently validated: the natively emitted values (0.6916530786551751 and 0.35591272614333413) match the rescoring tool’s reconstructions to the last digit. Remedy: cycle the daemon whenever a metric field is added; `rsync` of the binary is not sufficient. *Source: 2026-07-26-nexttrack-mmr-lambda-frontier.md.*

20 Autoregressive rollouts: a generation metric, and five sweeps that say no

Rationale. This campaign was provoked by a *user preference*, not by a metric. On a real 135-track playlist the user strongly preferred one engine’s 20-step journey over another’s, while `holisticness@10` scored those two journeys 0.003 apart. That gap is the point: every metric on record — Recall@10, `holisticness@10` and its four facets, `music@10`, even the multi-step continuation evaluator — scores *one* ranked list produced from a *real* prefix. None of them ever feeds a

model its own output. A 20-step journey does exactly that, so a trajectory failure — a rollout whose energy decays until a party seed has become a comedown — is structurally invisible to all of them. This is the first campaign in this document to build an instrument for the *deployment* regime, and it is the direct ancestor of the walk surface of Section 24.

What was built. Four things, all off the training path.

hold@N, a generation metric. Per legible axis a in {energy, valence, acousticness, tempo}, $\text{fade}_a = |\text{mean}(\text{2nd half}) - \text{mean}(\text{1st half})| / \text{band}_a$, and then

$$\text{hold@N} = 1 - \text{clamp}(\text{mean}_a(\text{fade}_a) + \text{stasis}, 0, 1).$$

The per-axis bands are calibrated on 2135 real *train* sessions (energy fade p10 0.0141, p50 0.0769, p90 0.1963; tempo 1.28, 8.05, 22.43), so axes with incompatible units combine and “a big fade” means big *relative to how this listener actually listens*. Real sessions are not flat — the median energy fade is 0.077 — so a metric rewarding zero fade would rank an unnaturally rigid generator above the listener it imitates.

stasis, the anti-gaming term. A pure fade metric is degenerate: twenty near-identical tracks fade by 0 and score 1.0, which is the same pathology that forces **holisticness@10** to carry an *external* recall floor. **stasis** puts the floor *inside* the metric by penalizing steps smaller than the listener’s own p10 jump, and it was verified on a synthetic flatline (moves-naturally 1.000, at-your-floor 1.000, half-floor 0.500, barely-moves 0.048).

An offline rollout evaluator. `seq_continuation_eval.py --rollout` scores sampled *test* sessions by **importing** the generation loop from the serving path rather than reimplementing it, so an offline score describes the journeys the lab actually serves; the refactor was verified byte-identical. It is on-demand and sampled because a rollout costs **steps** full-vocabulary passes per session against the primary path’s one ($\sim 20\times$).

A playlist-lab readout. A mood report card, a comparative journey-shape chart and per-step **passed_over** diagnostics, with the holisticness facets demoted to a collapsed detail block.

Results. Table 23 is five sweeps, 120 paired rollouts and 240 journeys, under a paired bootstrap over *seeds* plus a leave-one-seed-out jackknife. Rollout policy was frozen throughout at the lab defaults (20 steps, artist penalty 2, anchor 1, temperature 0 — deterministic, so one seed is one sample and there is no rollout variance).

Interpretation. **Rollout fade does not separate this family.** At matched training split the two architectures tie *twice*; on the user’s own playlists the pair they preferred ties; and the only surviving effect is **checkpoint-level** within one architecture (S4c), which does not transfer to the playlist regime. Mean **hold@20** over the identical 39 random seeds runs 0.6488 / 0.6309 for the two dual-tower checkpoints against 0.5983 / 0.5266 for the two single-GRU ones — so S3’s apparent architecture win decomposes into a non-significant architecture component (+0.05) and a significant *checkpoint* component (+0.072). Two **seq-nexttrack** models differ from each other more than dual-tower differs from single-GRU at matched split, and the checkpoint the playlist lab pre-selected by default happens to be the worst holder of the four. The user’s original comparison was the best holder against the worst.

Combined with the five prior architecture nulls this is a sixth, and it is the strongest form available, because the standing rebuttal to those nulls was “Recall@10 cannot see session quality”.

It cannot; and when an instrument built for the deployment regime looks, there is still nothing there.

Per-seed variance dominates model identity, and that is the operational finding.

Individual gaps on *one* model pair span +0.39 to -0.27, while the model-level mean difference on the user’s playlists is +0.004. Which seed you hand the generator matters far more than which of these models generates. Two consequences: prefer serving an *ensemble* and letting the listener choose over hunting a single better generator, and budget ≥ 40 seeds for any future rollout comparison — 8 to 12 cannot resolve anything at this variance.

hold@N is a working instrument but not a model ranker. It fired on a real checkpoint difference, it fired on the judged playlist (0.62 against 0.25) where **holisticness@10** saw a 0.003 gap, and it correctly returned TIE three times. The recommendation is therefore to keep it as a per-journey *lab readout* and **not** to wire it as a leaderboard column: its ground truth is listener preference, which stands at $n = 1$ blind vote. Promoting an unvalidated composite to primary is how the crown ended up needing an external recall floor, and replacing one unvalidated composite with another repeats the mistake.

Four methodological pitfalls, each of which cost a retracted positive. They are worth more than the null. **(1) A cross-split comparison masquerading as an architecture result.** The lab’s two default pre-selects are trained on *different* splits with identical vocab and item latents, so comparing them attributes checkpoint and split differences to architecture; pair from the same batch when the claim is architectural. **(2) Mood-extreme seed selection inflates effects** $\sim 2.5\times$ (S2’s +0.314 became S3’s +0.122 on a random sample). **(3) An anti-gaming term can be too weak to do its job:** *stasis* first normalized the shortfall by the band *width*, and because this listener’s jump band is wide a flatlined rollout scored 0.740 — *above* a real generator’s 0.477. Normalize by the *floor*; a guard beatable by the degeneracy it targets launders that degeneracy as a good score. **(4) A metric that returns nothing must drop the *pair*** and report the smaller n , or the pairing silently breaks. The jackknife earned its keep at $n < 40$ throughout: it killed two of three per-axis positives at $n = 8$ and flagged S4 and S5 as unsupported at 0/39 and 0/11.

What the per-axis read adds, including a failed negative control. At $n = 39$ only *energy* fade separates the arms (-0.039 [-0.069, -0.009], favouring dual-tower); valence, acousticness and tempo all straddle. But one pre-registered *negative* control moved: the energy *level* deviation from the seed favours the single GRU (+0.018 [+0.005, +0.032]). Those are different properties, so the claim had to be re-scoped from “holds the seed better” to “is more stationary” — and it shows that **hold@N**’s deliberate exclusion of a level term is a choice that *changes the ranking*, not a neutral simplification. Two other limits travel with the batch: 74% of the corpus carries the acoustic fields the metric needs, so seeds were pre-filtered for $\geq 60\%$ prefix coverage (a mild selection effect), and nothing here has been listened to.

Nothing was promoted, registered or wired; the champion, the crown board and the best-models pins are unchanged. For the record, the live-crown standing relevant to the dual-tower question is a *tie* rather than a loss — the best legitimate dual arm 0.02378 against the single GRU’s 0.02454, a 0.0008 gap against a ~ 0.0018 half-width. The family was refuted on Recall@10, ties on the live primary, and shows nothing on the rollout axis. The indicated next axis is therefore not topology but the *training regime*: all five architecture nulls held one-step teacher forcing byte-identical while the product rolls twenty steps on its own output, and that exposure-bias mismatch is the one thing those nulls could not have detected. *Source: 2026-07-27-rollout-hold-metric.md.*

21 A research memo, and what its provenance permits

Status, first. This section consolidates a document that is **not a campaign report**. Nothing was run on the server, nothing was promoted, nothing was registered; the champion, the leaderboard, the crown board and the best-models pins are all unchanged by it. It was produced by a research workflow (four scouts, five proposal lenses, five adversarial screeners, then synthesis) plus *offline* scoring against frozen checkpoints. Where it reports a number, that number came from an offline screen over stored artifacts, not from a native run. It is included here because it is the direct parent of the campaign in Section 22 and because its negative results — the graveyard below — are cheaper to reuse than to rediscover. It is reported at a lower register than the measured campaigns, and its figures are deliberately kept out of every board in this document.

The provenance partition, kept visible. The memo itself distinguishes two classes of number, and that distinction survives into this synthesis.

Independently re-derived by the orchestrator. The share of test queries whose last prefix item has *zero* train-bigram support: $795 \text{ of } 1431 = 0.5556$, reproduced by an independent script. The ungated control, which is self-validating because it must equal the champion: 0.21174, exact. The gated arm: 0.23201, a paired Δ of +0.02027 [+0.01048, +0.03075] with the CI above zero. And its bucket decomposition: the zero-support bucket moves 0.1660→0.2025 over 795 queries while the two warm buckets are **bit-identical** (0.3116 and 0.2326, unchanged).

Not verified by the orchestrator — agent-reported only, treated as provisional. The +0.01957 second-split figure, the replay channel’s +0.02865, the `latent_gamma` numbers, and the recomputed 0.0706 discordance with its ~ 0.0145 noise floor. None of these appears in Table 1 or on any crown board, and the first of them is precisely what the follow-up campaign was commissioned to establish natively.

One caveat the memo states about its own headline. The ten gating variants were screened **on the test set**, so quoting the best arm’s CI is optimistic by construction and the screen cannot itself be evidence. Three things mitigate it — four mechanistically related variants all land at 0.019 to 0.020, the bucket decomposition is a mechanical explanation rather than a fit, and the warm buckets do not move by a single hit — but the mitigation is an argument, not a measurement. The native two-split confirm is the real gate, and it is Section 22.

The honest headline: “a better topology” is the wrong question. The record had by then answered it six times — cell and loss on the AE-64 space; depth, width and direction; width on PCA-192; dual-tower fusion (Section 16); the MLP fusion head, an *active* regression recovered by deleting it; the per-step pre-encoder, a measured loss with the mechanism identified down to `frac_exact_zero = frac_negative_coords` (Section 17); and rollout fade (Section 20). Five lenses were pointed at the topology question and *every* architecture-shaped proposal came back weakened, sub-noise, or pre-killed. Two came back with measured negative results: a split Markov leg lost at every point of its own pre-registered grid, and a factored artist softmax head was killed by its own pre-kill run, which found that top-5 artist gating on the real checkpoint changes recall by **exactly** 0.0 because the trained GRU’s top-10 already lives inside its own top-5 artists.

Where the evidence points instead: the model is not the constraint. What it is asked to rank against, and which legs are allowed to speak on a given query, are. Every

proposal that survived adversarial screening is a candidate-set or evidence-conditioning change — an evidence-gated Markov leg, a causal cross-session replay channel, per-leg reliability scalars, a prefix-only negative mask, the never-swept softmax temperature. None is a topology. This is the same conclusion the record had reached from the representation side, arrived at from five independent directions, and it comes with a measurement rationale: eval-time and score-time levers are 3 to 33× cheaper to resolve than training-time ones (Section 19), so a shortlist of score-time proposals is the rational response to the instrument this lab has.

The two survivors worth naming. (1) **markov_gate** — gate the Markov leg to abstain where it has no bigram evidence, restoring the **trust** term that the blend’s z-normalization erases. Twelve lines, zero learned parameters, and confirmed natively in the next section. (2) **A causal cross-session replay channel** — 76.17 % of the held-out targets were played in a *strictly earlier* session and are not in the current prefix, and nothing in the family can see this, because the Markov leg is a train-frozen bigram and the recency scorer only ever scored prefix items, which the evaluator masks to $-\infty$ (its recorded 0.001 is dead code, not a measurement). Its hit-Jaccard with the GRU is 0.060 against 0.201 for the GRU/Markov pair that produced a working blend, i.e. it is genuinely different signal. Its catch is *governance*, not code: it is an information-set amendment that moves the rankable ceiling from 0.4493 to 0.7617, which needs its own board column and an explicit decision before any number from it is comparable with anything in this document. A third proposal, **latent_gamma** (a one-scalar spectral rescale of the item latents), is worth recording for its *reframe* rather than its delta: on the offline screen the champion’s entire recall contribution is reproducible by a 192-number diagonal rescale, which recasts “the single biggest win on record” as a conditioning fix that supervision happened to find, and predicts that further linear metric learning is near-exhausted. All of these figures are in the provisional class.

The graveyard, which is the reusable part. Each entry is a lane that should not be re-opened without new information. Full-catalogue or support-restricted cross-entropy is pre-empted by an existing arm at ~53 % support coverage (0.1223 against 0.1230) — the literature’s top candidate does not transfer because the item side is frozen. A factored artist softmax head is refuted by its own pre-kill (exactly 0.0). Splitting the Markov leg into track and back-off components was measured at every grid point and lost at all five (-0.0217 to -0.0384 , all CIs below zero), because separate z-normalization structurally destroys the **trust** term inside the warm regime. A pool-matched stage-2 re-ranker is unobtainable: the train pool’s positive density is 0.5975 against test’s 0.3368, so its own go-gate passes *because of* the confound. A mixture-of-vMF head loses to its own oracle decomposition. Diffusion, DreamRec and flow-matching are externally falsified for this setting, and generative retrieval with semantic IDs is strictly dominated when exact cosine over 15 MB is already affordable and the corpus is 55 % to 64 % cold. Popularity debiasing has nothing to debias (the popularity baseline is 0.0007). Hard-negative mining is harmful at low temperature here, because the hardest negatives are same-album and same-artist tracks — i.e. mining false positives. And the GRU’s pre-normalization output norm is not a confidence signal (correlation with correctness +0.0073, and the loss is exactly norm-invariant); the top-1-minus-next-10 score margin is, at +0.294.

The one durable theoretical result. The memo’s cross-cutting observation that **the learned-combiner anomaly is mechanically explained** is the part that changes how the record should be read. A prior oracle search over *global static scalar* leg weights returned +0.004 straddling zero, and it is now clear why that is compatible with a real +0.020 from a gate: dropping the Markov leg

globally gains in the zero-support bucket and loses in the warm bucket, and the two cancel almost exactly. No constant can express an intervention whose sign flips with a per-query data statistic, and z-normalization is scale-invariant so it provably cannot see one. The registered “four independent learned-combiner nulls” pitfall should therefore forbid *learned score-level fusion* — fitted weights, stackers, rank fusion, end-to-end representation fusion — and **not** per-query evidence gating, which is a third thing: nothing is learned, nothing co-adapts, and the condition is a train-side count. *Source: 2026-07-27b-nexttrack-model-topology-research-memo.md (research memo; see the provenance partition above).*

22 markov_gate: a leg that abstains where it has no evidence

Rationale. This is the native two-split confirm of the memo’s headline proposal (Section 21), and it produces the largest single recall gain in the whole next-track arc from a nine-line change with no new parameters.

The mechanism, and why it is not the refuted combiner lane. The Markov leg mixes its bigram row with an artist/genre back-off through a trust weight $\text{trust} = n_u / (n_u + 8)$, where n_u is the train-level bigram support of the query’s last prefix item u . At $n_u = 0$ the trust is zero and the leg is **pure back-off**: it carries no track-level evidence at all. But the blend’s candidate normalizer is *scale-invariant* by construction — it maps each leg’s score vector to unit variance so that three legs on wildly different scales are commensurable — so the evidence-free back-off is re-inflated and speaks at a full one-third weight anyway. On the canonical split that happens on 795 of 1431 **test queries**, 55.6%. The intervention is to make the leg *abstain* at $n_u = 0$ and renormalize the divisor from three legs to two. Nothing is learned; the condition is a train-side count. That is exactly the third category the registered learned-combiner pitfall did not cover, and the bucket decomposition below is a mechanical proof that no global static weight could find the effect: dropping the leg globally is a -0.0070 tie, because the zero-support gain and the warm-bucket loss cancel.

Design and the preflight gate. Six runs on the hub, all **seq-blend** from the champion definition, with the controls **re-trained in this batch** rather than substituted from the archive so that every paired comparison is code-matched to one generation. Two splits: the canonical **seq-20260715-131139** and the disjoint earlier-holdout **seq-20260718-211238** (64% cold, n_{test} 1431). The hard pre-registered stop was that the ungated control had to reproduce Recall@10 0.21174, exactly 303 of 1431, proving the new flag left the champion bit-identical. **It did, to the last digit** (0.21174004192872117).

Results. Table 24 is the batch, Table 25 the paired statistics, and Table 26 the decomposition that carries the argument. Figure 15 plots both.

Interpretation. **The defect is real, it belongs to the normalizer, and it is worth 29 hits.** Scale-invariance is exactly what makes the z-blend robust to legs on incommensurable scales, and exactly what destroys their *confidence*. A Markov leg that has never seen the current track emits a pure artist/genre back-off; the normalizer re-inflates it to the same footprint as a GRU that has actually modelled the prefix, and the blend averages the two as equals. Turning that third of the vote off where it is uninformed converts 43 misses into hits and 14 hits into misses — net +29, +9.6% relative, on the best model on record. Recall@10 0.23201 (332 of 1431) is **the best recall figure in this document**, with the paired CI above zero on two disjoint splits and MRR up on both. For scale: the champion’s own coronation margin over the prior crown was +0.0259

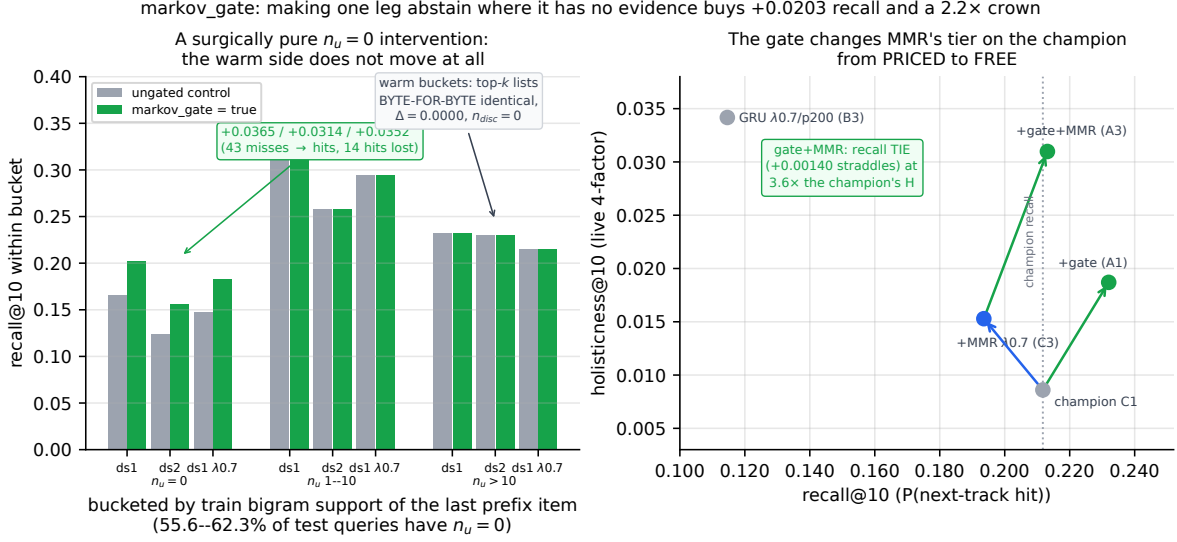


Figure 15: **Left:** the bucket decomposition on all three control/arm pairs (**ds1** canonical, **ds2** the 64%-cold second split, **ds1 $\lambda 0.7$** the same canonical pair under an MMR re-rank). The entire effect sits in the zero-support bucket; in both warm buckets the arms' top- k lists are byte-for-byte identical, which is why a global leg weight provably cannot express this intervention. **Right:** the same result on the recall $\times$ crown plane. MMR alone (blue) is a *priced* lever on the champion — it buys crown and loses recall with the CI below zero. The gate (green) buys *both*. Composed, gate+MMR lands at champion recall ($\Delta +0.00140$, straddling) with 3.6 $\times$ the champion's holisticness, so the gate changes MMR's tier on the champion from PRICED to FREE.

canonical and +0.0426 on the second split; this gate adds +0.0203 / +0.0196 *on top of that*, at no additional training cost.

The gate is also a crown lever, which was not predicted. ΔH is above zero in all three paired comparisons, and the facets say why: abstaining kills the *bigram-echo* component of the recommendation. Artist concentration falls 0.69165 $\rightarrow$ 0.59821 and adjacency 0.60971 $\rightarrow$ 0.57477 — the list stops repeating one artist — while *mood_coh* rises 0.39822 $\rightarrow$ 0.46226 and *music@10* rises 0.45379 $\rightarrow$ 0.49319. All four crown factors move the right way at once. This is the first lever on record that de-eagers the champion *without* paying in mood coherence, and it resolves the exact double-payment Section 19 identified as structural: there, de-eagering was a re-rank fighting the model's own scores, so walking away from the seed artist also walked the blend out of the prefix's mood neighbourhood. Here the de-eagering is the removal of a vote that had no business being cast.

Gate and MMR are additive, not substitutive, and their composition is the headline for the crown board. The gate-alone Δ (+0.02027) and the gate-under- $\lambda 0.7$ Δ (+0.01957) are indistinguishable, and the buckets show why: MMR re-ranks the top of a scored list, the gate changes which list gets scored, and they act on disjoint parts of the pipeline. Standing alone, MMR $\lambda 0.7$ on the champion is TIER-2 PRICED (−0.01817 recall, CI below zero). Add the gate and the recall cost is paid back exactly: A3 is +0.00140 [−0.01188, +0.01468], a statistical tie with the champion, at H 0.030980 — 3.6 $\times$ the champion's 0.008603, comfortably above the mandatory 0.108 recall floor. On the crown board that sits at **rank 2**, behind only the $\lambda 0.7$ single GRU, which scores its 0.034171 at recall 0.11461: A3 delivers 96% of the incumbent leader's crown at 1.86 $\times$ its recall.

The honest caveat, and why the out-of-sample structure redeems it. The memo screened ten gating variants on the test set, so its banked +0.02027 was selection-optimistic and could not be evidence. Four things redeem it here: the effect reproduces on a *second, disjoint, 64%-cold split that was not screened on*, within 0.0007 of the canonical figure; it reproduces under a third condition ($\lambda 0.7$) that was not screened either; four mechanistically related variants in the memo all landed at 0.019 to 0.020, so the screen was measuring a plateau rather than selecting a lucky corner; and the warm buckets do not move by a single hit, which is a *structural* prediction of the mechanism rather than a fitted outcome. One residual risk remains: the $n_u = 0$ boundary itself was chosen with test knowledge. The boundary-free version — scaling the leg by **trust before** normalization instead of hard-abstaining — is the top follow-up, and it is strictly more general. Nor is any of this a fluke of one split’s cold rate: the second split is 64% cold, its zero-support bucket grows to 62.3% of queries while the per-query gain shrinks slightly, and the net Δ comes out nearly identical. The lever’s value scales with how much of the query distribution has no bigram evidence, which is the majority on any split of this corpus.

Governance, and an infrastructure finding that nearly ate the result. Nothing was promoted, registered or tagged; by design boundary this campaign carries a promotion *recommendation* only, and the board row in Table 1 is labelled accordingly. The server’s deterministic top-12-by-holisticness recompute nevertheless auto-promoted four of the batch’s runs, which is server behaviour rather than a campaign action and left the group needing re-curation. Separately: the remote training worker was live with a **stale checkout containing zero occurrences of the new flag**. Had any gated arm been dispatched there it would have executed the *ungated* path and silently returned a null — the registered “remote checkout can silently diverge” pitfall, in its worst form, because the failure is a wrong number rather than a crash. It could not happen here only because remote dispatch is opt-in and every run in this batch was posted without it (confirmed by an unclaimed marker on all six). Until the checkout is freshened, any queued **seq-blend** run silently ignores the gate.

Record amendments. Two, both narrowing rather than overturning. (1) The learned-combiner pitfall should forbid *learned score-level fusion* and not *per-query evidence gating*; the prior oracle null and this +0.020 win are both true and are reconciled by the bucket decomposition, since a global static scalar cannot express a gate whose sign flips with n_u . (2) The normalizer’s erasure of per-leg *confidence* is a **general** defect, of which the Markov leg’s zero-support back-off is one instance. Any leg with a query-dependent notion of “I have no evidence here” is mis-weighted the same way, and the content and GRU legs have not been audited for it. *Source: 2026-07-27c-nexttrack-markov-gate-confirm.md.*

23 The eagerness regularizer at a calibrated margin: refuted, with a mechanism

Rationale. Section 15 left **eager_beta** — the only *training-time* de-eagering lever in the repository — on record as “a weak knob, not recommended”. That verdict deserved a re-test, because the probe behind it used a margin of zero at a dose worth 1% to 3% of the loss. This campaign asked whether the lever works when the margin is placed in the measured *right tail* of the transition-cosine distribution at 15 to 60× that dose. The stake was deployability: at the time, **mmr_lambda** entered only through the evaluation path and never reached serving, so the training-time regularizer was the one anti-eager candidate that could actually ship.

The design was written so that a null would still be informative. The pre-registered primary was the paired Δ on `artist_conc@10` against an in-batch control, with `holisticness@10` explicitly *demoted* to secondary because at the operating point $\partial H / \partial \text{music@10} = +0.2062$ against $\partial H / \partial \text{eager} = -0.0378$ — a $5.46\times$ ratio that would have scored a *successful* de-eagering as approximately zero. A relevance floor was pre-registered in two parts: absolute `music@10` ≥ 0.4300 and a paired $\Delta \text{music@10}$ lower bound above -0.030 . Eight `seq-dualgru` runs on the canonical split `seq-20260715-131139`, all remote, 60 min of compute, none failed. The bit-identity gate passed at exactly 170 hits with the reconstructed concentration matching the on-record arm to 15 decimal places, and a penalty-active gate was pre-registered as a silent-failure detector: it **passed on three arms and FAILED — reporting the penalty inert — on four**, which is the campaign’s result.

Results. Table 27 is the batch and Table 28 the paired statistics. Figure 16 shows the mechanism.

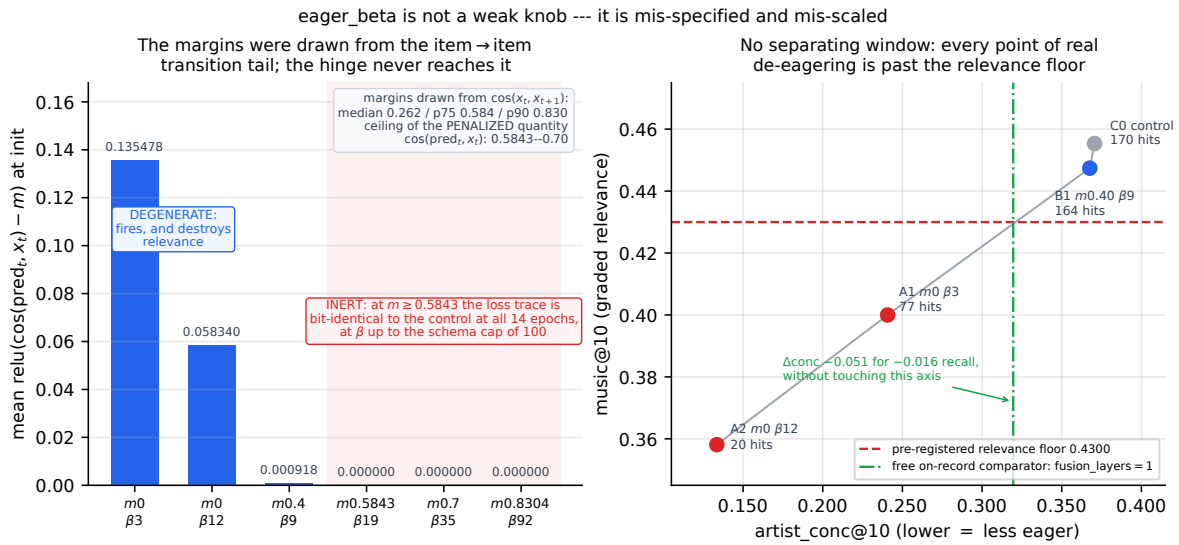


Figure 16: **Left:** the campaign’s central result. The hinge’s realized penalty at initialization, reconstructed from epoch-1 loss inflation at shared initialization (identical seed, so identical init and batch order). The margins were calibrated from the *item*→*item* transition cosine (median 0.262, p75 0.584, p90 0.830), but the penalty acts on *prediction*→*current*, whose ceiling over training sits between 0.5843 and 0.70: at $m \geq 0.5843$ the penalized quantity never enters the hinge’s support and the loss trace is bit-identical to the control at all 14 epochs, at β up to the schema cap. **Right:** where the knob *does* fire, eagerness and relevance are the same mechanism. Pushing the prediction off the current track’s latent also pushes it off the region where the true next track lives, so there is no window between real de-eagering and breaching the relevance floor. The dash-dotted line marks the free on-record comparator that no arm here matched inside the floor.

Interpretation. The refutation is expressive, not dosimetric, and that is the whole finding: the calibration measured the wrong distribution. The margins were drawn from $\cos(x_t, x_{t+1})$ — how far apart two consecutively played tracks are — but the hinge acts on $\cos(\text{pred}_t, x_t)$, how close the model’s *prediction* sits to the current track. Those distributions are not comparable. Reconstructing the hinge from epoch-1 loss inflation puts the ceiling of the penalized quantity between 0.5843 and 0.70, so p75 and p90 of the real-transition distribution lie

outside the support of the quantity being penalized, and arms B3 and B4 reproduced the control’s loss to 16 significant digits at every one of 14 epochs. The design’s hedge — that the realized penalty would *exceed* its nominal value — is empirically inverted by orders of magnitude. The reason is structural: an InfoNCE-trained prediction is a *hedged near-average* over many plausible continuations, so its absolute cosine to any single item is bounded low. **Eagerness is a ranking property — whether the current track’s neighbours occupy the top slots — and a hinge on an absolute cosine cannot express it.**

Where the knob does fire, eagerness and relevance turn out to be the same mechanism. At $m = 0$ the lever works spectacularly on the facet — A2 cuts concentration 0.371→0.134 and drives `ild` to 0.676 — while `music@10` collapses to 0.358 and recall to 20 of 1431. There is no separating window: 27% of real transitions are negative-cosine, so pushing the prediction away from the current track’s latent also pushes it out of the region where the true next track lives, and `mood_coh` collapses too (0.480→0.222). Already at $\beta = 3$ the absolute relevance floor is breached. The pre-registered failure mode fired exactly as written.

The prior is sharpened rather than retired, and the usable band is identified. The 2026-07-22 probe at $\beta \leq 0.2$, $m = 0$ reported “weak” — correct in band, wrong in interpretation. That setting sits just *under* the destruction threshold, and by $\beta = 3$ the same setting is a relevance destroyer. The entire usable dynamic range at $m = 0$ therefore lies between $\beta = 0.2$ and $\beta = 3$, a factor of 15, unexplored, bounded above by degeneration and below by no effect. The dose ladder run here was **not underpowered; it was aimed past the target.**

Two second-order results worth keeping. First, demoting H to secondary was load-bearing: A2 has by far the best facets and the *worst* H (0.003569), because the grounding factor annihilates it exactly as the $5.46\times$ derivative ratio predicted. Had H been primary this campaign would have read as a flat null instead of a mapped frontier. Second, the cheapest real de-eagering on record remains an *architecture* setting, not this knob: the free on-record comparator at `fusion_layers = 1` gives Δ concentration -0.051 at a -0.016 recall cost, and no arm here matched that inside the relevance floor — the floor-respecting arms delivered between -0.003 and $+0.0002$, and the arms that beat -0.051 paid 4 to $8\times$ the recall.

Why the approved later phases did not run. All three were gated on a phase-A win, and there is none. Phase C’s arms were additionally defined *by reference* to the winner’s margin and β , so with no winner they are undefined; applied literally to the arms that passed the floors, the selection rule would have spent three runs transferring a *provably inert* setting to another family. The runner stopped rather than improvise, which is the correct outcome: the next move is a re-aimed design, not the approved phase C. Concretely, re-specify the penalty rather than re-tune it — hinge on a *ranking* quantity such as $\cos(\text{pred}, x_t) - \cos(\text{pred}, x_{t+1})$, or an explicit same-artist mass term over the top- k — and, by elimination, make the eval-time re-rank reach the serving path, since with the training-time lever refuted the only lever that works was at that moment one the serving path could not reproduce. That last observation is what the next section acts on. *Source: 2026-07-30-eager-margin-calibrated-refutation.md.*

24 The walk surface: measuring what the product actually serves

Rationale. Every verdict up to this point — including the crown board of Table 1 and all three holistiness generations of Section 18.2 — is computed on a **one-shot top-10**: the model sees one held-out prefix, ranks the catalog once, and is scored against one target. But the artifact this lab ships, `clients/infinite-playlist`, serves an **autoregressive walk**: it picks a track, appends it

to the session, re-scores, and repeats for as many steps as the listener asks for. Those are different surfaces, and `domain.toml` has said so in prose the whole time — its `[metrics]` block states the goal as “mood-coherent, non-eager *sessions*, not guessing the literal next track.” The stated intent and the implemented metric had diverged. This section closes that gap: it defines four walk-level metrics calibrated against the user’s own listening, adds three *retrieval-time* controls, and runs a matched-sample head-to-head. It is also the one place in this document where a documented refutation is *reversed* on a new objective, which is why it is reported at length rather than folded into the campaign sections above.

24.1 The surface mismatch, and why drift was invisible

Two mechanical facts explain why nothing above could have caught a drifting journey.

First, the anchor grows. The `mood_coh` facet of `holisticness@10` measures each candidate’s cosine to the *prefix* centroid (`seq_common.py:558`). On a one-shot evaluation the prefix is fixed and that is a reasonable read of “does this list stay on vibe.” On a walk the prefix *is* the journey so far, so the reference point travels with the model: a journey that leaves its starting region by a thousand individually legal steps scores well *at every step*, because each step is compared to a centroid that has already moved most of the way there. **Drift is invisible by construction** — not under-weighted, but unmeasurable in this formulation.

Second, the composite is dominated by relevance. Differentiating the generation-3 composite at the champion’s operating point gives $\partial H / \partial \text{music@10} = +0.206$ against $\partial H / \partial \text{eager} = -0.038$: `holisticness@10` is $5.5\times$ more sensitive to its relevance factor than to its eagerness factor. So even the term that was *meant* to price session quality is mostly re-reporting graded relevance. This is the same asymmetry that Section 18.2 observed from the outside, where the grounding factor’s ~ 0.3 multiplier made every generation-2 value rank below every generation-1 value; here it is quantified at the derivative.

24.2 Four walk-level metrics, calibrated against the user’s own sessions

The new instrument (`tools/walk_eval.py`) generates journeys from held-out *real* sessions — a 5-track seed taken from the session, then 20 autoregressive steps, prefix items excluded, at most one track per artist unless stated — and scores them on four axes:

- 1. Transition match (primary).** Do the walk’s step-to-step distances look like this user’s *real* listening? This follows the playlist-lab finding directly: measure *transitions*, not averages, and calibrate against the user’s own sessions. Averaging a distance-to-centroid is the wrong shape — it penalizes a legitimate genre move and rewards standing still. Reported as the 1-Wasserstein distance (W_1 , lower better) between the journey’s consecutive-pair cosine distribution and the real one, plus two legible reads: the median step cosine and the share of steps inside the real p10–p90 band.
- 2. Vibe and drift.** The *seed-anchored* cosine: mean cosine of the generated tracks to the fixed seed-core centroid (`vibe`, higher better), and the end-quarter minus start-quarter difference (`drift`, nearer zero better). The anchor is fixed at the seed and does *not* grow, which is precisely the defect of `mood_coh` that Section 24.1 identifies.
- 3. Variety as credit.** Distinct artists and distinct genres in the journey, counted as a *credit* rather than as the cost that `artist_adj/artist_conc` price. Cohesion is not one genre:

different genres are welcome if they preserve the vibe, so a journey covering more of them at equal vibe is better, not worse.

4. **Grounding.** Graded relevance (best sonic cosine, in the balanced musical-distance space of Section 14) between each generated track and what the user *actually* played next in that session. This is the guard-rail: without it a pleasant unrelated drift wins, which is exactly the lesson the crown learned when `music_rel` was added.

The calibration, and the single number that reframed the project. The reference distribution is the user’s own consecutive-pair cosines over the 5723 *train* sessions of seq-20260715-131139: **73 632 pairs, median 0.261, p10–p90 [−0.192, 0.830], and 27.0 % of real steps *negative*.** Real listening moves — it changes direction more than a quarter of the time. Against that, the engine the showcase actually shipped (a single GRU as `gru.onnx` with a seed anchor of 0.4) walks with a median step of 0.661, roughly 2.5× tighter than the user does (Figure 17). The complaint that the showcase felt “flat despite lots of artists” now has a measurement behind it: many small, safe moves. The blend champion has the opposite failure — its median step, 0.317, sits essentially on the real median, but its *vibe* is only 0.187, because its InfoNCE projection (Section 11) encodes “what follows this,” not “what sounds like this,” so a mood anchor applied *in* that projected space caps at 0.44 while the same anchor in the raw PCA-192 space reaches 0.58. Table 29 is the full first read.

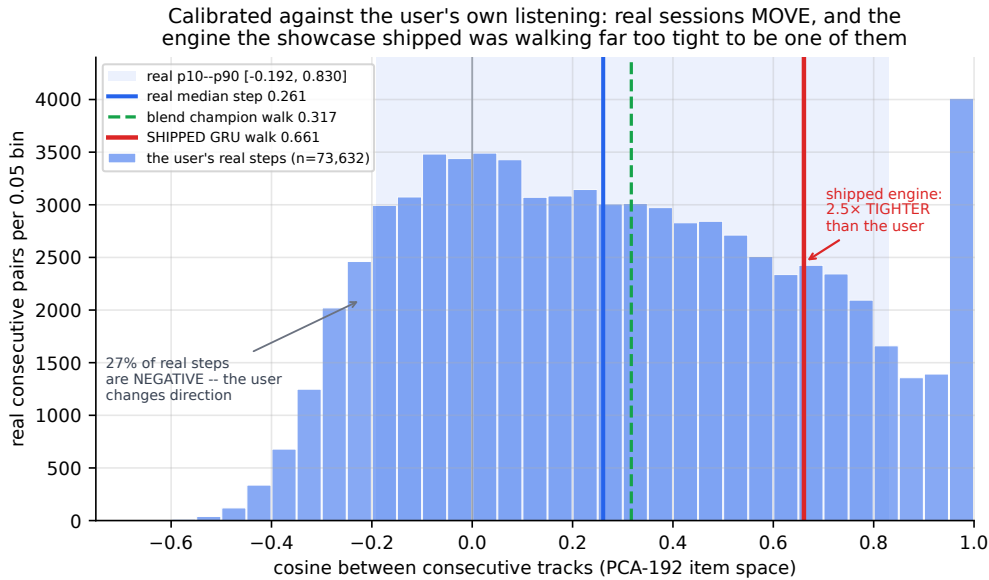


Figure 17: The core argument of this section, in one figure. The user’s own 73 632 consecutive-transition cosines (PCA-192 item space, train sessions only) are a *broad* distribution: median 0.261, p10–p90 [−0.192, 0.830], 27.0 % of steps negative. The engine the showcase shipped walks at a median step of 0.661 — about 2.5× tighter than the listener it is imitating — and no one-shot metric could see it, because `holisticness@10` scores a top-10 and its `mood_coh` anchors to a prefix centroid that grows with the walk. The blend champion (0.317) matches the real stride but cannot hold a vibe. Bin counts transcribed from the calibration block of `tools/walk_eval.py`.

W_1 is **not pairable, and that mattered**. The distributional W_1 is computed over a *pooled* sample of steps, so it has no per-session value and cannot enter a paired bootstrap — the statistic every other verdict in this document rests on. The first two sweeps therefore compared arms measured on *different* samples (20 sessions $\times$ 20 steps for the frontier, 12×15 for the dual-tower probe), and a vibe gap of 0.588 against 0.624 is well inside what that mismatch alone can produce: nothing was established. The head-to-head (Section 24.4) therefore replaces W_1 with **stride_err** = |median step - 0.261|, a per-session quantity that pairs, at the cost of describing only the distribution’s centre.

24.3 Three retrieval-time controls, and two refuted ones

None of the three controls below touches training. All are applied at *retrieval* time, on the scores the model already produces, which is what makes them cheap to ship and cheap to re-tune.

ANCHOR (*a*). Add $a \cdot \cos(\text{candidate}, \text{seed centroid})$ to every candidate’s score. It holds the geist by pulling the walk back toward the region the listener seeded it from. For the champion the term is z-normalized over the admissible candidates, because its base score is itself a z-blend of three legs and adding a raw cosine to a z-blend is scale-mismatched.

STRIDE (*s*). Subtract $s \cdot \cos(\text{candidate}, \text{previous track})$. It *enlarges* each step without letting the walk leave the seed’s neighbourhood, because the anchor still applies — the two terms are orthogonal by design, one pulling toward a fixed point and one pushing off the last point. It is deliberately *deterministic*: sampling would also loosen the stride, but the showcase ships permalinks, and “same recipe $\Rightarrow$ same journey” is now a promise, so a stochastic decoder was ruled out rather than untried.

CAP. A *hard* per-artist limit applied to the ordering (`seq_common.artist_cap_order`), after MMR, in both `eval_from_scores` and `rank_topk/predict_ranking`, so evaluation and serving agree by construction. Best-effort rather than absolute: where a prefix’s admissible candidate space is exhausted the remaining slots are backfilled with deferred items instead of returning fewer than k . Default **None**, so every pre-existing run and model is byte-identical; and because it needs no sonic vectors it is the only diversity lever here that ports to a thin client unchanged.

Why a cap and not a penalty — the two refuted controls. Both soft anti-eager levers of Section 15 fail for one shared reason: a soft penalty in score space loses to a steep relevance gradient. **eager_beta**, the training-time regularizer, is *mis-specified* (Section 23): it hinges on the absolute value of $\cos(\text{prediction}, \text{current track})$, which a hedged InfoNCE prediction never drives above ~ 0.6 , whereas eagerness is a *ranking* property — whether same-artist items occupy the top slots — not a property of an absolute similarity. **mmr_lambda**, the eval-time re-rank, was additionally *eval-only* until the serving path was fixed, so the λ frontier of Section 19 — genuine, confirmed, and the first crown lever on record — had never been a deployable lever at all; and even fixed it measures inert on the pathological case. On the real playlist **rock chabon ahre** the model ranks 59 of its top 60 candidates as one artist and the next artist first appears at **rank** 75, while MMR’s entire diversity budget there is worth $0.3 \times 0.26 \approx 0.078$ of normalized score — nowhere near enough to pay for a 70-rank relevance drop. A cap does not compete on score at all: it *cannot be outscored*, the same property `seq_extend`’s **artist_cooldown** already relied on. The showcase’s own soft artist penalty turned out to be effectively **binary** in the same way — at 0 it produced artist blocks with runs up to 13, and at *any* value ≥ 0.25 every 30-step walk returned

30 distinct artists, max run 1, concentration 0.000 — so the slider could not reach the middle at all, and a hard cap exposed as a journey shape (Explore / Balanced / Continue / Blocks) is what reaches it.

The cap on the one-shot surface, for the record. Table 30 is the cap measured where the rest of this document measures: the canonical split’s one-shot top-10, paired on bit-identical cached scores. It cuts `artist_conc@10` from 0.692 to 0.123 at cap 3 and raises the generation-3 `holisticness@10` from 0.008603 to 0.033862, at a cost of 73 of 1431 exact hits. The result that was *not* predicted is that **music@10 goes up, CI above zero, at every cap on both models tested** (+0.011 to +0.023): the pre-registered failure mode — de-eagering by degrading into generic filler — does not occur, because the items promoted from rank 75 and beyond are *more* musically apt to the true continuation than the same-artist tracks they displace. The cap trades *literal* relevance for *graded* relevance. On the champion plus `markov_gate` (the evidence gate of Section 22, which suppresses the Markov leg where the current item has no bigram support) the same cap 3 gives `artist_conc` 0.598→0.113 and `holisticness@10` 0.018699→0.048063, reproduced to every digit by three registered confirming runs (run-20260730-235524-0ec1d-seq-blend at cap off, run-20260730-235524-93d38-seq-blend at cap 3, run-20260730-234635-a6252-seq-blend at cap 5), and it **hardens on the disjoint 0.64-cold split** seq-20260718-211238 on two pre-registered legs: $\Delta\text{artist_conc}$ -0.52699 [-0.54546, -0.50825] CI below zero *and* $\Delta\text{music@10}$ +0.02644 [+0.01919, +0.03368] CI above zero, with the relevance gain *larger* on the colder window. Two cautions travel with it: exact recall is the only regressing metric at every cap, and the canonical split’s absolute `music@10` ≥ 0.43 floor does **not** transfer — the second split’s own control sits at 0.4295 on a legitimately colder window, so floors must be derived per baseline and pre-registration must be on the paired Δ .

The anchor $\times$ stride frontier. With the two score-space controls in hand the question becomes whether the *conjunction* is reachable at all: a real-listening stride *and* a held vibe. Table 31 sweeps 24 cells — 16 on the GRU, 8 on the champion, all at cap 1 — against a target window pre-registered before the sweep: median step inside the real band’s middle (0.20 to 0.38) *and* `vibe` ≥ 0.50 . Exactly two cells land inside it, one per engine (Figure 19). The structure of the grid is the finding: the two controls are genuinely orthogonal — along a fixed anchor, raising the stride walks the median step down monotonically while the vibe decays much more slowly — so the window is reachable only with both on, and neither shipped configuration was inside it. The champion’s anchor also has to be applied in the *raw* PCA-192 space rather than its own projected geometry, which is why the frontier reaches 0.503 where the first probe capped at 0.44.

24.4 The matched-sample head-to-head

Rationale. The two sweeps above each found a cell inside the window, but on different samples, and W_1 cannot be paired — so the finalists had to be re-run against one another on *identical* material before anything could be claimed. `tools/walk_headtohead.py` walks five arms over the same 40 held-out sessions from the same 5-track seeds, at cap 1, and reports paired bootstrap differences (2000 resamples, rng 1337) against the **shipped** configuration — the single GRU at anchor 0.4, stride 0 — on five per-session metrics: `vibe` (higher better), `stride_err` (lower better), `drift` (nearer zero), `genres` (higher) and `ground` (higher).

Results. Table 32 gives the means and Table 33 the paired differences; Figure 20 plots the latter one panel per metric, each on its own scale.

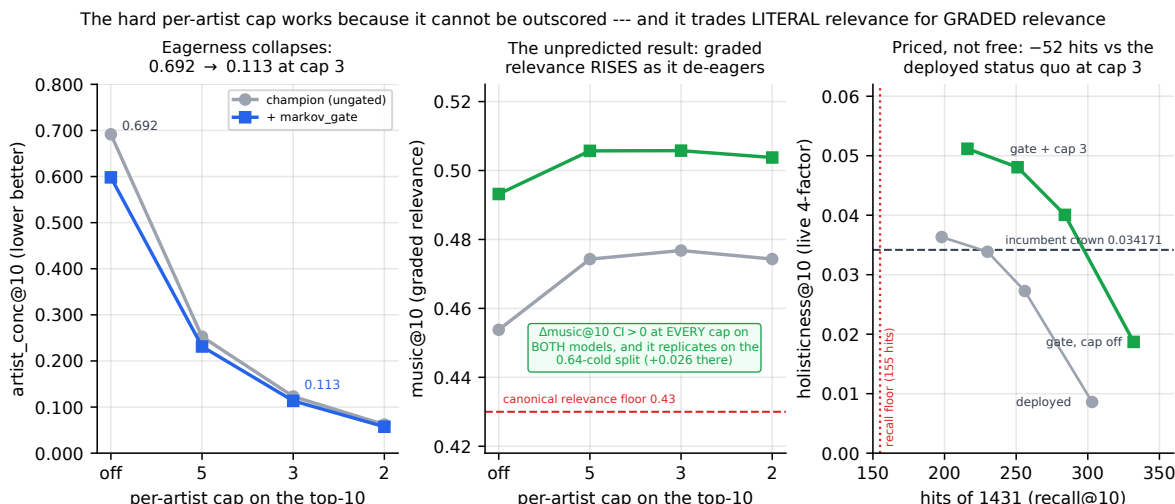


Figure 18: The cap, across both models measured. **Left:** artist concentration collapses, 0.692→0.113 at cap 3 on the gated champion. **Middle:** the result that was *not* predicted — graded relevance *rises*, with the paired CI above zero at every cap on both models, and it replicates on the disjoint 64%-cold split at +0.026. The pre-registered worry was that a cap would de-eager by degrading into generic filler; the opposite happens, because the items promoted from rank 75 and beyond are *more* musically apt to the true continuation than the same-artist tracks they displace. **Right:** the price, on the recall × crown plane. The cap is *priced*, not free: gate + cap 3 costs 52 of 1431 exact hits against the deployed status quo, while landing a generation-3 holisticness@10 about 40 % above the incumbent crown at 1.5× its recall. Every arm clears the 155-hit recall floor, so none is an instance of the “maximize holisticness by predicting worse” pathology.

Interpretation. Read down the `stride_err` column first: all four tuned arms cut it by ~ 0.25 with the CI below zero, which is simply the stride control working — the shipped engine’s 0.399 error was $\sim 2.6\times$ the next worst arm’s. The interesting variation is what each arm *pays* for it. Pushing the single GRU to a real stride **costs** *vibe* (-0.0409 , CI below zero) and the champion pays far more (-0.1327 , CI below zero): for these two engines stride and vibe are in tension, exactly as the frontier’s gentle-but-real slope predicted. Both dual-tower arms escape the trade-off. `latent/latent` at $a0.8\ s0.4$ *gains* *vibe* ($+0.0297$, CI above zero) and genres ($+1.05$, CI above zero) while matching the stride; `latent/cummean` at $a0.8\ s0.5$ holds *vibe* ($+0.0013$, straddling — held, not traded) and is the only arm with *three* wins, adding 1.5 genres (CI above zero) and the only significant drift improvement ($+0.0351$, CI above zero, i.e. the journey ends closer to where it started). On the objective as stated — mood-holding, genre-exploring sequences — `latent/cummean` is the winner, and the single GRU that was actually shipped is not on the frontier at all.

24.5 The dual-tower reversal, and what the record already knew

This is worth stating plainly, because it is the most consequential methodological result in this document: **the dual-tower family was closed by three campaigns on recall grounds and is the winner on the stated objective.** Section 16 refuted it across the complete $\{\text{latent}, \text{delta}, \text{cummean}\}^2$ view matrix; the `cummean/cummean` arm was even flagged *degenerate*

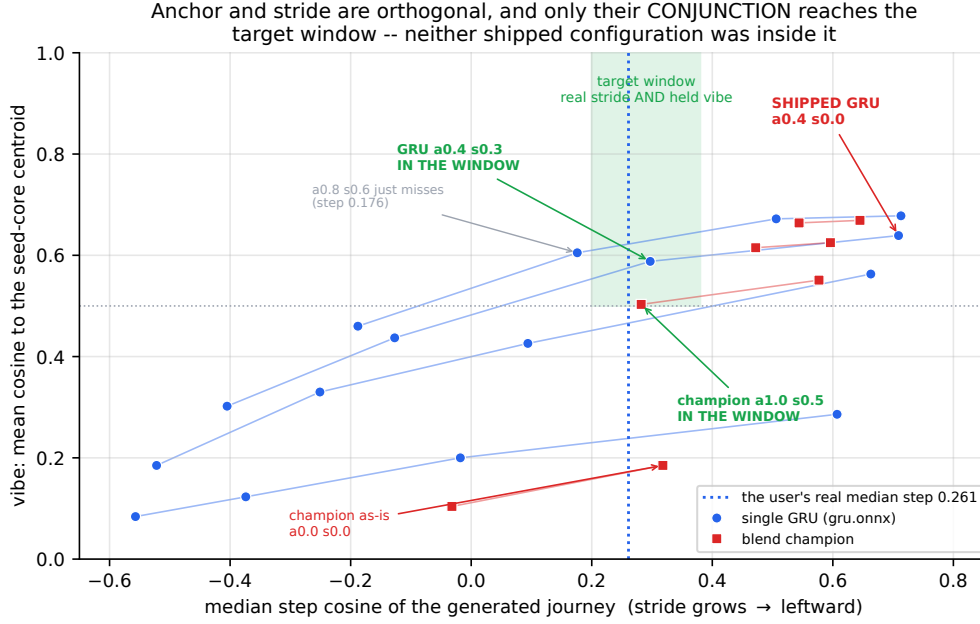


Figure 19: The two retrieval-time controls are orthogonal, and only their *conjunction* reaches the target window (shaded: median step $\in [0.20, 0.38]$ and $\text{vibe} \geq 0.50$). Each line joins a fixed anchor as the stride grows, which moves the journey leftward (larger steps) while the vibe decays slowly — so the trade-off is real but gentle. Neither shipped configuration is inside the window: the GRU at $a = 0.4$, $s = 0$ holds a vibe of 0.639 but strides 0.709, and the champion as deployed strides 0.318 at a vibe of 0.185. 20 held-out sessions $\times$ 20 steps, cap 1 (`tools/walk_frontier.py`).

for topping the generation-1 holisticness at Recall@10 0.066. Every one of those judgements was made on one-shot next-track accuracy, and none of them is wrong on that surface — the reversal is not a correction of the recall verdicts but a demonstration that they answered a different question. Section 18.1’s power analysis makes the two readings compatible rather than contradictory: what the dual-tower campaign established was the absence of a *large* recall gain, which is fully consistent with the family being better at something the recall metric does not measure.

The corroboration was already in the record, filed as a negative. The interpretability note at `predictors/seq_dualgru.py:113` records that the `cummean` tower “added real linear directions (CKA 0.458) but **zero** next-item concepts tower A lacked.” Read as a next-track result that is dead weight, and it was written down as such. Read on the walk surface it is the useful part: a tower whose content is orthogonal to next-item prediction is exactly a tower carrying *session state* rather than adjacency. Two structural facts complete the explanation. The family retrieves in the *raw* item-latent space with no learned projection, so a raw-space mood anchor is native to its geometry — unlike the champion, whose InfoNCE projection strips precisely the sound-similarity structure an anchor needs. And one promoted arm reads the *causal mean* of the prefix, which is structurally a running-average signal.

One sub-hypothesis is refuted. The natural prediction from the above — that the `cummean` tower holds a vibe *natively* — is false: with both controls off its vibe is 0.236, statistically indistinguishable from `latent/latent`’s 0.234. The architecture does not supply the mood hold on its own; it only becomes the best host for one once the anchor and stride are applied. That is a weaker and more interesting claim than the one the structural argument suggested, and it is the one the measurement supports.

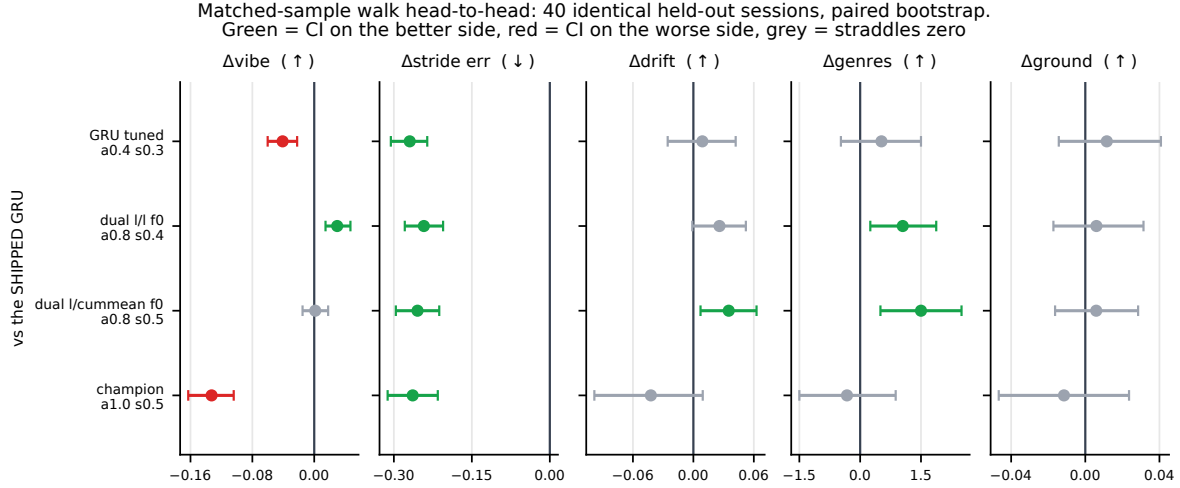


Figure 20: The matched-sample head-to-head, one panel per metric on its own scale — deliberately not a dual axis, which would imply a comparability these five quantities do not have. Green is a CI entirely on the metric’s better side, red entirely on the worse side, grey straddles zero. Every arm buys the real stride; only the dual-tower arms buy it without paying vibe, and **ground** straddles for *every* arm.

24.6 Deployment, and what is not established

The tuned dual-tower arm is now the **default engine** in the showcase, exported as `dualgru.onnx` (3.16 MB) with ONNX-versus-torch parity of 1×10^{-7} across every prefix length $T = 1 \dots 40$ — the `cummean` cumulative sum was the trace risk, and it is the reason the parity check swept T rather than spot-checking it. The single GRU is kept alongside for comparison, and the tuned anchor/stride live in the deployment manifest rather than in code, so re-tuning is a re-bake and not a code edit. The blend champion was removed from the showcase end-to-end: served from a Worker its walk cost ~ 0.94 s of CPU per batch of 8, roughly $100\times$ the 10 ms CPU cap of the free plan the showcase is deployed on, and batching cannot rescue a single step that already costs ~ 117 ms.

What this section does *not* establish. Four limits, stated plainly. **(1)** **ground** straddles zero for *every* arm (Table 33), so no arm is established as better at predicting the real continuation — the wins are about journey *shape* only, and the grounding guard-rail confirms only that no arm degraded into pleasant irrelevance. **(2)** One split, one seed, 40 sessions of 20 steps; the walk metrics have had no second-split hardening of the kind Section 12.1 applies to the recall crown, and the target window itself was chosen from the first sweep’s spread. **(3)** Nothing here has been *listened* to. Every claim is a geometric claim about cosines in a PCA-192 space, and “holds a mood” is operationalized as “stays near a seed centroid,” which is a hypothesis about perception, not a measurement of it. **(4)** The project’s primary metric is **unchanged**: `holisticness@10` remains `[metrics].primary` and this document’s board remains denominated in `Recall@10` (Section 18.2). Whether the walk metrics should become primary is an open decision, not a settled one — and until it is settled, the honest summary is that the walk surface is now *measurable*, and on it one refuted family wins. *Sources: 2026-07-30c-artist-cap-measurement.md, PROJECT-FACTS.md (roll-up 2026-07-31b), tools/walk_eval.py, tools/walk_frontier.py, tools/walk_dualgru.py, tools/walk_headtohead.py.*

25 Stage-stack topology: depth refuted on both surfaces

Rationale. The user’s question was direct: does an explicit *stack* of per-step MLP and recurrent stages beat a single GRU, on a corpus where five consecutive architecture campaigns have returned nulls? Four arrangements were specified — `ann+gru+ann+gru`, `ann+gru+gru+ann`, `ann+gru+gru+gru`, `gru+gru+gru+gru` — plus a fifth, `gru+ann+gru+gru`, chosen by the user to test *why* a leading MLP hurt at $N = 1$: Section 17 attributed that loss to half-wave rectification destroying signed prefix information that a GRU’s gated linear input transform preserves, so putting the MLP on a *gated hidden state* instead of raw signed PCA should make the harm vanish. This is also the first campaign in this arc to be judged on *two* surfaces by design rather than in retrospect, and it sets the standard the campaign after it inherits.

Design, and the equivalence proof that made a control possible. Seven runs of a new predictor, `seq-stack`, on the canonical split `seq-20260715-131139` (1431 test sessions), 90 min of compute, zero failures, zero contingency runs spent. Capacity was equalised at $\sim 1\,282\,000$ parameters across the five 4-stage arms (mutual spread 0.445 %), and all seven counts were re-verified by instantiating the model rather than trusted from the design. Everything but the stage string and the widths is held: dropout 0.1, InfoNCE, τ 0.07, 40 epochs, patience 6, lr 0.001, batch 128, seed 1337, MMR off, artist cap off.

Bit-identity with the family’s historical 176 hits is *not available* here and its absence is not a failure: the trainer re-seeds on entry, so handing it a pre-built model offsets the dropout stream even when the initial weights match. The campaign therefore gated on a **functional-equivalence proof** plus an **in-batch control**, and every verdict is stated against that control. The proof was run *before* the registry edit and the server restart, so a failure would have cost nothing: at `stages=["gru"]` the new model computes the same function as the incumbent single GRU, with $\max|\Delta|$ 0.00e+00 on both `forward` and `predict_next` at prefix lengths $T = 1, 2, 5, 17, 40$ and 394 944 parameters on both sides. The in-batch control then returned 178 of 1431, inside the pre-registered band [155, 198] and only 2 hits off the historical 176 — exactly the small offset the re-seeding argument predicts. Two further gates mattered: metric completeness was green on all seven runs (no worker-image key stripping of the kind Section 19 documents), and the walk instrument was verified against the shipped engine, whose Δvibe (−0.0033) and $\Delta\text{stride_err}$ (+0.0137) both straddle zero against the freshly-trained control. The walk harness additionally reproduced *all ten* previously recorded means exactly and stayed bit-stable as arms were added, which is what licenses pairing across separate invocations.

Results, the one-shot surface. Table 34.

No arm meets any win clause. Every 4-stage arm loses recall *and* holisticness by paired CI, and all five sit **below the mandatory** 0.108 **absolute recall floor** (0.0706 to 0.0860). Derived per-baseline from the control’s own `music@10` of 0.45519, every 4-stage arm is 5.0 % to 8.4 % below the relevance floor too, so none is even relevance-neutral. The losses are 2.5 to $3.7\times$ the measured half-width: this is not a noise-band question.

Results, the walk surface. Table 35, at the matched primary cell `a0.4/s0.3`, paired against the in-batch control on the walk metrics of Section 24.2. `vibe` is always reported with `stride_err` beside it, at matched stride, per the standing caveat that a vibe gain bought by taking shorter steps is not a gain.

Every 4-stage arm fires the refuted clause. D fires it *twice*, also failing the grounding floor (Δground −0.0250 [−0.0474, −0.0048]), and A additionally loses genre variety (−1.200 [−2.125, −0.300]). No arm approaches a win clause. Two robustness checks close the escape routes. A pre-

registered **stride probe** over $s \in \{0.2, 0.3, 0.4\}$ was run for the only two arms with any directional movement toward a walk win, and *both peak at $s0.3$ — the cell they were already measured in* (C: $\text{vibe}/\text{stride_err}$ 0.586/0.183, 0.548/0.113, 0.517/0.153; E: 0.576/0.155, 0.540/0.109, 0.516/0.174); at $s0.4$ both lose vibe with the CI below zero while **stride_err** gets *worse*. And the single best arm was re-measured at the **deployed** cell $a0.8/s0.5$, where it is refuted again ($\Delta\text{vibe} -0.0158$ [$-0.0267, -0.0059$], $\Delta\text{stride_err}$ straddling).

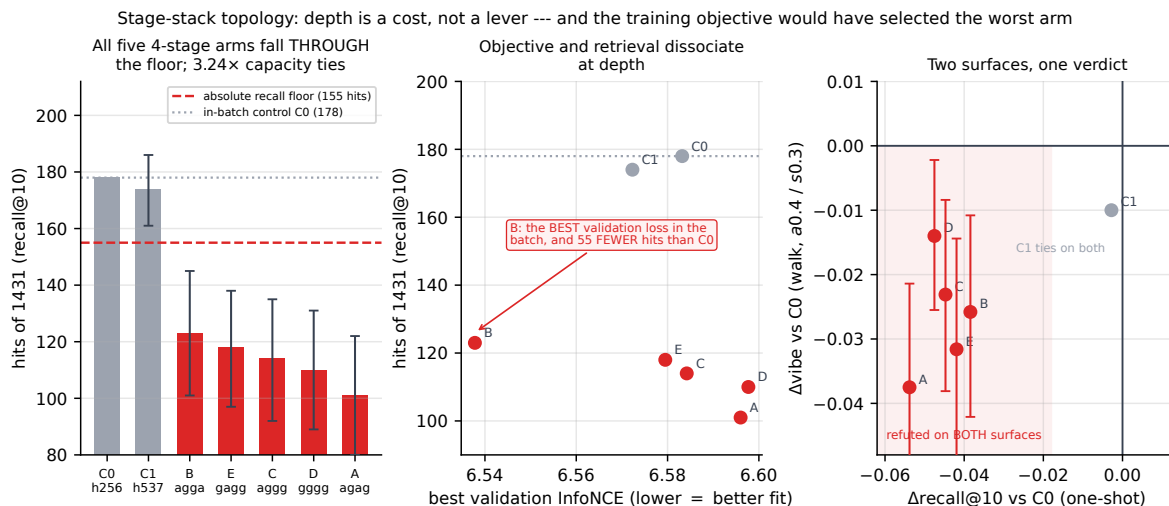


Figure 21: **Left:** all five 4-stage arms fall *through* the mandatory absolute recall floor, while the 3.24× capacity control ties the anchor — so the damage is depth, not capacity. Error bars are the paired bootstrap CI against the in-batch control, mapped onto the hit scale. **Middle:** the campaign’s most interesting number. Arm B reaches the *best* validation InfoNCE in the batch and retrieves 55 fewer hits than the control: the objective the whole family trains on and top-10 retrieval quality come apart at depth, so any architecture search selecting on validation loss would pick the wrong arm. **Right:** the two-surface closure. The walk read at the matched primary cell refutes the same five arms that the one-shot surface refutes; the capacity control ties on both. The axis does not merely fail to close on one surface — it closes on two.

Interpretation. **The axis is closed, and it is the first null in this arc to close on both surfaces.** All five 4-stage arms are refuted by paired CI on one-shot recall *and* on the walk with the instrument gate green, and the capacity control is indistinguishable on both. This is the sixth consecutive architecture null on this corpus and, unlike the five before it, it satisfies the standing rule that a one-shot verdict says nothing about the surface the product serves.

The user’s mechanism hypothesis is refuted, and that is the most reusable fact here. Arm E put the MLP on a *gated hidden state* rather than on raw signed PCA, which the rectifier story predicts should remove the harm. It does not: E reaches 118 of 1431, ties two other arrangements, beats only the worst, and comes nowhere near the control’s 178. So half-wave rectification of the *raw input* is not the mechanism — whatever is wrong survives moving the rectifier downstream of a gate. One run bought a clean refutation of a plausible and cheap-to-believe story.

The damage is depth itself, and three controls close the alternatives simultaneously. It is not capacity: the 3.24× control ties on both surfaces. It is not regularization or early stopping: *every* 4-stage arm trains *longer* than the control (21 to 40 epochs against 14), and two of five reach a *lower* best validation loss, so the pre-authorized dropout-count contingency failed both of its

trigger clauses and was not spent — the four-fold dropout application is exonerated. And it is not the MLP: a leading per-step MLP at depth 4 is recall-neutral and *holisticness-positive* against the pure-recurrent arrangement ($\Delta H +0.00291$, CI above zero). What remains is the depth of the recurrent chain, and its cost is large — 55 to 77 hits of 1431.

Bracketing beats interleaving at byte-identical capacity. `ann+gru+gru+ann` beats `ann+gru+ann+gru` by $+0.01537$ recall with the CI above zero at *the same parameter count by formula* — the tightest-controlled topology comparison the campaign could make. Interposing a per-step MLP *between* two recurrences is worse than putting MLPs on the outside, which is consistent with the depth reading: breaking the recurrent chain in the middle costs more than adding depth at the ends. Two further contrasts are flat: the number of recurrent stages (2 versus 3) straddles on recall, and the rectifier’s position buys nothing beyond beating the worst arrangement.

A leading per-step MLP is *not* harmful at depth 4, which narrows rather than confirms the $N = 1$ result. Section 17’s “leading MLP is harmful” was measured at $N = 1$ against a single GRU; at depth 4 the same addition is free-to-slightly-good. The pre-encoder pitfall should be scoped to the shallow case.

Objective and retrieval dissociate at depth, and this is the sharpest instance yet. Arm B achieves the *lowest* validation InfoNCE in the entire batch (6.5378 against the control’s 6.5832) while retrieving 55 **fewer hits** (123 against 178). Every prior architecture null on this corpus was consistent with “the extra machinery does nothing”; this one shows the extra machinery does something measurable to the loss and that it is the *wrong* something. The operational consequence is blunt: select on Recall@10 and the walk, never on the training objective. One caveat rides with it — B ran to the 40-epoch cap with patience never triggering, so its recall is a *lower bound*, and a longer re-run is the clean test of the dissociation (the prediction being a better validation loss and no better recall).

The eagerness facets improve monotonically as the arms get worse, and the floors caught it. Every 4-stage arm de-eagers substantially (concentration $0.359 \rightarrow 0.263$ to 0.318) and raises `ild`; under a naive holisticness-primary read some would look interesting. All are killed by the absolute recall floor and by the per-baseline relevance floor — the same mechanism that has now saved four campaigns. Note that the canonical *absolute* 0.43 relevance figure would have passed several of these arms, so the floor must be derived per baseline.

Two by-products, and the governance line. A previously closed $N = 2$ relative — a dual tower with a pre-MLP on one side, closed on a one-shot recall tie — was walk-evaluated for the first time here and is **refuted on the walk** ($\Delta \text{vibe} -0.0255 [-0.0432, -0.0096]$), its one real gain being less drift. That is the same signature a recurrent-cell variant showed earlier — better stride and drift, worse vibe — which now looks like a general property of adding pre-recurrence machinery rather than anything specific to either arm. Second, the grounding metric again discriminated almost nothing, separating one arm of nine, consistent with its recorded arithmetic (between-arm range 0.0060 against a per-session standard deviation of 0.1417, needing ~ 2140 sessions): it is a floor that catches degeneracy, not a ranking metric, and it was used only that way. Nothing was promoted as champion, nothing was registered as a crown claim and nothing was deployed; the seven arms were promoted *for measurement only*, each carrying the verbatim note “promoted to enable walk evaluation; not a crown claim”, which flooded the best-models group and was handed to the curation agent.

Two process findings worth keeping. First, the predictor-checksum gate fired *on a file the design did not name for copying* — it listed one module to copy and a shared module only to

verify, and the verification failed. The divergence turned out to be inert for this batch, but that is a *post-hoc* judgement that required reading the diff; the rule that caught it is the one worth keeping. **Verify the checksum of every file in the family’s import closure, not just the ones you copied.** Second, the approved launch body was rejected by the approved registry block: the design’s JSON sent six hyperparameters the block does not declare, so the first posts failed outright. It was resolved by dropping the six keys *after proving the drop is a no-op* — the predictor’s defaults supply exactly those values and the resolved hyperparameter dict is identical either way, verified by direct comparison — and the registry block was left exactly as approved. *Source: 2026-08-01-stage-stack-topology.md.*

26 Parallel tower banks: the N -axis closed on both surfaces

Rationale. The walk reversal of Section 24.5 left this corpus with exactly one measured architecture *win*, and with a stated mechanism for it: a dual tower whose second view is **cummean** holds a mood because it carries a **fixed reference never overwritten by recurrence**. A mechanism of that shape makes a testable prediction — it should be *dose-responsive*. If one fixed-reference tower buys a held vibe plus +1.5 genres, then N towers, enough of them on the raw **latent** view to hold the recall floor and the rest on **cummean**, should buy *more* of the same. This campaign runs that dose curve, and it is the follow-up the reversal itself asked for.

The vehicle is a new predictor, **seq-bank**: N *parallel* single-layer bare GRU towers over configurable causal views, their per-step hidden states concatenated and read by **one linear head**. Its design choice is to collapse tower count and view multiset into a *single views* hyperparameter, so N is not a separate axis that can drift out of sync with the views. What the predictor *cannot* express is as important as what it can: it declares exactly 15 hyperparameters and none of **fusion_layers**, **fusion_hidden**, **pre_hidden**, **pre_layers**, **layers**, **num_layers**, **arch**, **bidirectional**, **residual** or **eager_beta** is among them. Every mechanism this document has already refuted — the MLP fusion head (Section 16), the per-step pre-encoder (Section 17), the soft anti-eager regularizer (Section 15) — is therefore *structurally unreachable* rather than merely unused, and a **parse_views** guard rejects any spec without a **latent** tower, which puts the registered 0.058 to 0.087 **delta/cummean** pathologies out of reach as well.

Two hypotheses were pre-registered. The primary one is the *dose* hypothesis above. The independent sub-hypothesis is the mechanism the dual-tower scan had already established: if the raw **latent** view *subsumes* its siblings, then N identical **latent** towers should buy nothing, whatever N is.

Design and the two surfaces. Ten runs on the canonical split **seq-20260715-131139** (1431 test sessions, cold-item rate 0.5507), 84.2 min of compute, zero failures. Everything but **views / hidden / dropout** is held: **loss** InfoNCE, $\tau = 0.07$, 40 epochs, patience 6, lr 0.001, batch 128, seed 1337, MMR off, artist cap off. **Capacity is matched at 3.00× the h256 baseline** for every bank arm *and* for the single-GRU control C1 (h512), with the counts re-verified from the artifacts rather than trusted from the design — **seq_bank** emits **n_params** into **metrics.json**, and the mutual spread across the seven 3.00× arms including C1 is 0.405 %. The decision rule was written on *both* surfaces before launch, and both legs are load-bearing:

S1, one-shot recall@10. A win requires a paired Δ above zero against *both* the h256 anchor C0 *and* the parameter-matched single GRU C1. Two mandatory floors apply: an absolute recall floor ≥ 0.108 (155 of 1431) and a per-baseline relevance floor (paired Δ music@10 vs C0 must not fall below zero).

S2, the generated walk. 40 held-out sessions $\times$ 20 steps at artist cap 1, paired against the *in-batch* C0, on the walk metrics of Section 24.2. The primary cell is anchor 0.4 / stride 0.3; a saved definition additionally requires **the same sign at the deployed cell** $a0.8 / s0.5$.

The instrument was pinned before anything was measured. All ten pre-registered gates came back green, and three of them are worth recording because they make the comparison unusually tight. A *functional-equivalence* gate showed **seq-bank** to be a strict generalization of both incumbents rather than a new family: at `views="latent"` it reproduces **SeqNextLatent** and at `views="latent+latent"` it reproduces the `fusion_layers = 0` **DualTowerNextLatent**, with $\max|\Delta|$ 0.00e+00 on both `forward` and `predict_next` at prefix lengths $T = 1, 2, 5, 17, 40$ and identical parameter counts on both sides; an extra training-stream check matched per-epoch train *and* validation loss to the last digit over three epochs and all ten state-dict tensors at $\max|\Delta|$ 0.00e+00. Both bit-identity gates then hit exactly: C0 reproduced its on-record recall@10 0.12299091544374564 = 176 of 1431 (a **10th** reproduction, with `music@10`, `H@10` and `MRR` matching to the last digit) and R2 reproduced the dual-tower `latent/latent` f0 arm at 0.1187980433263452 = 170 of 1431, so the design’s band fallback was never needed. And the walk instrument was checked against the shipped `gru.onnx`: $\Delta\text{vibe} +0.0028$ and $\Delta\text{stride_err} +0.0005$ both straddle zero, so S2 verdicts against the in-batch C0 are not an artefact of the base.

Results, S1. Table 36 is the batch. **No arm is bolded as a winner because there is none:** no arm achieves even the first leg of the win rule, all nine non-anchor arms are ties against C0, and *five of the eight bank arms lose to the parameter-matched single GRU C1 with the CI below zero*. Both floors pass for every arm — the worst is V1 at 162 hits, clear of the 155 floor, and the most negative relevance Δ is V2’s -0.006109 , straddling.

The sharpest number in the batch is a zero. B4 — four independent h211 `latent` towers, 1 187 700 parameters — and C1 — one h512 GRU, 1 182 912 parameters — return the **identical** 186 **hits of** 1431, a paired Δ of precisely 0.000000 $[-0.00839, +0.00839]$. At a matched $\sim 3.00\times$ budget, *topology and width are interchangeable on this corpus*: whatever four towers with different random initializations discover, a single wider recurrence already had. Figure 22 (left) plots the N -axis at matched capacity, and the line through 186/170/172/186 is flat inside the split’s own measured resolution (Section 18.1).

This is also the cleanest demonstration on record of *why the parameter-matched control leg is mandatory*. Read against the h256 anchor alone, B4’s +10 hits would have looked like a topology gain and the five arms that lose to C1 would all have read as ties — the anchor cannot distinguish “the bank helped” from “the extra parameters helped, and the bank spent them worse than width would have”. The C1 leg does exactly that, and it is what turns a batch of nine ties into a closure. Note too what the flatness costs the *other* direction: C1’s +10 hits over C0 is itself not significant ($+0.006988$ $[-0.00210, +0.01607]$), the **seventh** parameter-matched width tie on record, so the campaign cannot claim h512 as an improvement either despite it holding the highest single-GRU hit count measured (186 against h454’s 179). The single-GRU width points on PCA-192 now read $\text{h256} \rightarrow 176$, $\text{h297} \rightarrow 162$, $\text{h425} \rightarrow 173$, $\text{h454} \rightarrow 179$, $\text{h512} \rightarrow 186$, $\text{h537} \rightarrow 174$ — non-monotone, spanning 0.0168, with no trend.

Results, S2. Table 37 is the walk. Two arms — B2 (`latent+latent` h334) and V3 (`latent+cummean+cummean`) — reached a genuine **TRADE** at the primary cell, each buying a drift improvement with the CI above zero while holding `vibe`. Both then **flip** at the deployed cell: the drift gain evaporates into a straddle and `vibe` — the non-negotiable line — goes below zero for both. Under the pre-agreed rule, **no definition was saved**.

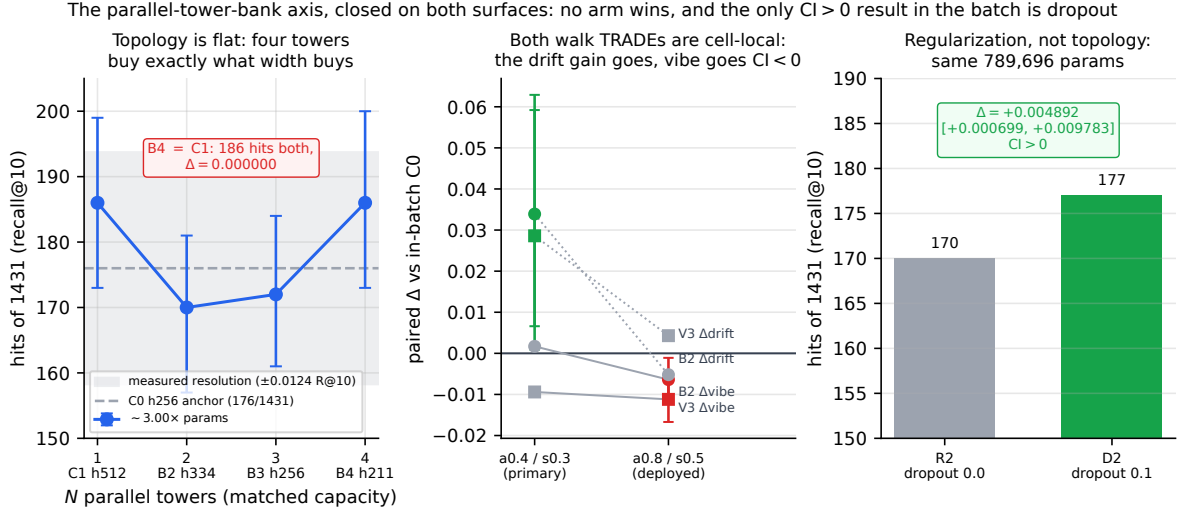


Figure 22: The parallel-tower-bank axis, closed on both surfaces. **Left:** the N -axis at matched capacity ($\sim 3.00\times$ the h256 baseline). Error bars are the paired bootstrap CI against C0 mapped onto the hit scale; the grey band is the split’s measured resolution (± 0.0124 recall@10 = ± 17.7 hits, Figure 13), and every arm sits inside it. Four h211 towers (B4) and one h512 GRU (C1) return the same 186 hits, paired Δ 0.000000. **Middle:** the two arms that earned a walk TRADE at the primary cell both flip at the deployed cell — the drift gain that bought the TRADE decays to a straddle while *vibe* goes below zero. Grey markers are point estimates whose CI straddles (the source report quotes bounds only for intervals clearing zero). **Right:** the campaign’s one positive result, and it is not about topology — at byte-identical parameters (789 696 both, `model.pt` 3 162 277 bytes both) dropout 0.1 beats dropout 0.0, CI above zero.

Interpretation. **The dose hypothesis is refuted, and it fails in the direction opposite to the prediction.** Adding fixed-reference `cummean` towers did not buy more of what $N = 2$ bought; it monotonically destroyed the walk. Ranked by Δ vibe at the primary cell the ordering is B4/B2 (all-latent, ≈ 0) > B3 > V3 (one `cummean`, -0.0094) > V2 (-0.0144 , CI below zero) > V1 (-0.0174 , CI below zero), and the same ordering holds on S1, where the three `cummean`-bearing arms occupy the *bottom three* places (162, 166, 170 hits) and are the only arms whose Δ music@10 point estimates fall past -0.005 . The irony is worth stating plainly: **the tower added in order to hold a mood is the tower that made the walk’s mood measure worse** — the only two arms whose *vibe* is refuted outright are the two that carry the most `cummean`. Whatever the deployed $N = 2$ latent/`cummean` engine is doing, “more `cummean` is more mood-holding” is not it, and the structural reading that motivated the design does not survive being dosed.

The sub-hypothesis is confirmed and extends cleanly from $N = 2$ to $N = 4$. B2 (170), B3 (172) and B4 (186) all straddle C0 and show no trend distinguishable from width noise. This is the same mechanism the dual-tower scan found (Section 16) — the raw *latent* view subsumes its siblings — now measured at three tower counts instead of two, which is what closing an axis by dose rather than by a single point requires.

The one positive finding is a by-product of the null, and it lands on the deployed engine. D2 and R2 differ in *one* hyperparameter: parameters are byte-identical (789 696 both, `model.pt` 3 162 277 bytes both), only `dropout` moves, and Δ recall@10 is $+0.004892$ [$+0.000699, +0.009783$], **CI above zero** — dropout 0.1 *beats* dropout 0.0 (177 against 170 hits). Read carefully this does not overturn the record’s three-times-confirmed “f0 beats f1” result;

it does something more awkward. Dropout enters `seq_dualgru` only through a pre-MLP, an inter-layer gap or the fusion MLP, and at `fusion_layers = 0` with one layer and no pre-encoder *all three are absent* — the f0 path constructs **zero** dropout modules. So every f0 arm on record, **including the deployed showcase engine** (Section 24.6), trained under-regularized and was leaving $\approx +0.005$ recall on the table; and the three recorded f0-vs-f1 gaps ($+0.016$ / $+0.020$ / $+0.0266$) were measured against an f1 that *had* regularization its f0 comparator lacked, so they are not like-for-like and the true topology advantage of f0 is if anything *larger* than recorded. Wherever those three figures are quoted as clean topology measurements they should be re-stated as not like-regularized. The cheapest actionable follow-up on the whole board falls out of this: retrain the deployed engine at `dropout = 0.1` and re-read both surfaces — a free improvement, not an architecture bet.

The validation-loss dissociation gets a fourth confirmation, now as a rank correlation (Figure 23). Across the ten arms, $\text{Spearman}(\text{best validation InfoNCE, hits}) = -0.2954$, $p = 0.407$ — no usable predictive power, and if anything the wrong sign. The sharp instances are brutal: B4 and B2 differ by 0.0003 in validation loss and by 16 hits; V1 holds the fourth-best validation loss in the batch and the *worst* hit count; C0 holds the *worst* validation loss and beats six of the eight bank arms; and R2 reaches a better validation loss than D2 while retrieving seven fewer hits, which makes the dissociation and the dropout finding two faces of one coin. The decision rule never read validation loss, and this is why.

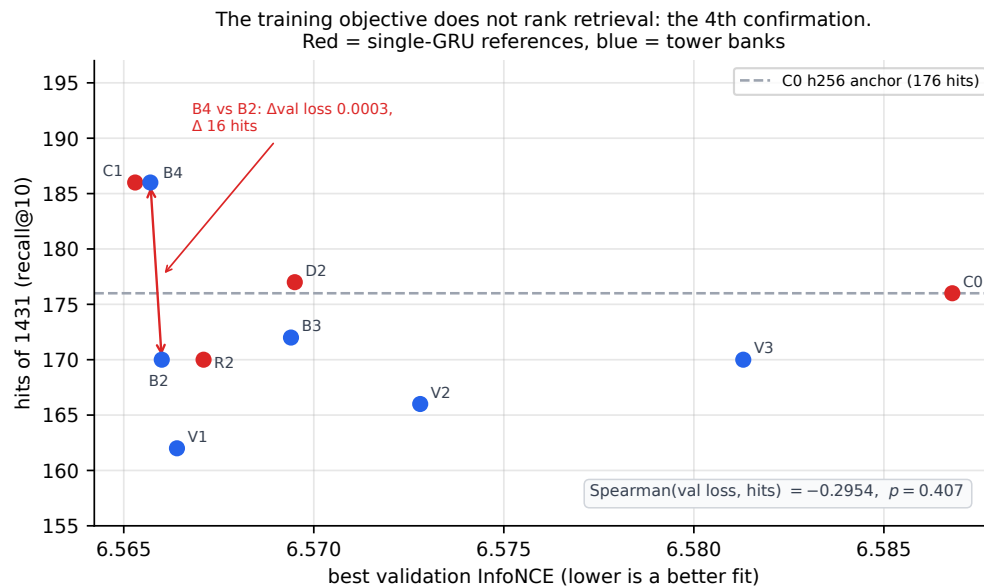


Figure 23: The training objective does not rank retrieval — fourth confirmation, and its sharpest instance yet. Ten arms of the tower-bank batch: best validation InfoNCE against hits of 1431. $\text{Spearman} = -0.2954$, $p = 0.407$. B4 and B2 are 0.0003 apart in validation loss and 16 hits apart; C0, the worst fit in the batch, beats six of the eight banks. Every arm early-stopped (epoch 14 seven times, then 21 and 24 twice), so no arm’s recall is a lower bound imposed by the epoch cap.

Both walk TRADEs were cell-local, which is a methodological result in its own right. B2 and V3 each earned a real Δdrift above zero at $a0.4/s0.3$ with vibe held, and both flipped to a vibe below zero at $a0.8/s0.5$ with the drift gain gone. Had the design not pre-registered the secondary-cell requirement, this campaign would have reported two positive walk results and saved two definitions. **The secondary-cell clause is the clause that did the work here,**

and it should stay mandatory in every future walk design. The pre-registered stride probe points the same way: at $s \in \{0.2, 0.3, 0.4\}$ all three `cummean` arms peak at $s0.3$, the cell they were already measured in — the third campaign in a row in which re-tuning the stride rescues nothing — and at $s0.2$ their Δvibe turns positive-straddling only by taking visibly shorter steps ($\Delta\text{stride_err}$ above zero for all three) and shedding genre variety ($-1.90 / -1.78 / -1.00$, all below zero). That is precisely the trade the walk win clause exists to reject.

The failure signature is the opposite of the preceding architecture campaign’s, and that is diagnostic. No arm here was refuted by a floor: the recall floor and the relevance floor both pass for all ten runs, and eight of the nine non-anchor arms sit at *higher* artist concentration than the anchor (0.3547 to 0.3799 against 0.3559), i.e. the arms did not de-eager. The floors were therefore not the operative mechanism — these arms are not degenerate, they are simply **flat**. That is a different failure mode from the immediately preceding depth-in-series campaign (Section 25), in which all five arms fell *through* the recall floor, and it is why this axis closes as a null rather than as a harm.

The scope of the closure, precisely. What closes is “ $N \geq 3$ parallel bare-GRU tower banks at `fusion_layers = 0`”, and only that, to the standard Section 25 set for a closure on this corpus: both surfaces, walk instrument verified. It does **not** re-close the dual-tower family, which stays reopened (Section 24.5) and remains the deployed engine. Nothing was promoted as champion, no definition was saved, and no dataset was built; the eight measurement arms were promoted for measurement only, each carrying the verbatim note “promoted to enable walk evaluation; not a crown claim”, and the resulting flooding of the best-models group (to 15 entries) was disclosed up front and handed to the curation agent. Zero of the three pre-authorized contingency runs fired, and the epoch trace says why: the dropout-confound extension needed R2 to early-stop later than D2 and both stopped at epoch 14; the epoch-cap contingency needed an arm to reach the 40 epoch cap and every arm early-stopped; the width-bracket contingency needed C1 to lose, and it did not.

One limitation is inherited rather than introduced. This campaign’s premise — the dose mechanism — was read off a finding that has *no campaign report of its own*: the walk-surface reopening of the dual-tower family (Section 24.5) exists only as a `PROJECT-FACTS.md` roll-up, a git commit, and the transcription in this document. The harness-reproduction gate re-derived all ten of that reopening’s recorded means exactly and re-derived its deltas, so the *measurement* is reproducible and the instrument has not moved; what is missing is the primary write-up, and it should be backfilled from the commit and the facts entries. *Source: 2026-08-02-tower-bank-parallel-towers.md.*

27 Conclusions

Twenty-one campaigns and one research memo (Jul–Aug 2026) on the next-track retrieval task converge on a clear story. (1) A session sequence model, even at its best (GRU-infonce), only *ties* the first-order Markov co-listening bar on Recall@10 and loses on MRR: predicting a next-item latent and retrieving by one cosine hop is a harder route to the exact next track than a bigram that directly memorizes adjacency. (2) The training *loss* is the dominant single-model lever (InfoNCE $\approx 2.3\times$ cosine); network *topology* — depth, width, residual, direction — is a spent lever, and no architecture beats the bar. (3) The breakthrough is *rank fusion*: an R+M z-blend of the GRU and Markov legs beats the bar by $\sim +61\%$ Recall@10 (three-seed confirmed), because the two legs rank *different* correct tracks. The win is about signal *diversity*, not model

count — adding a redundant neural leg does not help, and a learned XGB stacker *underperforms* the fixed z-blend, because its pointwise training does not capture the ranking objective. (4) An item-representation sweep found latent *dimension*, not the compressor, to be the lever: linear PCA beats the nonlinear autoencoder at every $\text{dim} \geq 64$ (the acoustic-preservation hypothesis is refuted), the curve peaks at PCA-192 and turns over by 256, and PCA-128 is the confirmed single-model base. (5) On the rich PCA space the one genuinely additive new signal — *content-kNN*, which only *tied* the champion on the AE-64 space — becomes a win: a three-leg R+M+C z-blend on PCA-192 is the current best-on-record row at Recall@10 0.1859, MRR 0.1025 (paired Δ over the prior champion +0.0133, CI above zero, confirmed across five reproductions). (6) Two honest caveats bound the crown: the reproduction confirm had zero spread because **seq-nexttrack** is deterministic on this CPU (so it verified determinism, not seed-robustness), and the paired- Δ lower bound clears zero by only +0.0007 on a single split — hardening it needs a second leak-free test split. (7) Every ranking predictor here is train-only and off-server, so the crown is a best-on-record *row*, not a promoted, app-exportable model; the prior champion R+M on AE-64 remains the productization anchor, and registering a three-leg blend with a **predict** subcommand is the standing top follow-up. (8) A subsequent literature-fit campaign then *superseded* the crown with the first genuinely-additive new signal since content: a supervised *learned content projection* ($R'+M+C'$), an InfoNCE metric map of the PCA-192 item space fit toward next-track adjacency in which both the GRU and content legs retrieve, at Recall@10 0.2117, MRR 0.1220 (paired Δ over R+M+C +0.0259, CI>0). It resolves both R+M+C caveats — it is seed-robust across three seeds and cold-reaching (its content leg scores cold targets as well as warm) — and it was *registered, promoted and servable* (**blend-gru-markov-content-proj**, best-models rank 1), closing the productization gap; an objective / rank / τ scan confirms InfoNCE $\gg$ BPR and holds the default, and the crown then hardened on a second disjoint 0.64-cold split (paired Δ +0.0426, CI>0, the margin *wider* on the colder split). Every borrowed literature-fit method (session-kNN, rank-fusion / learned combiners, GRU-side loss tricks, Markov smoothing) closed — the only additive gain came from new signal, not smarter fusion. (9) An item-vector representation exploration then confirmed *why* the projection is the champion: the equal-weighted PCA-192 geometry is a real handicap for a raw model (six of eight whitened / acoustic-dropped / metric spaces lift the frozen GRU, paired Δ CI>0), but the gain is fully *absorbed* by the learned projection (no space beats the champion on readout A) — whitening and the projection are the same fix, and the champion has already pulled it. (10) Finally, a continuous “sounds-alike” metric, *music@10*, was introduced and back-filled across all 53 runs: it correlates with Recall@10 ($r = 0.609$) but crowns a different winner (the content-regressing **seq-nexttrack** lands the musically-closest misses, best 0.482), corroborating the exact-recall findings without overturning them; it is display-only and the primary metric stays Recall@10. (11) Finally, a measurement campaign added a display-only *session-holisticness* suite (artist_adj@10 / mood_coh@10 / ild@10) and two anti-eager levers. It found the eagerness symptom *tracks* Recall@10 inside the blend family — the champion is the most artist-eager model (artist_adj@10 0.610) — but the eagerness is sourced from the *Markov* bigram leg, not the RNN (single GRU / LSTM artist_adj@10 ≈ 0.345), and the GRU and LSTM are statistically tied on both recall and eagerness (the “GRU wins because more eager” hypothesis is refuted). Of the two levers the eval-time MMR re-rank is the strong knob ($\lambda=0.9$ near-free: recall tie, ild@10 0.406 $\rightarrow$ 0.436; $\lambda \approx 0.7$ de-eagers for ~ 2 recall points; music@10 even rises as the list diversifies) while the training-time β regularizer is weak (β 0 to 0.2 moves artist_adj@10 only -0.0074); nothing was promoted, and $\lambda \approx 0.9$ is flagged for a future user holisticness decision (Section 15). (12) Two final architecture campaigns then closed the last open architecture axes, and both are nulls — but nulls with mechanism. A jointly-trained *dual-tower* GRU, crossing the complete

`{latent,delta,cummean}`² view matrix against a 0.098% parameter-matched single tower, fails to beat that tower on any of eight arms (Section 16). This is the *fourth* independent learned-combiner null and the one that matters most, because it answers the standing rebuttal to the earlier three (“of course a bolted-on combiner fails — train the fusion end-to-end”): trained end-to-end at the representation level, it still fails. The mechanism is that the raw `latent` view already *subsumes* the others — a GRU over raw latents integrates movement and running mean internally, and the one genuinely complementary pair is `delta+cummean`, which beats both its parents yet is hopeless in absolute terms. Two by-products are durable: the MLP fusion head is actively *harmful* (removing it wins in every regime measured, three times over, Table 19), and single-GRU width is now closed on PCA-192 as it already was on AE-64. A learned per-step *pre-encoder* then fails harder (Section 17): the decisive single-tower ablation *loses* outright (−0.0140, CI below zero, 156 hits against 176), the dose curve is flat to twice the input dimension, and placement is irrelevant. Because a *linear* pre-encoder is provably expressivity-neutral here, the rectifier is the only candidate cause — and a counterfactual interpretability read confirmed it, measuring the affine map neutral at $\sim 0.8\sigma$ and the ReLU costing $\sim 5.9\sigma$, with `frac_exact_zero` 0.5327 matching `frac_negative_coords` 0.5327 to four digits: half-wave rectification of a zero-centred latent deletes exactly its negative half. Across five architecture nulls the pattern is now hard to dismiss — on this single-user, 55%-cold corpus, adding representation machinery *around* a single GRU over a good latent space does not help, while every real gain came from improving the latent space itself or from the fixed z-blend. (13) Finally, two facts about the *instrument* rather than any model reframe the whole architecture arc (Section 18). The nulls are **power-limited**: at $n = 1431$ the paired-bootstrap CI half-width is 0.0124 on Recall@10 and ~ 0.0018 on *holisticness*@10, while a whole architecture family spans only 0.028 and 0.0031 — so of eight adjacent arms roughly two tiers are separable on recall and one on *holisticness*, and resolving a 0.005 difference would need ~ 8800 sessions. The honest reading is therefore that *large harms were detected reliably while moderate wins were never detectable at all*, which is a different claim from “these architectures do not work” and yields a prioritization rule: further architecture arms are a poor use of runs on this split whatever they return, because the axis’s whole dynamic range is barely twice the measurement error. And *holisticness*@10 exists in three mutually non-comparable code generations (3-factor; 4-factor grounded; grounded plus artist-concentration), whose differences dwarf every model difference and under all of which the servable champion ranks *last* — so this synthesis keeps its board denominated in Recall@10 and tags every *holisticness* column with its generation (Table 21). (14) Finally, a campaign about the *surface* rather than the model finds that the board above had been scoring the wrong one (Section 24). Everything else in this document is a one-shot top-10, while the shipped product serves an autoregressive *walk* — and *holisticness*@10’s `mood_coh` anchors to the *prefix* centroid, which on a walk grows with the journey, so a run of individually legal steps that ends in the opposite region of the space scores well at every step: drift was invisible by construction, and the composite is $5.5\times$ more sensitive to relevance than to eagerness besides. Calibrated against the user’s own 73 632 consecutive transitions (median step cosine 0.261, 27.0% negative), the engine the showcase shipped walks $2.5\times$ too tight (median step 0.661) — “flat despite lots of artists” was many small safe moves — while the blend champion matches the real stride but cannot hold a mood, because its InfoNCE projection encodes what *follows* rather than what *sounds alike*. Three deterministic retrieval-time controls fix the reachability: a seed anchor, a stride term (deterministic on purpose — sampling would break the permalink guarantee), and a *hard* per-artist cap, which succeeds exactly where the two soft anti-eager levers failed because it cannot be outscored (on the pathological playlist 59 of the top 60 candidates are one artist and the next artist sits at rank 75, against an MMR diversity budget worth ≈ 0.078 of normalized score; `eager_beta` is mis-specified

on an absolute cosine when eagerness is a *ranking* property). The cap hardens on two disjoint splits and, unexpectedly, *raises music@10* at every setting: it trades literal for graded relevance. And on 40 identical held-out sessions under paired bootstrap the **dual-tower family, closed by three campaigns on recall grounds, is the winner on the stated objective** — latent/cummean at anchor 0.8 stride 0.5 takes stride error -0.2545 , genres $+1.50$ and drift $+0.0351$ with every CI clear of zero while *holding* vibe, where the single GRU (-0.0409) and the champion (-0.1327) both pay vibe for the same stride. The corroboration was already in the record as a *negative* (seq_dualgru.py:113: the cummean tower “added real linear directions (CKA 0.458) but zero next-item concepts tower A lacked”) — dead weight for next-track prediction is exactly session state for mood-holding, though the stronger reading is refuted: cummean does *not* hold a vibe natively (0.236 against 0.234), it only hosts one once the controls are applied. It is deployed as the showcase’s default engine; and ground straddles zero for every arm, so what is established is journey *shape*, not next-track truth, on one split, one seed and 40 sessions, with nothing yet listened to and the primary metric unchanged. (15) A closing architecture campaign then tested the reversal’s own stated mechanism — “holding a mood needs a fixed reference never overwritten by recurrence” — as a *dose*, and closed the axis on both surfaces (Section 26). A bank of N parallel bare-GRU towers under one linear head, matched at $3.00\times$ the h256 baseline and dosed from $N = 2$ to $N = 4$, reaches no win on either surface: every arm *ties* the baseline on one-shot recall while **five of eight lose to a parameter-matched single GRU** with the CI below zero, and the two arms that earn a walk trade at the primary cell both flip to *vibe* below zero at the deployed cell, so no definition was saved. The decisive datapoint is a zero: four h211 towers (1187700 params) and one h512 GRU (1182912 params) return the *identical* 186 hits of 1431, paired Δ 0.000000 — *topology and width are interchangeable at this budget*, which is also the cleanest demonstration on record of why the parameter-matched control leg is mandatory: against the anchor alone, five losing arms would have read as ties. The dose hypothesis is refuted *ironically* — the cummean towers added in order to hold a mood are the worst arms in the batch on both surfaces and the only two whose walk *vibe* is refuted outright — and the one CI-clear result in the whole batch is a by-product about regularization, not topology: at byte-identical parameters dropout 0.1 beats dropout 0.0 by $+0.0049$ recall@10. That is consequential because the fusion_layers = 0 path constructs *zero* dropout modules, so every f0 arm on record, **including the deployed showcase engine**, trained under-regularized, the three recorded f0-vs-f1 gaps are not like-regularized, and the cheapest improvement on the board is a single retrain.

(16) Three score-time campaigns then produced the arc’s first genuine *holisticness* lever and its largest *recall* gain, neither of them a model change. The eval-time MMR λ frontier (Section 19) is confirmed natively on both anchors: every one of eight $\lambda < 1$ arms lifts holisticness@10 with the CI above zero, three single-GRU arms clear the prior all-time bar 0.024872 at a recall tie (best 0.030355 at $\lambda 0.7$, pool 50), and the gain is legitimate rather than composite-gaming because *three* of the four crown factors — diversity, eagerness and *music@10* — improve together. Its pre-registered mechanism question resolves unanimously to the responsive hypothesis (8 of 8 arms, $|\Delta_{\text{artist_conc}}|/|\Delta_{\text{artist_adj}}|$ between 1.59 and 3.06), so same-artist tracks *are* near-neighbours in the content metric, sonic MMR *is* a de-eagering lever, and the explicit artist-penalty sequel is cancelled. Two by-products are durable: the binding facet *switches* from concentration to adjacency as λ falls, which locates the end of the lever; and the shared-weights pairing is 3 to $33\times$ sharper than the architecture-family noise floor (half-widths from 0.000382), so eval-time levers must budget their minimum detectable effect from the pairing, not from Section 18.1’s figure. (17) The largest single recall gain in the whole arc is then **markov_gate** (Section 22): make the Markov leg *abstain* where the last prefix item has zero train-bigram support — 55.6% of queries — and

Recall@10 rises to 0.23201 (332 of 1431), paired +0.02027 [+0.01048, +0.03075] canonically and +0.01957 on a disjoint 64%-cold split, MRR up on both. The defect is the blend’s *scale-invariant* normalizer, which re-inflates an evidence-free artist/genre back-off to a full one-third vote; the fix has zero learned parameters and nine lines. The bucket decomposition is the argument: the whole effect sits in the zero-support bucket while both warm buckets are **byte-for-byte identical** on three independent control/arm pairs, which mechanically proves no global static leg weight could express it (dropping the leg globally is a -0.0070 tie, the two buckets cancelling) and therefore *narrows* the four-times-confirmed learned-combiner pitfall to *learned score-level fusion* rather than per-query evidence gating. Unpredicted, it is also a crown lever — all four factors move the right way at once, and it is the first lever that de-eagers the champion *without* paying mood coherence, because it removes a vote that had no business being cast rather than fighting the model’s own scores. Composed with MMR $\lambda 0.7$ it is recall-neutral against the champion (+0.00140, straddling) at holisticness 0.030980, $3.6\times$ the champion’s. Nothing was promoted: this is a recommendation, and the one residual concern is that the zero-support *boundary* was chosen with test knowledge. (18) Against those two positives, the training-time anti-eager lever is refuted with a mechanism, and the refutation is expressive rather than dosimetric (Section 23). The margins were calibrated from the item→item transition cosine (p75 0.584, p90 0.830) but the hinge acts on prediction→current, whose ceiling over training lies between 0.5843 and 0.70 — so the calibrated margins fall *outside the support of the penalized quantity* and four arms are provably inert, reproducing the control’s loss to 16 significant digits at every epoch at β up to the schema cap. Where the knob fires at margin zero it de-eagers spectacularly and destroys relevance in the same motion, because an InfoNCE prediction is a hedged near-average and eagerness is a *ranking* property that an absolute cosine cannot express. The prior “weak knob” reading is sharpened rather than retired: the entire usable range at margin zero lies in an unexplored factor of 15 between $\beta 0.2$ and $\beta 3$. (19) Two campaigns about *instruments* rather than models bracket these. A purpose-built rollout metric (Section 20) — fade per legible axis, calibrated on 2135 real train sessions, with the anti-gaming term *inside* the metric rather than bolted on as an external recall floor — returns a sixth architecture null in the deployment regime, which is the strongest form available because the standing rebuttal to the earlier five was that Recall@10 cannot see session quality. Its durable finding is that *per-seed* variance dominates model identity (± 0.3 per seed against a model-level difference of +0.004), so budget ≥ 40 seeds for any rollout comparison and prefer serving an ensemble; and its three retracted positives are a catalogue of how a generation metric goes wrong (a cross-split comparison read as architecture, mood-extreme seed selection inflating effects $\sim 2.5\times$, and an anti-gaming term normalized by band width that scored a flatline *above* a real generator). A research memo (Section 21) is reported at a lower register — nothing in it was run on the server, and its verified and provisional numbers are kept partitioned — but its convergent thesis, that every proposal surviving adversarial screening is an *evidence-conditioning* change rather than a capacity change, is what points at (17), and its graveyard is cheaper to reuse than to rediscover. (20) Finally, a stage-stack depth campaign (Section 25) closes the *serial* branch of the architecture axis to the two-surface standard the tower-bank campaign then inherits. Five arrangements of per-step MLP and recurrent stages, capacity-equalised at $\sim 1\,282\,000$ parameters, all fall *through* the mandatory recall floor (0.0706 to 0.0860 against 0.108) and are refuted again on the walk at the matched cell, while a $3.24\times$ capacity control ties on both surfaces — so the damage is depth itself, not capacity, not regularization (every deep arm trains *longer* and two reach a *lower* validation loss) and not the MLP (a leading per-step MLP is recall-neutral and holisticness-positive at depth 4, which *narrows* the $N = 1$ pre-encoder result to the shallow case). Two facts outlive the null: *bracketing beats interleaving* at byte-identical capacity (+0.01537, CI above zero), and the training objective

dissociates from retrieval — the arm with the best validation InfoNCE in the batch retrieves 55 fewer hits than the control, so any architecture search selecting on validation loss on this family will select the wrong arm.

28 Source reports

This synthesis consolidates twenty-two next-track source documents from **experiments/** — twenty-one campaign reports and one research memo, the latter flagged as such because nothing in it was run on the server:

- **2026-07-13-nexttrack-sequence-sweep.md** — the founding next-track campaign: a GRU/LSTM sequence sweep against the first-order Markov bar; the best GRU-infonce ties on R@10 but loses on MRR (Section 4).
- **2026-07-13-nexttrack-topology-scan.md** — the depth / width / residual / bidirectional topology scan; architecture is a spent lever, all arms cap below the Markov bar (Section 5).
- **2026-07-14-nexttrack-blend-and-ensemble.md** — the GRU×Markov R+M z-blend beats the bar +61% R@10 (the first next-track champion); no richer ensemble improves on it and the learned XGB stacker underperforms (Section 6).
- **2026-07-14-nexttrack-graded-relevance-and-new-legs.md** — first-class graded-relevance metrics (artist@k / genre@k / artist_mrr) and new leak-free base learners; content-kNN is the only additive signal, tying the champion on AE-64 (Section 7).
- **2026-07-15-nexttrack-compression-representation-bakeoff.md** — the item-representation compression bake-off (PCA vs autoencoder × latent dim 32/64/128) under a frozen GRU readout; PCA-128 is the single-GRU winner but the win does not carry into the blend (Section 8).
- **2026-07-15b-nexttrack-compression-dim-extension.md** — the PCA/AE dimension sweep extended to 192/256; the curve peaks at PCA-192 and turns over, so PCA-128 stays the Phase-2 base (Section 9).
- **2026-07-15c-nexttrack-phase2-model-sweep.md** — the model-family sweep on the rich PCA spaces; the fixed z-blend beats the learned stacker, LSTM is the best single model, and R+M+C on PCA-192 is crowned the new best-on-record row (Section 10).
- **2026-07-18-nexttrack-literature-fit-campaign.md** — a bibliography-driven exploration; a supervised learned content projection ($R'+M+C'$) is the new best-on-record row (R@10 0.2117, paired Δ over R+M+C +0.0259 CI>0, seed-robust, cold-reaching), and every borrowed method (session-kNN, rank fusion, learned combiner, Markov smoothing, GRU importance negatives) closes (Section 11).
- **2026-07-18b-nexttrack-projection-registration-and-scan.md** — the projection reproduced on-server, registered + promoted + servable as **blend-gru-markov-content-proj** (best-models rank 1); an objective / rank / τ scan (InfoNCE $\gg$ BPR; default held) and second-split hardening on a disjoint 0.64-cold split (paired Δ +0.0426 CI>0) (Section 12).

- 2026-07-18c-nexttrack-representation-exploration.md — an item-vector representation audit; the equal-weighted PCA-192 geometry handicaps the raw GRU (six of eight alt spaces lift it $CI > 0$) but the gain is absorbed by the projection (no space beats the champion on readout A) (Section 13).
- 2026-07-20-musical-distance-metric.md — the continuous “sounds-alike” music@10 metric and its 53-run offline backfill; it correlates with Recall@10 ($r = 0.609$) but crowns a different winner, the content-regressing seq-nexttrack (Section 14).
- 2026-07-22-session-holisticness-and-antieager-levers.md — a display-only session-holisticness metric suite (artist_adj@10 / mood_coh@10 / ild@10) and two anti-eager levers (eval-time MMR mmr_lambda, training-time eager_beta); the champion is the most artist-eager model, the eagerness is sourced from the Markov leg, $GRU \approx LSTM$, and MMR λ is the strong knob (Section 15).
- 2026-07-25-nexttrack-dual-tower-fusion-scan.md — the dual-tower view-pairing $\times$ fusion-depth scan against a 0.098% parameter-matched single tower; **REFUTED** (the fourth learned-combiner null, and the one that closes the “train the fusion end-to-end” rebuttal), with the raw-latent view shown to subsume delta/cummean, the MLP fusion head shown actively harmful, single-GRU width closed on PCA-192, and holisticness/recall shown anti-correlated across the batch (Section 16).
- 2026-07-26-nexttrack-pre-encoder-scan.md — the learned per-step pre-encoder scan (presence $\times$ placement $\times$ dose) with the decisive single-tower ablation; **REFUTED and actively harmful** (-0.0140 Recall@10, CI below zero), the rectifier identified analytically and then measured (affine map neutral at $\sim 0.8\sigma$, ReLU $\sim 5.9\sigma$, frac_exact_zero = frac_negative_coords = 0.5327), plus the measured power analysis and the three holisticness generations (Sections 17 and 18).
- 2026-07-26-nexttrack-mmr-lambda-frontier.md — the native eval-time MMR λ frontier on both anchors; **CONFIRMED** as the first genuine holisticness lever on record (three TIER-1 “free” single-GRU arms above the prior all-time bar at a recall tie, best H 0.030355), with the pre-registered mechanism question resolved unanimously to the responsive hypothesis (8 of 8 arms, ratio 1.59 to 3.06), the binding facet shown to switch as λ falls, the candidate-pool axis closed as a fine-tuning knob, and the shared-weights pairing measured 3 to 33 $\times$ sharper than the architecture-family noise floor (Section 19).
- 2026-07-27-rollout-hold-metric.md — hold@N, a generation metric for the autoregressive rollout regime (per-axis fade calibrated on 2135 real train sessions, with the anti-gaming term inside the metric), plus five sweeps and 240 journeys; **REFUTED** as a model discriminator — a sixth architecture null, obtained in the deployment regime — with per-seed variance shown to dominate model identity and three retracted positives documented as a catalogue of generation-metric pitfalls (Section 20).
- 2026-07-27b-nexttrack-model-topology-research-memo.md — **a research memo, not a campaign report**: nothing was run on the server, nothing promoted, nothing registered, and its numbers come from offline screens over frozen checkpoints. Consolidated at a lower register, with its own verified/provisional partition preserved and its test-set screening caveat stated; its reusable content is the convergent thesis that every surviving proposal is an *evidence-conditioning* change rather than a capacity change, the mechanical explanation of the learned-combiner anomaly, and the graveyard (Section 21).

- 2026-07-27c-nexttrack-markov-gate-confirm.md — the native two-split confirm of `markov_gate`; **CONFIRMED** and the largest single recall gain in the arc ($R@10$ 0.23201 = 332 of 1431, paired +0.02027 canonically and +0.01957 on a disjoint 64%-cold split, MRR up on both), with the warm evidence buckets *byte-for-byte identical*, an unpredicted crown gain in which all four factors move the right way at once, a first recall-neutral crown arm when composed with MMR, and the learned-combiner pitfall narrowed accordingly (Section 22).
- 2026-07-30-eager-margin-calibrated-refutation.md — the training-time eagerness regularizer at margins calibrated from the measured transition tail; **REFUTED across the whole axis, with a mechanism**: the margins were drawn from the item→item cosine while the hinge acts on prediction→current, whose ceiling sits between 0.5843 and 0.70, so four arms are provably inert (loss traces bit-identical at β up to the schema cap) and the arms that fire destroy relevance before they help (Section 23).
- 2026-07-30c-artist-cap-measurement.md — the *hard per-artist cap*, the first de-eagerness lever that works, and it works because it cannot be outscored: `artist_conc@10` 0.692→0.123 with `music@10` *rising* (CI above zero) at every setting, hardened on a second disjoint 0.64-cold split, together with the refutations of the two soft levers it replaces (Section 24.3). The walk-level material of Section 24 — the surface mismatch, the four walk metrics, the anchor × stride frontier, the matched-sample head-to-head and the dual-tower reversal — is transcribed from the PROJECT-FACTS.md roll-up of 2026-07-31b and from the four measurement tools it cites (`tools/walk_eval.py`, `walk_frontier.py`, `walk_dualgru.py`, `walk_headtohead.py`), whose module docstrings are the authority for how each metric is computed; that material has no campaign report of its own yet.
- 2026-08-01-stage-stack-topology.md — the stage-stack topology campaign (arrangements of per-step MLP × recurrence at equalised capacity, via the new `seq-stack` predictor); **REFUTED on both surfaces** and the first closure on this corpus to that standard: all five 4-stage arms fall *through* the recall floor and are refuted again on the walk, the 3.24× capacity control ties on both, the user’s rectifier-position hypothesis is refuted, bracketing beats interleaving at byte-identical capacity, and the best-fitting arm in the batch retrieves 55 fewer hits than the control (Section 25).
- 2026-08-02-tower-bank-parallel-towers.md — the parallel-tower-bank dose campaign (N bare-GRU towers under one linear head, via the new `seq-bank` predictor), **CLOSED on both surfaces**: no arm wins, five of eight lose to a parameter-matched single GRU, four h211 towers and one h512 GRU return the identical 186 hits (paired Δ 0.000000), the `cummean` dose makes the walk strictly worse, both walk trades are cell-local, and the batch’s one CI-clear result is that dropout 0.1 beats dropout 0.0 at byte-identical parameters — which implies every `fusion_layers` = 0 arm on record, the deployed engine included, trained under-regularized (Section 26).

Table 1: Next-track best-on-record board, on the leak-free next-distinct split (1431 test sessions; bootstrap 95% CI over 2000 resamples). R@10, MRR, artist@10, genre@10, music@10 higher-better; the first-order Markov model is the incumbent reference bar. music@10 values are the offline backfill (Section 14); rows without a mapped run are left blank. Champion row and prior champions bolded; best cell per metric bolded. The three refuted architecture rows added 2026-07-25/26 are shown to *bound* their axes and are excluded from the best-cell bolding; they carry no marginal CI, only a paired Δ against the single-GRU baseline (Tables 15 and 17). The top row, added 2026-07-27c, is the evidence-gated champion of Section 22: it is a promotion *recommendation* rather than a registered model, so its R@10 is bolded as the best on record but it is otherwise excluded from the best-cell bolding, and it carries a paired Δ (Table 25) rather than a marginal CI.

Model	R@10 [95% CI]	MRR	art@10	gen@10	mus@10	Note
$R'+M+C'$ + gate (best recall)	0.2320	0.1335	0.367	0.463	0.493	not promoted
$R'+M+C'$, PCA-192 (champion, servable)	0.2117 [0.191, 0.233]	0.1220	0.333	0.401	0.454	new champion
R+M+C, PCA-192 (prior fusion champ.)	0.1859 [0.166, 0.207]	0.1025	0.331	0.394	0.449	content additive
R+M+C, PCA-128	0.1824 [0.162, 0.203]	0.0993	0.332	0.396	—	content additive
R+M, AE-64 (prior champion)	0.1726 [0.154, 0.192]	0.0963	0.322	0.397	0.441	prod. anchor; 3-seed
R+M+C, AE-64 (graded-preferred)	0.1740 [0.154, 0.194]	0.1014	0.330	0.400	—	content only tied
R+M, PCA-192	0.1712 [0.151, 0.192]	0.0952	0.329	0.398	0.443	$\approx$ prior champion
R+M, PCA-128	0.1670 [0.148, 0.187]	0.0920	0.329	0.398	0.444	GRU gain not carried
R+A+M, AE-64	0.1684	0.0943	—	—	—	ANN leg redundant
GRU single h512, PCA-192	0.1300	—	—	—	0.458	width tie #7; not sig
tower bank lat $\times$ 4 h211, PCA-192	0.1300	—	—	—	0.459	bank = width; closed
LSTM single, PCA-128	0.1272 [0.110, 0.145]	0.0495	0.335	0.458	0.451	best single; $\ll$ blend
GRU single h454, PCA-192	0.1251	0.0539	0.338	0.454	0.457	width closed
GRU single, PCA-192	0.1230 [0.106, 0.140]	0.0534	0.338	0.450	0.456	dim point-optimum
dual-tower GRU f0, PCA-192	0.1188	0.0563	0.337	0.456	0.455	dual tower: refuted
GRU single, PCA-128 (base)	0.1146 [0.099, 0.132]	0.0489	0.333	0.452	0.457	confirmed base space
GRU single, PCA-256	0.1167 [0.101, 0.134]	0.0502	0.338	0.458	0.456	past the dim peak
pre-MLP $\rightarrow$ GRU, PCA-192	0.1090	0.0446	0.324	0.469	0.450	pre-encoder harmful
B2 first-order Markov (bar)	0.1069 [0.092, 0.124]	0.0694	0.298	0.389	0.407	app incumbent
GRU single, AE-64 (control)	0.0957 [0.081, 0.111]	0.0411	0.294	0.450	0.432	bake-off control
ANN pooled-MLP, PCA-128	0.0901 [0.076, 0.105]	0.0394	0.299	0.435	0.426	loses to the GRU
XGB stacker (M+G / M+G+A)	0.080–0.087	0.042–0.049	0.31	0.38	0.41	combiner underperform
content-kNN (song-AE)	0.0636 [0.050, 0.077]	0.0321	0.272	0.405	—	the additive C leg
item2vec (train-only SGNS)	0.0517	0.0214	0.166	0.312	—	drags the blend
B0 popularity / B1 recency	0.0007	—	—	—	0.185	floors

Table 2: The fusion campaign: the R+M z-blend beats the Markov bar decisively, and no richer ensemble improves on it. R@10 and MRR higher-better; champion bolded.

Model	R@10 [95% CI]	MRR	Note
R+M z-blend $\alpha=0.5$ (champion)	0.1726 [0.154, 0.192]	0.0963	Δ vs Markov +0.066 [+0.05, +0.08]
R+A+M z-blend	0.1684	0.0943	ANN leg redundant with GRU
B2 Markov bar (M)	0.1069 [0.092, 0.123]	0.0694	the reference
XGB stacker M+R	0.1020	0.0561	learned combiner $\ll$ z-blend
GRU alone (R)	0.0957	0.0411	ties the bar on R@10 only
R+A z-blend	0.0894	0.0377	$\approx$ GRU alone
XGB stacker M+R+A	0.0887	0.0572	barely clears Markov

Table 3: The item-representation compression bake-off: six item spaces (linear PCA vs the nonlinear AE $\times$ latent dim 32 / 64 / 128), each scored under the *same* frozen **gru-infonce-h256** readout (1431 test sessions; paired- Δ vs the in-batch AE-64 control). R@10, MRR, artist@10, genre@10 higher-better. Winner (PCA-128) bolded; best cell per metric bolded.

Space	Run id	R@10 [95% CI]	Δ vs AE-64	MRR	art@10	gen@10
PCA-128 (winner)	run-2026071 5-042100-a1 d9d	0.1146 [0.099, 0.132]	+0.019 [+0.004, +0.034]	0.0489	0.3326	0.4521
PCA-64 (runner-up)	run-2026071 5-042100-88 51d	0.1041 [0.088, 0.121]	+0.008 (tie)	0.0416	0.3117	0.4598
AE-64 (control)	run-2026071 5-042059-dd 261	0.0957 [0.081, 0.111]	— (control)	0.0411	0.2942	0.4500
AE-128	run-2026071 5-042059-e9 6ae	0.0929 [0.078, 0.108]	−0.003 (tie)	0.0398	0.3026	0.4640
AE-32	run-2026071 5-042059-d3 f9e	0.0545 [0.043, 0.066]	−0.041	0.0274	0.2376	0.3955
PCA-32	run-2026071 5-042059-b1 242	0.0412 [0.031, 0.052]	−0.055	0.0224	0.2187	0.3816

Table 4: Phase-2 model-family sweep on the rich PCA spaces, ranked by R@10 (higher-better). The fixed z-blend is the combiner of record; the learned XGB stacker underperforms it badly. Champion row and prior champion bolded; best cell per metric bolded. Paired Δ is per-session against the named reference.

Model	Run id / source	R@10 [95% CI]	MRR	paired Δ
R+M+C, PCA-192 (champ. row)	analyze_rmc (A4)	0.1859 [0.166, 0.207]	0.1025	vs champ +0.0133 [+0.001, +0.026]
R+M+C, PCA-128	analyze_rmc (A2)	0.1824 [0.162, 0.203]	0.0993	vs champ +0.0098 (tie)
R+M, AE-64 (prior champion)	run-20260714- 170928-645ae	0.1726 [0.154, 0.192]	0.0963	— (reference)
R+M, PCA-192	run-20260715- 142440-96f53	0.1712 [0.151, 0.192]	0.0952	vs champ −0.0014 (tie)
R+M, PCA-128	run-20260715- 044920-36dce	0.1670 [0.148, 0.187]	0.0920	vs champ −0.0056 (tie)
LSTM single, PCA-128	run-20260715- 142440-546d0	0.1272 [0.110, 0.145]	0.0495	vs GRU-128 +0.0126 (> 0)
GRU single, PCA-192	run-20260715- 142440-89d6a	0.1230 [0.106, 0.140]	0.0534	vs GRU-128 +0.0084 (tie)
ANN pooled-MLP, PCA-128	run-20260715- 142440-7e528	0.0901 [0.076, 0.105]	0.0394	vs GRU-128 −0.0245
XGB stacker M+G, PCA-128	run-20260715- 142440-371b1	0.0867 [0.073, 0.101]	0.0493	vs GRU-128 −0.0280
XGB stacker M+G+A, PCA-192	run-20260715- 142440-f111d	0.0853 [0.071, 0.099]	0.0490	vs GRU-192 −0.0377

Table 5: Tier A1 of the literature-fit campaign: the learned content projection on PCA-192, versus the R+M+C champion (1431 test sessions; bootstrap 95% CI; paired Δ per session vs the champion). Winner ($R'+M+C'$) bolded; best cell per metric bolded.

Model	R@10 [95% CI]	MRR	art@10	gen@10	paired Δ vs champion
R+M+C champion (thirds)	0.1859 [0.166, 0.207]	0.1025	0.331	0.394	— (reference)
$R'+M+C'$ (both projected)	0.2117 [0.191, 0.233]	0.1220	0.333	0.401	+0.0259 [+0.012, +0.041] > 0
R+M+C+ C' (4-leg)	0.2061 [0.185, 0.227]	0.1148	0.345	0.410	+0.0203 [+0.013, +0.029] > 0
R+M+ C' (swap $C \rightarrow C'$)	0.1971 [0.177, 0.218]	0.1107	0.334	0.400	+0.0112 [−0.000, +0.023] cand.
C' proj-content (sym, r192)	0.1516	0.0708	0.305	0.401	vs unsup. C 0.0685: +0.083
C' dir. two-tower (r192)	0.1349	0.0552	0.291	0.393	symmetric wins

Table 6: The literature-fit lanes that closed: every borrowed method fails to beat the fixed z-blend champion (1431 test sessions; blend paired Δ per session vs the champion).

Lane	Standalone R@10	Blend Δ vs champion	Verdict
Markov KN-smoothing (Tier A2)	0.1069→0.1104	−0.0021 (straddles 0)	null in blend
Ranking combiner, <i>oracle</i> (B1)	0.1901 (blend)	+0.0042 (straddles 0)	no honest headroom
RRF rank fusion (B1)	0.1572 (blend)	−0.029 (CI<0)	hurts recall
V-SKNN session-kNN (B2)	0.0671	−0.038 (CI<0)	drags (warm-only)
GRU importance negatives (C)	0.1230→0.1279	−0.0014 (tie)	washes out

Table 7: The on-server objective / rank / τ ablation scan (6 cells, sorted by recall@10). All overlay the promoted definition; hit@10 = recall@10 by construction. Promoted champion (default, held) bolded; best cell per metric bolded (all held by the sub-noise τ -0.10 cell).

Config	recall@10	MRR	art@10	gen@10	run id
S2 infonce / full / τ 0.10	0.2229	0.1273	0.3424	0.4032	run-20260718-193 111-a781c
infonce / full / τ 0.07 (promoted, held)	0.2117	0.1220	0.3326	0.4011	run-20260718-191 539-1b89c
S3 infonce / rank 64 / τ 0.07	0.2055	0.1169	0.3326	0.3990	run-20260718-193 650-66c96
S1 infonce / full / τ 0.05	0.1929	0.1111	0.3354	0.3997	run-20260718-192 443-57432
S4 bpr / full / τ 0.07	0.1579	0.0939	0.3187	0.3913	run-20260718-194 228-4cc06
S5 bpr / rank 64 / τ 0.07	0.1523	0.0909	0.3194	0.3906	run-20260718-194 832-d070f

Table 8: Second-split hardening on the disjoint 0.64-cold earlier-holdout split seq-20260718-211238 (1431 test sessions; validation-only, nothing promoted). Point metrics (top) and the paired per-session Δ (bottom, 2000-resample bootstrap, seed 1337). Projection champion bolded.

Config	recall@10	MRR	art@10	gen@10	run id
$R'+M+C'$ projection champion	0.1705	0.1050	0.2991	0.3417	run-20260718-212639-50e1d
R+M+C crown (primary reference)	0.1279	0.0816	0.2956	0.3410	run-20260718-213156-cf1dd
R+M 2-leg (context reference)	0.1174	0.0791	0.2844	0.3368	run-20260718-214114-93bad
<i>Paired Δ recall@10 (projection – reference):</i>					
PROJ vs R+M+C (primary)	+0.0426	[+0.0280, +0.0573]	— entirely > 0; MRR		
PROJ vs R+M (context)	+0.0531	[+0.0391, +0.0678]	— entirely > 0		

Table 9: Item-vector representation exploration: eight alternative 192-d item spaces vs the PCA-192 champion, under readout A (champion projection) and readout C (frozen GRU). Paired Δ on exact hit@10 (2000-resample bootstrap, rng 1337) vs the same-readout baseline; a lift needs the CI entirely > 0. Baseline rows and the best readout-C lift (V5) bolded.

Variant (space)	dataset id	Readout A (projection)		Readout C (frozen GRU)	
		R@10	Δ hit@10 [CI]	R@10	Δ hit@10 [CI]
full — PCA-192 (baseline)	seq-20260715-131139	0.2117	(baseline)	0.1230	(baseline)
V1 noaco	seq-20260718-222742	0.2089	−0.0028 [−0.013, +0.008]	0.1481	+0.0252 [+0.009, +0.041]
V2 std (whiten all)	seq-20260718-222754	0.2082	−0.0035 [−0.015, +0.008]	0.1530	+0.0300 [+0.011, +0.049]
V3 std-noaco	seq-20260718-222807	0.2166	+0.0049 [−0.008, +0.017]	0.1572	+0.0342 [+0.015, +0.055]
V4 textcat	seq-20260718-222820	0.1950	−0.0168 [−0.029, −0.006]	0.1565	+0.0335 [+0.015, +0.051]
V5 balanced-pca192	seq-20260718-222834	0.2159	+0.0042 [−0.006, +0.015]	0.1586	+0.0356 [+0.019, +0.052]
V7 std-txtcatup	seq-20260718-222848	0.2055	−0.0063 [−0.019, +0.006]	0.1488	+0.0259 [+0.006, +0.046]
V10 text-only	seq-20260718-222901	0.1999	−0.0119 [−0.025, +0.001]	0.1377	+0.0147 [−0.005, +0.034]
V6 sonic-64 (64-d)	seq-20260718-222905	0.1936	−0.0182 [−0.031, −0.005]	0.1216	−0.0014 [−0.017, +0.013]

Table 10: music@10 by predictor family, from the offline backfill over all 53 next-track runs (one shared 1431-case split). The recall@10 champion (**seq-blend**) is *not* the music@10 champion (**seq-nexttrack**). Best cell per column bolded.

Predictor family	runs	best recall@10	best music@10	mean music@10
seq-blend	23	0.223	0.461	0.441
seq-nexttrack	24	0.159	0.482	0.448
seq-markov	1	0.107	0.407	0.407
seq-ann	1	0.090	0.426	0.426
seq-stacker	3	0.087	0.418	0.411
seq-popularity	1	0.001	0.185	0.185

Table 11: Phase A holisticness fingerprint across the model family, sorted by Recall@10 (1431 test sessions). All runs share the prefix `run-20260722-165345-`; the id column lists the distinguishing suffix. Recall@10 (raw hits of 1431), MRR, artist@10, genre@10, music@10, mood_coh@10, ild@10 higher-better; **artist_adj@10 lower-better**. Champion row and best cell per metric bolded. [†] The popularity floor’s extreme artist_adj@10 / ild@10 are *degenerate* (it ignores the seed): it lands ≈ 0 recall and negative mood_coh@10, proving the metrics are not gamed low/high by a good model.

Config	id	R@10 (raw)	MRR	art@10	gen@10	mus@10	artadj@10↓	mood@10↑	ild@10↑
A1 champion $R'+M+C'$	82a8f	0.2117 (303)	0.122	0.333	0.401	0.454	0.610	0.398	0.406
A2 R+M (blend)	28dbb	0.1712 (245)	0.095	0.329	0.398	0.443	0.579	0.399	0.403
A3 GRU alone ($\beta=0$)	e39c9	0.1230 (176)	0.053	0.338	0.450	0.456	0.347	0.479	0.454
B LSTM (else matched to A3)	bb387	0.1188 (170)	0.049	0.328	0.455	0.449	0.344	0.458	0.463
A4 markov (first-order)	047cc	0.1069 (153)	0.069	0.298	0.389	0.407	0.543	0.310	0.462
A5 popularity	34614	0.0007 (1)	0.001	0.014	0.085	0.185	0.003[†]	−0.041	0.898[†]

Table 12: Phase C MMR λ sweep on the champion, computed offline from its checkpoint (`mmr_pool 200`; 1431 test sessions). Recall@10 (raw hits), MRR, artist@10, music@10, mood_coh@10, ild@10 higher-better; **artist_adj@10 lower-better**. Best cell per metric bolded; the champion default ($\lambda=\text{off}$) recall / MRR bolded.

λ	R@10 (raw)	Δ vs off	MRR	art@10	mus@10	artadj↓	mood↑	ild↑
off (1.0)	0.2117 (303)	—	0.122	0.333	0.454	0.610	0.398	0.406
0.9	0.2103 (301)	−0.0014 (TIE)	0.121	0.341	0.462	0.603	0.391	0.436
0.7	0.1936 (277)	−0.0181	0.116	0.347	0.476	0.571	0.364	0.520
0.5	0.1642 (235)	−0.0475	0.108	0.359	0.487	0.486	0.315	0.653
0.3	0.1097 (157)	−0.1020	0.100	0.358	0.492	0.293	0.238	0.815

Table 13: Phase D anti-eager regularizer β sweep on the single GRU (`eager_margin 0.0`; 1431 test sessions). Recall@10, artist@10, mood_coh@10, ild@10 higher-better; **artist_adj@10 lower-better**. Best cell per metric bolded.

run id	β	R@10 (raw)	MRR	art@10	artadj↓	mood↑	ild↑
run-20260722-165345-e39c9	0.0	0.1230 (176)	0.053	0.338	0.347	0.479	0.454
run-20260722-165345-353b5	0.05	0.1237 (177)	0.053	0.337	0.346	0.478	0.455
run-20260723-002044-af11a	0.1	0.1202 (172)	0.053	0.335	0.344	0.477	0.456
run-20260723-002044-17cf8	0.2	0.1202 (172)	0.051	0.336	0.340	0.475	0.458

Table 14: Phase E multi-step continuation (`harness -m 3 -k 10`, 1172 cases) on `seq-20260715-131139`: the GRU (A3 checkpoint) beats the first-order Markov bigram on both metrics (higher-better). Winner bolded.

Model	suffix_recall@10	cont_prec@10
markov (first-order bigram)	0.0869	0.0236
GRU (gru-infonce-h256, A3)	0.0963	0.0273

Table 15: Dual-tower view-pairing $\times$ fusion-depth scan, ten runs on seq-20260715-131139 (1431 test sessions), sorted by Recall@10. All arms share the run-id prefix run-20260725-143502-; C0 is run-20260725-143143-97c1e. Paired Δ is a per-session paired bootstrap against C0 (2000 resamples, rng 1337). Recall@10 raw = successes of 1431; CI bounds are given to three decimals, and a CI straddling zero is a *tie*. Arm codes carry the view pair (lat = **latent**, del = **delta**, cum = **cummean**) and the fusion depth (-f0 = linear read, otherwise a 256-wide MLP head). R@10, MRR, mus@10, ild@10 higher-better; **artadj@10 lower-better**; H is the 3-factor holisticness generation (Section 18). Best cell per metric bolded. [†] D-CC’s extreme artadj@10 / H are *not* a real crown: they are bought at 95 of 1431 hits, 0.0566 below the baseline, failing the ± 0.015 recall floor by $3.8\times$.

Arm	id	params	R@10 (raw)	Δ vs C0 [95% CI]	MRR	mus@10	artadj↓	ild↑	H↑
C0 h256 <i>base</i>	97c1e	394 944	0.12299 (176)	—	0.05341	0.4558	0.3474	0.4536	0.14172
C1 h425 <i>ctrl</i>	be11b	871 017	0.12089 (173)	-0.0021 [-.011, +.006]	0.05215	0.4597	0.3640	0.4506	0.13850
D-LL-f0 <i>lat/lat</i>	18b4a	789 696	0.11880 (170)	-0.0042 [-.013, +.004]	0.05631	0.4553	0.3630	0.4506	0.13774
D-LC-f0 <i>lat/cum</i>	3e5d4	789 696	0.11740 (168)	-0.0056 [-.017, +.005]	0.05087	0.4458	0.3557	0.4630	0.13648
D-LL <i>lat/lat</i>	12fe6	871 872	0.10273 (147)	-0.0203 [-.035, -.007]	0.04235	0.4446	0.3092	0.4873	0.14635
D-LC <i>lat/cum</i>	a95c5	871 872	0.09713 (139)	-0.0259 [-.039, -.013]	0.03901	0.4380	0.2997	0.4913	0.15000
D-LD <i>lat/del</i>	8d214	871 872	0.09154 (131)	-0.0315 [-.045, -.018]	0.04029	0.4385	0.2999	0.4988	0.14673
D-DC <i>del/cum</i>	e9cfb	871 872	0.08665 (124)	-0.0363 [-.050, -.022]	0.03379	0.4125	0.2319	0.5413	0.16340
D-CC <i>cum/cum</i>	88a4c	871 872	0.06639 (95)	-0.0566 [-.072, -.041]	0.02664	0.3935	0.1899[†]	0.5235	0.19715[†]
D-DD <i>del/del</i>	4324f	871 872	0.05800 (83)	-0.0650 [-.082, -.049]	0.02819	0.4016	0.2363	0.5457	0.13335

Table 16: View attribution: each mixed cell against *both* of its single-view parents (paired Δ Recall@10, 1431 sessions). Only one of three mixed pairs is genuinely complementary — and it is the one with no raw-latent tower.

Mixed cell	vs parent 1	vs parent 2	Verdict
D-LC <i>lat/cum</i>	D-LL: -0.0056 [-.018, +.007]	D-CC: +0.0308 [+0.018, +.045]	<i>not</i> complementary
D-LD <i>lat/del</i>	D-LL: -0.0112 [-.023, +.001]	D-DD: +0.0335 [+0.020, +.047]	<i>not</i> complementary
D-DC <i>del/cum</i>	D-DD: +0.0287 [+0.015, +.043]	D-CC: +0.0203 [+0.008, +.034]	complementary: both CI>0

Table 17: Pre-encoder presence $\times$ placement $\times$ dose, ten runs on seq-20260715-131139 (1431 test sessions), sorted by Recall@10. All arms share the prefix run-20260726-161801-; C0 is run-20260726-161429-bdcbcb. Paired Δ is a per-session paired bootstrap against C0 (2000 resamples, rng 1337); CI bounds are given to three decimals, and a CI straddling zero is a *tie*. Dual-arm codes give the two towers’ pre-encoder widths (A0/B256 = tower A bare, tower B rectified at width 256) and the fusion depth. H here is the 4-factor (grounded) holisticness generation and is **not** comparable to Table 15’s 3-factor column (Section 18). Best cell per metric bolded; **artadj@10 lower-better**. [†] T4’s minimal artadj and T1’s maximal ild are the familiar de-eagering-by-predicting-worse trade: both sit ~ 0.025 below C0 on recall.

Arm	id	params	R@10 (raw)	Δ vs C0 [95% CI]	MRR	mus@10	artadj↓	ild↑	H↑
C2 h454 <i>ctrl</i>	7646d	969 936	0.12509 (179)	+0.0021 [-.007, +.011]	0.05389	0.4572	0.3668	0.4466	0.045765
T1-f0 A0/B256, f0	a6423	888 256	0.12369 (177)	+0.0007 [-.012, +.013]	0.05470	0.4575	0.3627	0.4616	0.044541
C0 h256 <i>base</i>	bdcbcb	394 944	0.12299 (176)	—	0.05341	0.4558	0.3474	0.4536	0.047086
C1 h297 <i>ctrl</i>	7bab3	494 697	0.11321 (162)	-0.0098 [-.017, -.003]	0.05199	0.4513	0.3602	0.4474	0.045285
S1 <i>preMLP256</i>	ad53f	493 504	0.10901 (156)	-0.0140 [-.027, -.001]	0.04463	0.4499	0.3240	0.4815	0.047132
C3 <i>bare dual, f1</i>	0df6f	871 872	0.10273 (147)	-0.0203 [-.035, -.007]	0.04235	0.4446	0.3092	0.4873	0.046612
T2 A0/B128, f1	dc7d2	847 424	0.09993 (143)	-0.0231 [-.037, -.009]	0.04490	0.4415	0.3039	0.4939	0.046431
T3 A0/B384, f1	ca0b8	1 093 440	0.09993 (143)	-0.0231 [-.037, -.009]	0.04333	0.4441	0.3068	0.4961	0.045895
T4 A256/B256, f1	bd5b5	1 068 992	0.09783 (140)	-0.0252 [-.039, -.011]	0.03829	0.4343	0.2845[†]	0.4986	0.043985
T1 A0/B256, f1	9450b	970 432	0.09713 (139)	-0.0259 [-.040, -.013]	0.04176	0.4369	0.2962	0.5016[†]	0.045769

Table 18: The dose curve, as measured: paired Δ Recall@10 against the bare dual-tower arm C3 at `fusion_layers = 1`. Flat by the pre-registered test — but nominally downward, never upward.

pre_hidden_b	Arm	R@10 (raw)	Δ vs C3 (bare)
0	C3	0.10273 (147)	—
128	T2	0.09993 (143)	−0.0028 [−0.0147, +0.0091] straddles
256	T1	0.09713 (139)	−0.0056 [−0.0189, +0.0077] straddles
384	T3	0.09993 (143)	−0.0028 [−0.0154, +0.0105] straddles

Table 19: The `fusion_layers` regression, three independent measurements. Paired Δ Recall@10 of the linear-read arm minus its MLP-head twin; positive means removing the fusion MLP *helps*. The ΔH column is available only for the two same-batch 3-factor pairs. *Sources:* `2026-07-25-nexttrack-dual-tower-fusion-scan.md` (rows 1–2), `2026-07-26-nexttrack-pre-encoder-scan.md` (row 3).

Pair (f0 – f1)	Regime	Δ R@10 [95% CI]	ΔH [95% CI]
D-LL-f0 – D-LL	redundant views, no pre-encoder	+0.0161 [+0.0021, +0.0301]	−0.0086 [−0.0116, −0.0055]
D-LC-f0 – D-LC	complementary views, no pre-encoder	+0.0203 [+0.0063, +0.0335]	−0.0135 [−0.0171, −0.0100]
T1-f0 – T1	with a per-step pre-encoder	+0.0266 [+0.0140, +0.0391]	—

Table 20: Measured resolution of the test split. Half-widths are means over the nine paired-bootstrap comparisons of the pre-encoder batch (2000 resamples, $n = 1431$); the family spread is that batch’s full range on each metric. “Tiers” is spread/half-width, i.e. how many genuinely distinguishable levels eight adjacent arms can occupy. *Sources:* `2026-07-25-nexttrack-dual-tower-fusion-scan.md`, `2026-07-26-nexttrack-pre-encoder-scan.md`.

Metric	CI half-width at $n = 1431$	family spread	tiers	n for $\Delta = 0.005$
Recall@10	0.0124	0.0280	~2 of 8	~8800
holisticness@10	0.0018	0.0031	~1 of 8	—

Table 21: The same models under all three holisticness generations, sorted by Recall@10. The generation effect dwarfs the model effect, and the ordering is roughly *inverse* to Recall@10 throughout — which is why this document’s leaderboard stays denominated in Recall@10. Dashes are runs whose facets could not be re-derived under that generation (remote-only runs with no materialized predictions file). *Sources:* `2026-07-25-nexttrack-dual-tower-fusion-scan.md`, `2026-07-26-nexttrack-pre-encoder-scan.md`.

Model (run)	R@10 (raw)	gen-1 (3-f)	gen-2 (4-f)	gen-3 (current)
$R' + M + C'$ blend (1c1a3)	0.21174 (303)	0.0632	0.020830	0.00860
single GRU h454 (7646d)	0.12509 (179)	—	0.045765	0.02399
single GRU h256 (97c1e, bdcbc)	0.12299 (176)	0.14172	0.047086	0.02454
single GRU h425 (be11b)	0.12089 (173)	0.13850	0.046677	0.02487
S1 pre-MLP→GRU (ad53f)	0.10901 (156)	0.14499	0.047132	0.02410
D-CC <i>cum/cum</i> (88a4c)	0.06639 (95)	0.19715	0.050436	—
mood-session (degen.)	0.02380 (34)	0.23319	0.043881	0.00000

Table 22: The MMR λ frontier on both anchors, canonical split seq-20260715-131139 ($n = 1431$), sorted by holisticness@10. Every run id is run-20260726- plus a launch time and hex suffix, in arm order: 184115-bf395 (B3), 184115-1dc77 (A4), 184116-20d9b (B4), 184115-147a6 (B2), 184115-e1539 (B1), 183333-9a0b2 (B0), 184115-7a2f7 (A3), 184115-36edb (A2), 184115-6cb38 (A1), 183333-1c361 (A0); the A arms are seq-blend, the B arms seq-nexttrack. Every Δ is a paired bootstrap (2000 resamples, rng 1337) against the arm’s own $\lambda = 1.0$ control, with all five facets re-aggregated on the same resampled index set before the product; CI bounds are given to three decimals for width. Tier codes: *t1* TIER-1 FREE, *t2* TIER-2 PRICED, *t3* TIER-3 REJECTED, *ctrl* the arm’s own $\lambda = 1.0$ control. The binding facet — the one entering the eagerness factor through the max() — is **conc** on A0/A1/A2, B0 and **adj** everywhere else. H@10, R@10, mus (= music@10) and **ild** higher-better; **adj**, **conc** lower-better. Best cell per metric bolded; the promotable arm B4 is bolded in full. mood_coh is omitted for width and read in the interpretation: B0→B3 0.4788→0.4585 (−4 %) against A0→A4 0.3982→0.2378 (−40 %), the champion’s collapse under diversification. *Source: 2026-07-26-nexttrack-mmr-lambda-frontier.md.*

Arm	λ /pool	H@10	ΔH [95% CI]	R@10 (hits)	$\Delta R@10$ [95% CI]	mus	ild	adj / conc
B3 <i>t2</i>	0.7 / 200	0.034171	+0.0096 [+0.008, +0.011]	0.11461 (164)	−0.0085 [−.016, −.001]	0.4747	0.5314	0.313 / 0.271
A4 <i>t3</i>	0.3 / 200	0.031626	+0.0230 [+0.021, +0.026]	0.10971 (157)	−0.1019 [−.120, −.084]	0.4925	0.8146	0.293 / 0.190
B4 <i>t1</i>	0.7 / 50	0.030355	+0.0058 [+0.005, +0.007]	0.12020 (172)	−0.0029 [−.008, +0.003]	0.4670	0.5017	0.327 / 0.302
B2 <i>t1</i>	0.8 / 200	0.029458	+0.0049 [+0.004, +0.006]	0.11950 (171)	−0.0036 [−.009, +0.001]	0.4648	0.4961	0.330 / 0.307
B1 <i>t1</i>	0.9 / 200	0.026761	+0.0022 [+0.002, +0.003]	0.12089 (173)	−0.0021 [−.006, +0.002]	0.4597	0.4716	0.341 / 0.337
B0 <i>ctrl</i>	1.0 / 200	0.024539	—	0.12299 (176)	—	0.4558	0.4536	0.347 / 0.356
A3 <i>t3</i>	0.5 / 200	0.023576	+0.0150 [+0.013, +0.017]	0.16422 (235)	−0.0473 [−.061, −.033]	0.4871	0.6527	0.486 / 0.461
A2 <i>t2</i>	0.7 / 200	0.015289	+0.0067 [+0.006, +0.008]	0.19357 (277)	−0.0181 [−.029, −.008]	0.4759	0.5201	0.571 / 0.608
A1 <i>t1</i>	0.9 / 200	0.010435	+0.0018 [+0.001, +0.002]	0.21034 (301)	−0.0013 [−.008, +0.005]	0.4625	0.4362	0.603 / 0.671
A0 <i>ctrl</i>	1.0 / 200	0.008603	—	0.21174 (303)	—	0.4538	0.4064	0.610 / 0.692

Table 23: Five hold@20 sweeps, paired bootstrap over seeds (2000 resamples, rng 1337) with a leave-one-seed-out jackknife (“LOO” is the number of refits preserving the verdict). Positive Δ favours the dual-tower arm. The pre-registered rule was CONFIRMED iff the CI lies entirely above zero *and* the verdict survives $\geq 90\%$ of LOO refits; TIE iff the CI straddles. S2 and S3 met that rule and were nonetheless *retracted* — see the interpretation. *Source: 2026-07-27-rollout-hold-metric.md.*

#	Sweep	n	$\Delta \text{hold@20}$ [95% CI]	LOO	Verdict
S1	pilot, mood-extreme seeds, 10-track prefix	8	−0.034 [−0.138, +0.079]	0/8	TIE
S2	pilot, mood-extreme seeds, 30-track prefix	8	+0.314 [+0.107, +0.529]	8/8	<i>retracted</i>
S3	replication, random seeds	39	+0.122 [+0.044, +0.201]	39/39	<i>confounded</i>
S4	same-split control	39	+0.033 [−0.030, +0.093]	0/39	TIE
S4b	same-split, other dual checkpoint	39	+0.051 [−0.012, +0.118]	0/39	TIE
S4c	within-architecture	39	−0.072 [−0.141, −0.002]	—	significant
S5	the user’s own 11 playlists	11	+0.004 [−0.088, +0.104]	0/11	TIE
—	the single judged playlist	1	+0.379 (0.623 / 0.245)	—	outlier

Table 24: The `markov_gate` batch, six runs from the champion definition `blend-gru-markov-content-proj`. `ds1 = seq-20260715-131139` (canonical, 55% cold), `ds2 = seq-20260718-211238` (disjoint, 64% cold); `ds2`’s absolute recall is lower *by construction*, so read it only through the paired Δ . Every run id is `run-20260728-` plus the value in the `run` column. The graded metrics track the gate too: `artist@10` 0.333→0.367 and `genre@10` 0.401→0.463 from C1 to A1, and 0.382 / 0.497 on A3. Rows sorted by `Recall@10`. `H@10` is generation 3 (Section 18.2). `R@10`, `MRR`, `H@10`, `mus@10`, `ild`, `mood`, `art@10`, `gen@10` higher-better; `adj`, `conc` lower-better. Best cell per metric bolded among the canonical-split arms; the new best-recall row A1 is bolded in full. *Source: 2026-07-27c-nexttrack-markov-gate-confirm.md.*

id	run	ds	overrides	R@10 (hits)	MRR	H@10	mus@10	ild	adj	conc	mood
A1	005349-507bb	ds1	gate	0.23201 (332)	0.13352	0.018699	0.49319	0.43377	0.57477	0.59821	0.46226
A3	012030-6d1b7	ds1	gate + $\lambda 0.7$	0.21314 (305)	0.12852	0.030980	0.51649	0.56770	0.51279	0.47325	0.41885
C1	004444-b32ea	ds1	— (champion)	0.21174 (303)	0.12199	0.008603	0.45379	0.40637	0.60971	0.69165	0.39822
C3	011245-bfc67	ds1	$\lambda 0.7$ only	0.19357 (277)	0.11560	0.015289	0.47592	0.52010	0.57135	0.60823	0.36433
A2	010658-39e20	ds2	gate	0.19008 (272)	0.11801	0.008414	0.42953	0.41401	0.58672	0.64847	0.42618
C2	010113-e2599	ds2	— (champion)	0.17051 (244)	0.10498	0.003392	0.39616	0.40177	0.61013	0.71945	0.35364

Table 25: Paired bootstrap over the 1431 test sessions (2000 resamples, rng 1337, paired by row id). ΔH is computed *product-of-means per resample* — all five facets re-aggregated on the same resampled index set for both arms, then multiplied — not as a mean of per-row products. n_{disc} counts sessions whose `hit@10` changed at all, split by sign. A verdict of WIN means the recall CI lies entirely above zero, TIE that it straddles; bounds are abbreviated to four decimals for width. *Source: 2026-07-27c-nexttrack-markov-gate-confirm.md.*

Comparison	$\Delta R@10$ [95% CI]	Verdict	ΔH [95% CI]	n_{disc} (+/-)
A1 – C1 (gate, ds1)	+0.02027 [+0.0105, +0.0308]	WIN	+0.010096 [+0.0085, +0.0117]	57 (43/14)
A2 – C2 (gate, ds2)	+0.01957 [+0.0105, +0.0301]	WIN	+0.005022 [+0.0039, +0.0062]	56 (42/14)
A3 – C3 (gate, $\lambda 0.7$)	+0.01957 [+0.0091, +0.0301]	WIN	+0.015691 [+0.0136, +0.0179]	64 (46/18)
A3 – C1 (gate+MMR / champ.)	+0.00140 [-0.0119, +0.0147]	TIE	+0.022377 [+0.0199, +0.0249]	96 (49/47)
C3 – C1 (MMR $\lambda 0.7$ alone)	−0.01817 [−0.0294, −0.0077]	costs recall	+0.006686 [+0.0057, +0.0077]	60 (17/43)

Table 26: The bucket decomposition, reproduced *natively* through the real transition builder and bucketed by the train-level bigram support n_u of each query’s last prefix item — the exact quantity the Markov scorer computes. `Recall@10` within bucket, hits in parentheses. In both warm buckets the two arms’ top- k lists are **byte-for-byte identical**, so the gate is a surgically pure $n_u = 0$ intervention with literally zero warm-side cost, on three independent control/arm pairs. *Source: 2026-07-27c-nexttrack-markov-gate-confirm.md.*

Pair	Bucket	n	control	gated	Δ	top- k identical?
<i>Canonical split, A1 vs C1</i>						
	$n_u = 0$	795	0.1660 (132)	0.2025 (161)	+0.0365	no
	n_u 1–10	292	0.3116 (91)	0.3116 (91)	+0.0000	YES
	$n_u > 10$	344	0.2326 (80)	0.2326 (80)	+0.0000	YES
<i>Second split, A2 vs C2</i>						
	$n_u = 0$	892	0.1244 (111)	0.1558 (139)	+0.0314	no
	n_u 1–10	322	0.2578 (83)	0.2578 (83)	+0.0000	YES
	$n_u > 10$	217	0.2304 (50)	0.2304 (50)	+0.0000	YES
<i>Canonical split under MMR $\lambda 0.7$, A3 vs C3</i>						
	$n_u = 0$	795	0.1472 (117)	0.1824 (145)	+0.0352	no
	n_u 1–10	292	0.2945 (86)	0.2945 (86)	+0.0000	YES
	$n_u > 10$	344	0.2151 (74)	0.2151 (74)	+0.0000	YES

Table 27: The `eager_beta` $\times$ margin batch, eight `seq-dualgru` runs on `seq-20260715-131139` ($n = 1431$). Every run id is `run-20260730-` plus the value in the `run` column. m is the hinge margin, β the penalty weight. H@10 is generation 3 (Section 18.2). “best ep/ran” is the best epoch over the epochs actually run; Recall@10 is `hits`/1431 throughout (C0 0.118798, A1 0.053809, A2 0.013976, B1 0.114605, and 0.118798 on every inert arm). `ild` is omitted for width and moves only on the arms that fire (0.4506 on C0, 0.5572 on A1, 0.6762 on A2, 0.4530 on B1, 0.4504 to 0.4506 on the inert arms). `mood`, `mus@10`, H@10 higher-better; `conc`, `adj` lower-better. The lowest-eagerness cells are bolded and belong to *degenerate* arms, which is what the relevance floor exists to catch. *Source: 2026-07-30-eager-margin-calibrated-refutation.md.*

Arm	run	m	β	hits	conc	adj	mood	mus@10	H@10	ep	verdict
C0	175402-cd006	0	0	170	0.370588	0.362963	0.479846	0.455347	0.023785	8/14	CONTROL
A1	181405-53088	0	3	77	0.240717	0.206010	0.346867	0.399997	0.013339	18/24	DEGENERATE
A2	181405-0bc1a	0	12	20	0.133535	0.107687	0.221756	0.358132	0.003569	37/40	DEGENERATE
B1	181405-84da4	0.40	9	164	0.367606	0.364151	0.463440	0.447386	0.021601	15/21	no effect
B2	175402-7c304	0.5843	19	170	0.370634	0.363103	0.479955	0.455210	0.023751	8/14	INERT
B3	184556-b08c1	0.70	35	170	0.370588	0.362963	0.479846	0.455347	0.023785	8/14	INERT (bit-id.)
B4	175402-0d310	0.8304	92	170	0.370588	0.362963	0.479846	0.455347	0.023785	8/14	INERT (bit-id.)
B5	184556-47d7b	0.5843	57	170	0.370743	0.363103	0.479955	0.455379	0.023771	8/14	INERT

Table 28: Paired statistics against the in-batch control C0 (2000 resamples, rng 1337, $n = 1431$). n_{disc} counts sessions whose hit@10 changed. Measured half-widths on the *active* arms (0.0080 to 0.0173) match the design’s predicted 0.0102, so the instrument was correctly powered — the axis failed anyway. *Source: 2026-07-30-eager-margin-calibrated-refutation.md.*

Arm	$\Delta_{\text{artist_conc}}$ [95% CI]	$\Delta_{\text{music@10}}$ [95% CI]	ΔH	n_{disc}	mean $ \Delta_{\text{conc}} $
A1	−0.12987 [−0.14354, −0.11605]	−0.05535 [−0.06606, −0.04465]	−0.010445	125	0.191304
A2	−0.23705 [−0.25493, −0.22026]	−0.09721 [−0.11107, −0.08333]	−0.020215	166	0.282895
B1	−0.00298 [−0.01085, +0.00519]	−0.00796 [−0.01411, −0.00194]	−0.002183	62	0.093734
B2	+0.00005 [−0.00028, +0.00039]	−0.00014 [−0.00037, +0.00000]	−0.000034	0	0.000419
B3	+0.00000 [+0.00000, +0.00000]	+0.00000 [+0.00000, +0.00000]	+0.000000	0	0.000000
B4	+0.00000 [+0.00000, +0.00000]	+0.00000 [+0.00000, +0.00000]	+0.000000	0	0.000000
B5	+0.00016 [−0.00031, +0.00068]	+0.00003 [−0.00036, +0.00045]	−0.000014	0	0.000901

Table 29: First walk-level read of the two shipped engines. 40 held-out sessions, 5-track seed, 20 autoregressive steps each; `cap` is the per-artist limit inside the journey. W_1 and $|\text{median step} - 0.261|$ lower-better; `vibe`, in-band share, artists, genres and `ground` higher-better; `drift` nearer zero better. The last row is the reference the metrics are calibrated against (its in-band share is 80 % by construction). Best cell per column bolded among the model arms. *Source: tools/walk_eval.py.*

Arm	W_1	med. step	in-band	vibe	drift	artists	genres	ground
GRU +anchor 0.4, cap 1 (shipped)	0.320	0.661	80.6 %	0.609	−0.089	20.0	9.93	0.345
GRU +anchor 0.4, cap 3	0.341	0.681	76.7 %	0.610	−0.075	13.2	6.53	0.393
GRU no anchor, cap 1	0.263	0.574	85.2 %	0.302	−0.277	20.0	8.78	0.276
Champion ($R' + M + C'$), cap 1	0.078	0.317	90.3 %	0.187	−0.186	20.0	10.33	0.289
Champion ($R' + M + C'$), cap 3	0.147	0.452	84.8 %	0.193	−0.224	11.1	5.33	0.306
<i>the user’s real sessions</i>	0.000	0.261	80.0 %	—	—	—	—	—

Table 30: The hard per-artist cap on the one-shot surface, on the deployed champion ($R'+M+C'$, ungated), canonical split seq-20260715-131139, $n = 1431$. Off-run measurement: scores are computed once per model and memoized, so every cap arm sees bit-identical score vectors and the pairing is exact rather than merely matched (2000 resamples, rng 1337). *holisticness@10* is generation 3 (Section 18.2). *artist_conc*, *artist_adj* lower-better; the rest higher-better. *Source: 2026-07-30c-artist-cap-measurement.md.*

cap	R@10	hits	MRR	artist_conc	artist_adj	ild	music@10	H@10 (gen-3)
off	0.211740	303	0.121985	0.691653	0.609713	0.406374	0.453790	0.008603
5	0.178896	256	0.119693	0.252613	0.337247	0.563417	0.474291	0.027258
3	0.160727	230	0.116915	0.122975	0.215304	0.613641	0.476787	0.033862
2	0.138365	198	0.112313	0.061946	0.150384	0.639717	0.474317	0.036331

Table 31: The anchor (a) $\times$ stride (s) frontier, 20 held-out sessions $\times$ 20 steps, cap 1, both engines. W_1 lower-better; “med. step” is compared against the user’s real 0.261; *vibe*, genres, *ground* higher-better; *drift* nearer zero better. The window is median step $\in [0.20, 0.38]$ and *vibe* ≥ 0.50 , pre-registered before the sweep; the two rows that satisfy it are bolded. The shipped GRU configuration is $a = 0.4$, $s = 0$; the champion as deployed is $a = 0$, $s = 0$. *Source: tools/walk_frontier.py.*

Engine	a / s	W_1	med. step	vibe	drift	genres	ground
GRU	0.0 / 0.0	0.287	0.607	0.286	-0.328	9.70	0.270
GRU	0.0 / 0.3	0.273	-0.018	0.200	-0.157	8.95	0.291
GRU	0.0 / 0.6	0.564	-0.374	0.123	-0.102	12.00	0.244
GRU	0.0 / 1.0	0.702	-0.557	0.084	-0.036	15.80	0.224
GRU	0.2 / 0.0	0.323	0.663	0.563	-0.124	9.40	0.345
GRU	0.2 / 0.3	0.186	0.094	0.426	-0.098	11.05	0.335
GRU	0.2 / 0.6	0.473	-0.251	0.330	-0.067	13.25	0.303
GRU	0.2 / 1.0	0.675	-0.522	0.185	-0.077	15.05	0.253
GRU	0.4 / 0.0 (<i>shipped</i>)	0.361	0.709	0.639	-0.090	9.95	0.349
GRU	0.4 / 0.3	0.104	0.297	0.588	-0.076	11.05	0.358
GRU	0.4 / 0.6	0.356	-0.127	0.437	-0.044	12.55	0.303
GRU	0.4 / 1.0	0.588	-0.405	0.302	-0.027	15.20	0.287
GRU	0.8 / 0.0	0.364	0.713	0.678	-0.067	11.00	0.352
GRU	0.8 / 0.3	0.190	0.506	0.672	-0.070	12.15	0.351
GRU	0.8 / 0.6	0.122	0.176	0.605	-0.038	12.60	0.358
GRU	0.8 / 1.0	0.420	-0.188	0.460	-0.016	14.65	0.335
Champion	0.0 / 0.0 (<i>deployed</i>)	0.062	0.318	0.185	-0.203	11.00	0.275
Champion	0.0 / 0.5	0.240	-0.032	0.104	+0.002	12.15	0.266
Champion	1.0 / 0.0	0.239	0.577	0.551	-0.138	8.95	0.327
Champion	1.0 / 0.5	0.072	0.282	0.503	-0.130	9.55	0.324
Champion	2.0 / 0.0	0.276	0.596	0.625	-0.060	8.90	0.359
Champion	2.0 / 0.5	0.170	0.472	0.615	-0.105	9.45	0.338
Champion	4.0 / 0.0	0.300	0.645	0.669	-0.051	9.85	0.338
Champion	4.0 / 0.5	0.236	0.544	0.664	-0.085	9.70	0.340

Table 32: Walk-metric means over the 40 identical held-out sessions (20 steps, cap 1). **stride_err** = |median step - 0.261| (lower better); **vibe**, **genres**, **ground** higher-better; **drift** nearer zero better. Best cell per column bolded. *Source: tools/walk_headtohead.py.*

Arm	vibe	stride_err	drift	genres	ground
GRU shipped, <i>a</i> 0.4 <i>s</i> 0.0	0.609	0.399	-0.089	9.925	0.345
GRU tuned, <i>a</i> 0.4 <i>s</i> 0.3	0.568	0.129	-0.080	10.450	0.357
dual latent/latent f0, <i>a</i> 0.8 <i>s</i> 0.4	0.638	0.157	-0.063	10.975	0.351
dual latent/cummean f0 , <i>a</i> 0.8 <i>s</i> 0.5	0.610	0.144	-0.053	11.425	0.351
Champion (R' +M+ C'), <i>a</i> 1.0 <i>s</i> 0.5	0.476	0.135	-0.131	9.600	0.334

Table 33: Paired per-session differences against the **shipped** GRU (*a*0.4 *s*0.0) over the same 40 sessions, 2000 bootstrap resamples, rng 1337. “Verdict” reads the CI relative to the metric’s better direction: **stride_err** is lower-better, so a CI below zero is a *win*; **drift** starts negative for every arm, so a positive Δ moves it toward zero. Wins bolded. *Source: tools/walk_headtohead.py.*

Arm	Metric	Δ [95% CI]	Verdict
GRU tuned <i>a</i> 0.4 <i>s</i> 0.3	vibe	-0.0409 [-0.0602, -0.0221]	CI<0: <i>costs vibe</i>
	stride_err	-0.2695 [-0.3058 , -0.2358]	CI<0: win
	drift	+0.0089 [-0.0255, +0.0420]	straddles
	genres	+0.525 [-0.475, +1.500]	straddles
	ground	+0.0116 [-0.0143, +0.0408]	straddles
dual latent/latent f0 <i>a</i> 0.8 <i>s</i> 0.4	vibe	+0.0297 [+0.0146 , +0.0468]	CI>0: win
	stride_err	-0.2423 [-0.2790 , -0.2053]	CI<0: win
	drift	+0.0259 [-0.0011, +0.0521]	straddles
	genres	+1.050 [+0.250 , +1.876]	CI>0: win
	ground	+0.0060 [-0.0172, +0.0314]	straddles
dual latent/cummean f0 <i>a</i> 0.8 <i>s</i> 0.5	vibe	+0.0013 [-0.0151, +0.0180]	straddles: <i>held</i>
	stride_err	-0.2545 [-0.2962 , -0.2127]	CI<0: win
	drift	+0.0351 [+0.0071 , +0.0627]	CI>0: win
	genres	+1.500 [+0.500 , +2.500]	CI>0: win
	ground	+0.0059 [-0.0163, +0.0285]	straddles
Champion (R' +M+ C') <i>a</i> 1.0 <i>s</i> 0.5	vibe	-0.1327 [-0.1629, -0.1041]	CI<0: <i>costs vibe</i>
	stride_err	-0.2638 [-0.3120 , -0.2155]	CI<0: win
	drift	-0.0422 [-0.0983, +0.0093]	straddles
	genres	-0.325 [-1.500, +0.876]	straddles
	ground	-0.0115 [-0.0467, +0.0236]	straddles

Table 34: The stage-stack batch on the one-shot surface, seven runs on seq-20260715-131139 ($n_{\text{test}} = 1431$), sorted by Recall@10. Run ids are run-20260801- plus the chunk time and hex suffix, in full: 003536-eaf81 (C0), 003536-bcedf (C1), 003646-07994 (B), 003646-8cc2d (E), 003646-dbe68 (C), 003536-d612e (D), 003646-6e3bf (A). Stage codes: a = per-step MLP, g = GRU. MRR is truncated at 10 and recomputed from the stored predictions (the stored MRR is over the full ranking and is therefore not pairable). Every H@10 was recomputed from predictions.json under the live definition and matches the stored value exactly. The paired Δ is against the in-batch control C0 (2000 resamples, rng 1337). Truncated MRR@10 and mood_coh are omitted for width and track recall closely: MRR@10 0.04555 (C0), 0.04579 (C1), then 0.02080 to 0.02894 across the five 4-stage arms; mood_coh 0.4788 (C0), 0.4856 (C1), then 0.3821 to 0.3964. R@10, H@10, mus@10 higher-better; conc, adj lower-better — and the lowest-eagerness cells belong to *refuted* arms, which is what the recall floor is for. ild rises monotonically as the arms get worse (0.4549 on C0, 0.4424 on C1, then 0.5112 to 0.5373 across the five 4-stage arms) and is omitted here for width. *Source: 2026-08-01-stage-stack-topology.md.*

Arm	id	stages / width	params	R@10 (hits)	Δ R@10 [95% CI]	H@10	conc	adj	mus@10
C0 <i>ctrl</i>	eaf81	gru h256	394 944	0.12439 (178)	—	0.024378	0.3589	0.3482	0.4552
C1 <i>cap.</i>	bcedf	gru h537	1 280 937	0.12159 (174)	−0.00280 [−.0119, +.0056]	0.023418	0.3773	0.3653	0.4556
B	07994	agga h299	1 281 407	0.08595 (123)	−0.03843 [−.0538, −.0231]	0.017558	0.3176	0.2639	0.4264
E	8cc2d	gagg h259	1 277 321	0.08246 (118)	−0.04193 [−.0566, −.0280]	0.021053	0.2704	0.2376	0.4310
C	dbe68	aggg h256	1 283 008	0.07966 (114)	−0.04472 [−.0601, −.0301]	0.021442	0.2626	0.2375	0.4324
D	d612e	gggg h229	1 282 821	0.07687 (110)	−0.04752 [−.0622, −.0328]	0.018529	0.2644	0.2326	0.4222
A	6e3bf	agag h299	1 281 407	0.07058 (101)	−0.05381 [−.0685, −.0391]	0.017374	0.2730	0.2335	0.4168

Table 35: The stage-stack batch on the generated walk: 40 held-out sessions $\times$ 20 steps at artist cap 1, paired against the *in-batch* control C0 (2000 resamples). The refuted clause is “ Δ vibe CI below zero and Δ stride_err not CI below zero”. vibe, genres, ground higher-better; stride_err lower-better; drift nearer zero better. CI bounds are quoted where the interval clears zero. *Source: 2026-08-01-stage-stack-topology.md.*

Arm	vibe	stride_err	drift	genres	ground	Δ vibe [95% CI]	V2
GRU shipped	<i>a0.4 s0.0</i>	0.609	0.399	−0.089	9.925	0.345 +0.0376 CI>0	reference
shipped engine	<i>a0.4 s0.3</i>	0.568	0.129	−0.080	10.450	0.357 −0.0033 straddles	instrument gate
C0	<i>a0.4 s0.3</i>	0.571	0.116	−0.080	10.325	0.344 base	base
C1	gru h537	0.561	0.122	−0.082	10.625	0.359 −0.0100 straddles	indistinguishable
D	gggg	0.557	0.122	−0.071	10.225	0.319 −0.0140 [−.0255, −.0022]	REFUTED ×2
C	aggg	0.548	0.113	−0.064	10.250	0.340 −0.0231 [−.0381, −.0084]	REFUTED
B	agga	0.545	0.126	−0.061	9.675	0.335 −0.0258 [−.0421, −.0108]	REFUTED
E	gagg	0.540	0.109	−0.084	9.225	0.335 −0.0316 [−.0494, −.0144]	REFUTED
A	agag	0.534	0.117	−0.085	9.125	0.336 −0.0375 [−.0559, −.0214]	REFUTED

Table 36: The tower-bank batch on S1, ten runs on `seq-20260715-131139` (1431 test sessions), sorted by hits. The batch was launched in four chunks; every run id is `run-20260802-` plus the chunk time and the hex suffix in the id column, in full: `182050-91b54`, `182050-2e6f2`, `183404-91e31`, `183404-62604`, `183404-67185`, `190415-1d662`, `190415-90866`, `190415-b5928`, `192627-17502`, `192627-836b1`. Paired Δ is a per-session paired bootstrap (2000 resamples, rng 1337) computed from the stored per-row `predictions.json`; a CI straddling zero is a *tie*, and * marks a Δ vs C1 whose CI lies entirely *below* zero. `d0.1 / d0.0` is the dropout of the byte-identical D2 / R2 pair. Capacity: the seven arms C1, B2, B3, B4, V1, V2 and V3 span 2.996 to 3.007 $\times$ the h256 baseline (mutual spread 0.405 %), D2 and R2 sit at exactly 2.000 $\times$ and C0 at 1.000 $\times$. View codes: `lat=latent`, `cum=cummean`, `del=delta`. H@10 is the *generation-3* holisticness (Section 18.2), in raw next-track units. R@10, H@10 and mus@10 higher-better; a `_conc` is lower-better but display-only and is deliberately *not* bolded — no arm’s eagerness is a claim in this campaign. Best cell per metric bolded; C1 and B4 tie exactly on R@10 and both are bolded. *Source: 2026-08-02-tower-bank-parallel-towers.md.*

Arm	id	views / width	params	R@10 (raw)	Δ vs C0 [95% CI]	Δ vs C1	H@10	mus@10	a_conc
C1 <i>ctrl</i>	91e31	1 $\times$ GRU h512	1 182 912	0.12998 (186)	+0.0070 [−.002, +.016]	—	0.02380	0.4581	0.3825
B4	1d662	lat $\times$ 4 h211	1 187 700	0.12998 (186)	+0.0070 [−.002, +.017]	+0.0000	0.02446	0.4594	0.3716
D2	62604	lat $\times$ 2 h256 d0.1	789 696	0.12369 (177)	+0.0007 [−.008, +.008]	−0.0063	0.02426	0.4567	0.3698
C0 <i>base</i>	91b54	1 $\times$ GRU h256	394 944	0.12299 (176)	—	−0.0070	0.02454	0.4558	0.3559
B3	90866	lat $\times$ 3 h256	1 184 448	0.12020 (172)	−0.0028 [−.010, +.006]	−0.0098*	0.02358	0.4554	0.3764
R2	2e6f2	lat $\times$ 2 h256 d0.0	789 696	0.11880 (170)	−0.0042 [−.013, +.004]	−0.0112*	0.02378	0.4554	0.3706
B2	67185	lat $\times$ 2 h334	1 186 560	0.11880 (170)	−0.0042 [−.013, +.003]	−0.0112*	0.02336	0.4554	0.3799
V3	17502	lat+cum+cum h256	1 184 448	0.11880 (170)	−0.0042 [−.016, +.006]	−0.0112	0.02320	0.4497	0.3581
V2	836b1	lat+cum+del h256	1 184 448	0.11600 (166)	−0.0070 [−.020, +.005]	−0.0140*	0.02245	0.4497	0.3547
V1	b5928	lat $\times$ 2+cum h256	1 184 448	0.11321 (162)	−0.0098 [−.022, +.001]	−0.0168*	0.02299	0.4500	0.3572

Table 37: The tower-bank batch on S2, the generated walk: 40 held-out sessions $\times$ 20 steps at artist cap 1, paired against the *in-batch* C0 (2000 resamples). The upper block is the primary cell `a0.4 / s0.3`; the lower block re-measures the two TRADE arms at the *deployed* cell `a0.8 / s0.5`, where the `save` clause requires the same sign. `vibe`, Δ drift, Δ genres and Δ ground higher-better; `stride_err` lower-better. CI bounds are quoted only where the interval clears zero, as in the source report, and * marks every such interval — a *gain* on Δ drift, a *refutation* on Δ vibe and Δ genres. *Source: 2026-08-02-tower-bank-parallel-towers.md.*

Arm	vibe	stride_err	Δ vibe	Δ stride_err	Δ drift	Δ genres	Δ ground
<i>Primary cell a0.4 / s0.3</i>							
C0 (base)	0.566	0.130	—	—	—	—	—
B2 TRADE	0.567	0.139	+0.0017	+0.0099	+0.0339 [+0.007, +.063]*	−0.225	+0.0044
V3 TRADE	0.556	0.124	−0.0094	−0.0045	+0.0286 [+0.001, +.059]*	−0.450	−0.0024
C1	0.563	0.130	−0.0020	+0.0015	+0.0194	−0.375	+0.0008
D2	0.559	0.129	−0.0065	+0.0005	+0.0104	−0.750	+0.0003
B4	0.566	0.139	+0.0006	+0.0103	+0.0164	−0.775	+0.0004
B3	0.557	0.135	−0.0078	+0.0060	+0.0146	−1.025*	+0.0154
V2 <i>refuted</i>	0.551	0.127	−0.0144 [−.029, −.001]*	−0.0019	+0.0259	−0.025	−0.0148
V1 <i>refuted</i>	0.548	0.119	−0.0174 [−.035, −.001]*	−0.0099	+0.0227	−1.725*	+0.0017
<i>Deployed cell a0.8 / s0.5 — both TRADEs collapse</i>							
B2 <i>refuted</i>	—	—	−0.0064 [−.012, −.001]*	−0.0107	−0.0052	−0.175	+0.0006
V3 <i>refuted</i>	—	—	−0.0112 [−.017, −.006]*	−0.0074	+0.0043	−0.400	−0.0039